

# Emergent Mathematics

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## Contents

|            |                          |           |
|------------|--------------------------|-----------|
| <b>I</b>   | <b>Symmetry</b>          | <b>2</b>  |
| 1          | Spectral Theory          | 2         |
| 2          | Fourier Analysis         | 10        |
| 3          | Peter–Weyl Theory        | 15        |
| 4          | Harish–Chandra Theory    | 20        |
| 5          | Automorphic Forms        | 23        |
| <br>       |                          |           |
| <b>II</b>  | <b>Counting</b>          | <b>26</b> |
| 6          | Graded Characters        | 26        |
| 7          | Modular Characters       | 28        |
| 8          | Transfer Matrices        | 31        |
| 9          | Determinantal Ensembles  | 33        |
| 10         | Schur Measures           | 36        |
| <br>       |                          |           |
| <b>III</b> | <b>Quantization</b>      | <b>39</b> |
| 11         | Geometric Quantization   | 39        |
| 12         | Deformation Quantization | 43        |
| 13         | Quantum Groups           | 51        |
| 14         | Integrable Systems       | 61        |

## Part I

# Symmetry

## 1 Spectral Theory

**Definition 1.1** (Vector spaces and linear maps). Let  $k$  be a field. A *vector space* over  $k$  is an abelian group  $V$  with scalar multiplication satisfying

$$a(v + w) = av + aw, \quad (a + b)v = av + bv, \quad (ab)v = a(bv), \quad 1v = v.$$

A map  $T : V \rightarrow W$  is *linear* when  $T(av + bw) = aT(v) + bT(w)$ . Its kernel and image are

$$\ker T = \{v : T(v) = 0\}, \quad \operatorname{im} T = \{T(v) : v \in V\}.$$

For  $T : V \rightarrow V$ , a scalar  $\lambda$  is an *eigenvalue* when  $\ker(T - \lambda I) \neq 0$ ; a nonzero vector in this kernel is an *eigenvector*.

**Definition 1.2** (Dual spaces and tensor products). The *dual space* of  $V$  is  $V^\vee = \operatorname{Hom}_k(V, k)$ . Its canonical evaluation pairing is

$$\operatorname{ev}_V : V^\vee \times V \longrightarrow k, \quad (\varphi, v) \longmapsto \varphi(v).$$

The algebraic transpose of  $T : V \rightarrow W$  is  $T^\vee : W^\vee \rightarrow V^\vee$ , defined by  $T^\vee(\varphi) = \varphi \circ T$ .

The *tensor product*  $V \otimes W$  is characterized by a bilinear map  $(v, w) \mapsto v \otimes w$  with the following universal property: every bilinear map  $b : V \times W \rightarrow X$  factors uniquely as

$$V \times W \longrightarrow V \otimes W \xrightarrow{\tilde{b}} X.$$

The evaluation pairing induces a linear map  $\operatorname{ev}_V : V^\vee \otimes V \rightarrow k$ .

**Definition 1.3** (Finite-dimensional vector spaces). A vector space  $V$  is *finite-dimensional* when there is a coevaluation map

$$\operatorname{coev}_V : k \longrightarrow V \otimes V^\vee$$

satisfying the snake identities

$$(\operatorname{id}_V \otimes \operatorname{ev}_V) \circ (\operatorname{coev}_V \otimes \operatorname{id}_V) = \operatorname{id}_V,$$

$$(\operatorname{ev}_V \otimes \operatorname{id}_{V^\vee}) \circ (\operatorname{id}_{V^\vee} \otimes \operatorname{coev}_V) = \operatorname{id}_{V^\vee}.$$

**Exercise 1.1** (Coevaluation uniqueness). Let  $c, d : k \rightarrow V \otimes V^\vee$  both satisfy the snake identities with  $\operatorname{ev}_V$ . Apply

$$\operatorname{id}_V \otimes \operatorname{ev}_V \otimes \operatorname{id}_{V^\vee}$$

to  $d \otimes c : k \rightarrow V \otimes V^\vee \otimes V \otimes V^\vee$  and use the snake identity for  $d$  to identify the result with  $c$ . Use the snake identity for  $c$  to identify the same result with  $d$ , and deduce  $c = d$ .

**Definition 1.4** (Exterior powers, dimension, and determinant). For  $m \geq 0$ , the  $m$ th *exterior power* of a vector space  $V$  is

$$\Lambda^m V = V^{\otimes m} / \langle v_1 \otimes \cdots \otimes v_m : v_i = v_j \text{ for some } i \neq j \rangle,$$

and the image of  $v_1 \otimes \cdots \otimes v_m$  is written  $v_1 \wedge \cdots \wedge v_m$ . An endomorphism  $T$  induces

$$\Lambda^m T(v_1 \wedge \cdots \wedge v_m) = Tv_1 \wedge \cdots \wedge Tv_m.$$

The products

$$\Lambda^r V \otimes \Lambda^s V \longrightarrow \Lambda^{r+s} V, \quad (u_1 \wedge \cdots \wedge u_r) \otimes (v_1 \wedge \cdots \wedge v_s) \longmapsto u_1 \wedge \cdots \wedge u_r \wedge v_1 \wedge \cdots \wedge v_s$$

make  $\Lambda^\bullet V = \bigoplus_{m \geq 0} \Lambda^m V$  the *exterior algebra* of  $V$ . A finite-dimensional vector space  $V$  has *dimension*  $n$  when  $\Lambda^n V \neq 0$  and  $\Lambda^{n+1} V = 0$ . Its *determinant* is the scalar  $\det(T)$  defined by the action on the highest nonzero exterior power:

$$\Lambda^n T = \det(T) \operatorname{id}_{\Lambda^n V}.$$

**Definition 1.5** (Trace). For a finite-dimensional vector space  $V$ , let  $s_{V, V^\vee} : V \otimes V^\vee \rightarrow V^\vee \otimes V$  exchange the factors. For  $T \in \operatorname{End}(V)$ , define

$$\operatorname{tr}(T) = \operatorname{ev}_V \circ s_{V, V^\vee} \circ (T \otimes \operatorname{id}_{V^\vee}) \circ \operatorname{coev}_V.$$

**Exercise 1.2** (Dimension, trace, and determinant). Prove that every finite-dimensional vector space has a unique dimension  $n$  and that

$$\operatorname{tr}(\operatorname{id}_V) = n \mathbf{1}_k.$$

Prove

$$\dim \Lambda^m V = \binom{n}{m}, \quad \Lambda^m V = 0 \quad (m > n).$$

Prove that the exterior algebra is graded-commutative:

$$\xi \wedge \eta = (-1)^{rs} \eta \wedge \xi, \quad \xi \in \Lambda^r V, \quad \eta \in \Lambda^s V.$$

Use functoriality of the highest exterior power to prove

$$\det(ST) = \det(S) \det(T), \quad T \text{ is invertible} \iff \det(T) \neq 0.$$

For  $T \in \operatorname{End}(V)$ , extend scalars to  $k[z]$  and prove

$$\det(I + zT) = \sum_{m=0}^n \operatorname{tr}(\Lambda^m T) z^m.$$

Recover  $(1 + z)^n = \sum_m \binom{n}{m} z^m$  by taking  $T = I$ .

**Definition 1.6** (Rising factorial and binomial series). For  $a \in \mathbb{C}$  and  $m \in \mathbb{Z}_{\geq 0}$ , define

$$(a)_0 = 1, \quad (a)_m = a(a+1) \cdots (a+m-1) \quad (m \geq 1).$$

For  $|z| < 1$ , define the *binomial series*

$$B_a(z) = \sum_{m=0}^{\infty} \frac{(a)_m}{m!} z^m.$$

**Exercise 1.3** (Generalized binomial theorem). Prove that the binomial series converges for  $|z| < 1$  and satisfies

$$(1 - z)B'_a(z) = aB_a(z), \quad B_a(0) = 1.$$

For the branch normalized to equal 1 at  $z = 0$ , deduce

$$B_a(z) = (1 - z)^{-a}.$$

For  $a = -n$ ,  $n \in \mathbb{Z}_{\geq 0}$ , prove that the series terminates and, after replacing  $z$  by  $-z$ , recover

$$(1 + z)^n = \sum_{m=0}^n \binom{n}{m} z^m.$$

**Definition 1.7** (Hilbert spaces and orthogonality). An *inner product* on a complex vector space is conjugate-linear in its first argument, linear in its second, positive definite, and satisfies  $\langle x, y \rangle = \overline{\langle y, x \rangle}$ . Its norm is  $\|x\| = \sqrt{\langle x, x \rangle}$ . A *Hilbert space* is an inner-product space complete for this norm.

Vectors are *orthogonal* when their inner product is zero. For a subspace  $M \subseteq H$ , define

$$M^\perp = \{x : \langle m, x \rangle = 0 \text{ for every } m \in M\}.$$

**Definition 1.8** (Bounded operators and adjoints). A linear map  $T : H \rightarrow K$  between Hilbert spaces is *bounded* when

$$\|T\| = \sup_{\|x\| \leq 1} \|Tx\| < \infty.$$

Write  $B(H, K)$  for the bounded operators and  $B(H) = B(H, H)$ . An *adjoint* of  $T$  is an operator  $T^* : K \rightarrow H$  satisfying

$$\langle x, Ty \rangle = \langle T^*x, y \rangle.$$

An operator is self-adjoint when  $T = T^*$ , unitary when  $T^*T = TT^* = I$ , and normal when  $T^*T = TT^*$ . An *orthogonal projection* is an operator  $P$  satisfying  $P = P^* = P^2$ . A bounded operator is *compact* when it maps the closed unit ball to a subset with compact closure.

**Exercise 1.4** (Hilbert-space foundations). Prove the Cauchy–Schwarz inequality, the projection theorem, and the Riesz representation theorem. Deduce the existence of adjoints and prove

$$\|T\|^2 = \|T^*T\|, \quad (\text{im } T)^\perp = \ker T^*.$$

For every closed subspace  $M \subseteq H$ , prove  $H = M \oplus M^\perp$  and construct the orthogonal projection  $P_M : H \rightarrow M$ .

**Definition 1.9** (Measurable spaces). A  $\sigma$ -algebra  $\mathcal{F}$  on a set  $X$  contains  $\emptyset$  and is closed under complements and countable unions. A *measure* is a countably additive map  $\mu : \mathcal{F} \rightarrow [0, \infty]$  with  $\mu(\emptyset) = 0$ , and  $(X, \mathcal{F}, \mu)$  is a *measure space*. A map is *measurable* when inverse images of measurable sets are measurable.

**Definition 1.10** ( $L^p$  spaces). For a measure space  $(X, \mathcal{F}, \mu)$  and  $1 \leq p < \infty$ , define

$$L^p(X, \mu) = \left\{ f : X \rightarrow \mathbb{C} \text{ measurable} : \int_X |f|^p d\mu < \infty \right\} / \text{equality a.e.}$$

Its norm is

$$\|f\|_p = \left( \int_X |f|^p d\mu \right)^{1/p}.$$

Let  $L^\infty(X, \mu)$  consist of essentially bounded measurable functions modulo equality almost everywhere, with

$$\|f\|_\infty = \text{ess sup}_{x \in X} |f(x)|.$$

**Exercise 1.5** (Lebesgue integration and convergence). Construct the Lebesgue integral from nonnegative simple functions. Prove the monotone convergence theorem and Fatou's lemma.

**Exercise 1.6** (Dominated convergence). Prove the dominated convergence theorem by applying Fatou's lemma to an integrable dominating function. Deduce convergence in  $L^1$  and justify interchanging a convergent series with an integral under an integrable majorant.

**Exercise 1.7** (Product integration). Construct product measure from measurable rectangles. Prove Tonelli's theorem for nonnegative functions and Fubini's theorem for integrable functions, stating the hypotheses that distinguish them. Derive the change-of-variables formula for a  $C^1$  diffeomorphism between open subsets of finite-dimensional Euclidean spaces.

**Exercise 1.8** ( $L^p$  geometry). Prove Hölder's and Minkowski's inequalities and deduce that  $L^p(X, \mu)$  is complete for  $1 \leq p \leq \infty$ . For  $1 < p < \infty$  with  $p^{-1} + q^{-1} = 1$ , assume that  $(X, \mathcal{F}, \mu)$  is  $\sigma$ -finite and prove that

$$L^q(X, \mu) \longrightarrow L^p(X, \mu)^*, \quad g \longmapsto \left( f \mapsto \int_X \bar{g}f \, d\mu \right)$$

is an isometric isomorphism.

**Definition 1.11** (Gamma and beta integrals). For  $s, t > 0$ , define

$$\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} \, dx, \quad B(s, t) = \int_0^1 x^{s-1} (1-x)^{t-1} \, dx.$$

**Exercise 1.9** (Beta-gamma identities). Use Tonelli's theorem and the change of variables  $x = ru$ ,  $y = r(1-u)$  to prove

$$B(s, t) = \frac{\Gamma(s)\Gamma(t)}{\Gamma(s+t)}.$$

Use integration by parts to prove  $\Gamma(s+1) = s\Gamma(s)$ , and use the pushforward of Euclidean measure under the norm map to obtain

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \quad \int_{\mathbb{R}^n} e^{-|x|^2/2} \, dx = (2\pi)^{n/2}.$$

State the uses of Tonelli, Fubini, monotone convergence, and dominated convergence in the derivation.

**Definition 1.12** ( $C^*$ -algebras, spectrum, and positivity). A  $C^*$ -algebra is a complete normed complex algebra  $A$  with a conjugate-linear involution satisfying

$$(ab)^* = b^*a^*, \quad \|ab\| \leq \|a\|\|b\|, \quad \|a^*a\| = \|a\|^2.$$

For a unital  $C^*$ -algebra, define

$$\text{spec}_A(a) = \{\lambda \in \mathbb{C} : a - \lambda 1 \text{ is not invertible}\}.$$

An element is *positive*, written  $a \geq 0$ , when  $a = b^*b$  for some  $b \in A$ . It is *self-adjoint* when  $a = a^*$  and *normal* when  $a^*a = aa^*$ .

**Definition 1.13** (Gelfand spectrum). For a commutative  $C^*$ -algebra  $A$ , let  $\hat{A}$  be the space of nonzero homomorphisms  $\chi : A \rightarrow \mathbb{C}$  with its weak-\* topology. The *Gelfand transform* is

$$\Gamma : A \longrightarrow C_0(\hat{A}), \quad \Gamma(a)(\chi) = \chi(a).$$

**Definition 1.14** (Positive functionals and states). A linear functional  $\varphi : A \rightarrow \mathbb{C}$  on a unital  $C^*$ -algebra is *positive* when  $\varphi(a^*a) \geq 0$  for every  $a \in A$ . A *state* is a positive functional satisfying  $\varphi(1) = 1$ .

**Exercise 1.10** (Spectrum and spectral radius). Prove that  $\text{spec}_A(a)$  is nonempty and compact and that

$$r(a) = \lim_{n \rightarrow \infty} \|a^n\|^{1/n}.$$

For normal  $a$ , prove  $r(a) = \|a\|$ .

**Exercise 1.11** (Commutative Gelfand theory). Prove that every character of a commutative  $C^*$ -algebra is continuous and that the Gelfand transform is an isometric  $*$ -isomorphism  $A \cong C_0(\hat{A})$ .

**Exercise 1.12** (Continuous functional calculus). For every normal element  $a$  of a unital  $C^*$ -algebra, apply commutative Gelfand theory to  $C^*(1, a)$  and prove that there is a unique unital isometric  $*$ -homomorphism

$$C(\text{spec}(a)) \longrightarrow C^*(1, a), \quad f \longmapsto f(a),$$

sending the coordinate function to  $a$ , and prove the continuous spectral mapping theorem. Deduce that positive elements have unique positive square roots and that  $a \geq 0$  if and only if  $a = a^*$  and  $\text{spec}_A(a) \subseteq [0, \infty)$ .

**Definition 1.15** (Projection-valued measures). A *projection-valued measure* on a measurable space  $X$  is a map  $E$  from measurable subsets of  $X$  to orthogonal projections on  $H$  satisfying

$$E(\emptyset) = 0, \quad E(X) = I, \quad E\left(\bigsqcup_{j=1}^{\infty} B_j\right) = \sum_{j=1}^{\infty} E(B_j)$$

for pairwise disjoint measurable sets, where the sum converges in the strong operator topology. The spectral integral  $\int_X f dE$  for bounded measurable  $f$  is defined by uniform approximation from simple functions. For a topological space  $X$ , the measure is *regular* when every scalar measure  $B \mapsto \langle x, E(B)y \rangle$  is a regular Borel measure.

**Exercise 1.13** (Positive Riesz–Markov representation). Let  $X$  be locally compact Hausdorff. Prove that every positive linear functional  $\varphi : C_c(X) \rightarrow \mathbb{C}$  that is bounded on functions supported in each compact set has a unique regular Borel measure  $\mu_\varphi$  satisfying

$$\varphi(f) = \int_X f d\mu_\varphi.$$

**Exercise 1.14** (Complex Riesz–Markov representation). Prove that the real and imaginary parts of a bounded functional on  $C_0(X)$  are differences of positive functionals. Apply the positive Riesz–Markov theorem to prove that every  $\varphi \in C_0(X)^*$  has a unique finite complex regular measure  $\mu_\varphi$ , and prove  $\|\varphi\| = |\mu_\varphi|(X)$ .

**Exercise 1.15** (Spectral distributions of states). Let  $A$  be a unital  $C^*$ -algebra, let  $a = a^* \in A$ , and let  $\varphi$  be a state. Apply the continuous functional calculus and Riesz–Markov theorem to prove that there is a unique Borel probability measure  $\mu_{\varphi,a}$  on  $\text{spec}(a)$  such that

$$\varphi(f(a)) = \int_{\text{spec}(a)} f(\lambda) d\mu_{\varphi,a}(\lambda), \quad f \in C(\text{spec}(a)).$$

Deduce

$$\varphi(a) = \int \lambda d\mu_{\varphi,a}(\lambda), \quad \varphi(a^2) - \varphi(a)^2 = \int (\lambda - \varphi(a))^2 d\mu_{\varphi,a}(\lambda).$$

**Exercise 1.16** ( $C_0(X)$  representations). Let  $X$  be locally compact Hausdorff. A representation  $\pi : C_0(X) \rightarrow B(H)$  is *nondegenerate* when  $\overline{\pi(C_0(X))H} = H$ . Prove that every nondegenerate  $*$ -representation determines scalar measures by applying the preceding exercise to  $f \mapsto \langle x, \pi(f)y \rangle$ . Use polarization and multiplicativity to assemble them into a unique regular projection-valued measure  $E_\pi$  satisfying

$$\pi(f) = \int_X f(x) dE_\pi(x), \quad f \in C_0(X).$$

**Exercise 1.17** (Bounded normal spectral theorem). For a normal operator  $T \in B(H)$ , apply the preceding representation theorem to the continuous functional calculus  $C(\text{spec}(T)) \rightarrow B(H)$  and prove that there is a unique projection-valued measure  $E_T$  on  $\text{spec}(T)$  satisfying

$$T = \int_{\text{spec}(T)} z dE_T(z), \quad f(T) = \int_{\text{spec}(T)} f(z) dE_T(z).$$

Deduce that a normal operator on a finite-dimensional Hilbert space has the intrinsic spectral decomposition

$$H = \bigoplus_{\lambda \in \text{spec}(T)} E_T(\{\lambda\})H, \quad T|_{E_T(\{\lambda\})H} = \lambda I.$$

**Exercise 1.18** (Compact spectral theorem). For a compact normal operator  $K$ , prove that every nonzero spectral value is an eigenvalue of finite multiplicity, that the nonzero eigenvalues can accumulate only at zero, and that

$$K = \sum_{\lambda \in \text{spec}(K) \setminus \{0\}} \lambda E_K(\{\lambda\})$$

in operator norm, and prove that the spectral subspaces in the displayed sum are mutually orthogonal.

**Definition 1.16** (Unbounded self-adjoint operators). An unbounded operator is a linear map  $A : \mathcal{D}(A) \rightarrow H$  on a dense domain. Its adjoint has domain

$$\mathcal{D}(A^*) = \{y \in H : x \mapsto \langle y, Ax \rangle \text{ is bounded on } \mathcal{D}(A)\}.$$

For  $y \in \mathcal{D}(A^*)$ , the vector  $A^*y$  is determined by  $\langle A^*y, x \rangle = \langle y, Ax \rangle$  for  $x \in \mathcal{D}(A)$ . The operator is *self-adjoint* when  $A = A^*$ , including equality of domains, and *closed* when its graph is closed in  $H \oplus H$ .

**Exercise 1.19** (Cayley transform). For a self-adjoint operator  $A$ , prove that  $A \pm iI$  are bijective from  $\mathcal{D}(A)$  to  $H$  with bounded inverses and that the Cayley transform

$$C_A = (A - iI)(A + iI)^{-1}$$

is unitary with  $\ker(I - C_A) = 0$ . Prove conversely that every unitary  $C$  with  $\ker(I - C) = 0$  determines a self-adjoint operator through the inverse Cayley transform.

**Exercise 1.20** (Unbounded spectral theorem). Apply the bounded normal spectral theorem to  $C_A$  and transport its projection-valued measure by the inverse Cayley map. Prove that there is a unique projection-valued measure  $E_A$  on  $\mathbb{R}$  satisfying

$$A = \int_{\mathbb{R}} \lambda dE_A(\lambda), \quad \mathcal{D}(A) = \left\{ x : \int_{\mathbb{R}} \lambda^2 d\langle x, E_A(\lambda)x \rangle < \infty \right\}.$$

**Exercise 1.21** (Stone's theorem). Use the measurable functional calculus of  $E_A$  to prove that  $e^{-itA}$  is a strongly continuous unitary group with generator  $-iA$ . Conversely, let  $(U_t)_{t \in \mathbb{R}}$  be a strongly continuous unitary group and define

$$\mathcal{D}(A) = \left\{ x : \lim_{t \rightarrow 0} i \frac{U_t x - x}{t} \text{ exists} \right\}, \quad Ax = \lim_{t \rightarrow 0} i \frac{U_t x - x}{t}.$$

Prove that  $\mathcal{D}(A)$  is dense and that  $A$  is symmetric. Use the strong integrals

$$i \int_0^\infty e^{-t} U_t dt, \quad -i \int_{-\infty}^0 e^t U_t dt$$

to prove that  $A - iI$  and  $A + iI$  are surjective. Deduce that  $A$  is self-adjoint, prove  $U_t = e^{-itA}$ , and prove uniqueness of  $A$ .

**Exercise 1.22** (Spring and string frequencies). Let  $V$  be a finite-dimensional real vector space with positive-definite mass form  $m$  and nonnegative stiffness form  $k$ . Use  $m$  to define the self-adjoint operator  $A$  by  $k(x, y) = m(x, Ay)$ . Prove that solutions of  $\ddot{x} + Ax = 0$  decompose intrinsically through the spectral projections of  $A$  and that the observed angular frequencies are  $\sqrt{\lambda}$  for  $\lambda \in \text{spec}(A) \setminus \{0\}$ .

For a stretched string of length  $L$ , wave speed  $c$ , and fixed endpoints, use the self-adjoint Dirichlet realization of  $-c^2 d^2/dx^2$  to prove

$$\omega_n = \frac{n\pi c}{L}, \quad u_n(x) = \sin\left(\frac{n\pi x}{L}\right).$$

Deduce the harmonic resonance ratios and the nodal locations measured by driving the string at each  $\omega_n$ .

**Exercise 1.23** (Spectral lines and time evolution). Let  $A$  be self-adjoint, let  $B \in B(H)$ , and set  $B(t) = e^{itA/\hbar} B e^{-itA/\hbar}$ . For eigenvalues  $\lambda, \mu$  of  $A$ , write  $P_\lambda = E_A(\{\lambda\})$  and prove

$$P_\mu B(t) P_\lambda = e^{it(\mu-\lambda)/\hbar} P_\mu B P_\lambda.$$

Use the functional calculus to prove the factorized weak identity

$$B(t) = \left( \int_{\mathbb{R}} e^{it\mu/\hbar} dE_A(\mu) \right) B \left( \int_{\mathbb{R}} e^{-it\lambda/\hbar} dE_A(\lambda) \right).$$

When  $A$  has pure-point spectrum, deduce in the weak operator topology that

$$B(t) = \sum_{\lambda, \mu} e^{it(\mu-\lambda)/\hbar} P_\mu B P_\lambda.$$

For an initial state  $\psi$ , define the line strength

$$I_{\mu\lambda}(\psi) = \|P_\mu B P_\lambda \psi\|^2.$$

Prove that a line at  $(\mu - \lambda)/\hbar$  occurs only when this quantity is nonzero, and identify vanishing spectral blocks as selection rules.

**Definition 1.17** (Hermite polynomials and functions). Define the physicists' Hermite polynomials by

$$e^{2xt-t^2} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}.$$

The normalized *Hermite functions* are

$$h_n(x) = \frac{e^{-x^2/2} H_n(x)}{(2^n n! \sqrt{\pi})^{1/2}}.$$



**Exercise 1.24** (Hermite orthogonality). Use the generating function and the Gaussian integral to prove that  $(h_n)_{n \geq 0}$  is an orthonormal family in  $L^2(\mathbb{R})$ . Derive the Hermite differential equation and recurrence relations.

**Exercise 1.25** (Mehler kernel). For  $|r| < 1$ , use the Hermite generating function and Gaussian integration to derive Mehler's identity

$$\sum_{n=0}^{\infty} r^n h_n(x) h_n(y) = \frac{1}{\sqrt{\pi(1-r^2)}} \exp\left(-\frac{(1+r^2)(x^2+y^2)-4rxy}{2(1-r^2)}\right).$$

**Exercise 1.26** (Hermite completeness). Let  $K_r$  be the integral operator with the Mehler kernel for  $0 < r < 1$ . Use its explicit Gaussian form to prove

$$K_r f \longrightarrow f \quad \text{in } L^2(\mathbb{R}) \quad (r \uparrow 1).$$

Prove that the image of  $K_r$  lies in the closed linear span of the Hermite functions and deduce that  $(h_n)_{n \geq 0}$  is complete in  $L^2(\mathbb{R})$ .

**Exercise 1.27** (Oscillator spectroscopy). On Schwartz functions let

$$(Q\psi)(x) = x\psi(x), \quad (P\psi)(x) = -i\hbar\psi'(x).$$

For

$$\hat{H} = -\frac{\hbar^2}{2M} \frac{d^2}{dx^2} + \frac{M\omega^2 x^2}{2},$$

use the Hermite functions to prove that

$$\psi_n(x) = \left(\frac{M\omega}{\hbar}\right)^{1/4} h_n\left(\sqrt{\frac{M\omega}{\hbar}}x\right), \quad E_n = \hbar\omega\left(n + \frac{1}{2}\right)$$

give the complete spectral decomposition of  $\hat{H}$ . Construct the creation and annihilation operators and compute

$$\langle \psi_k, Q\psi_n \rangle = \sqrt{\frac{\hbar}{2M\omega}} (\sqrt{n+1} \delta_{k,n+1} + \sqrt{n} \delta_{k,n-1}).$$

Deduce the spectral-line prediction  $|E_{n\pm 1} - E_n|/\hbar = \omega$  and the relative line strengths for a linearly coupled oscillator. For  $\alpha \in \mathbb{C}$ , prove convergence and normalization of

$$\psi_\alpha = e^{-|\alpha|^2/2} \sum_{n \geq 0} \frac{\alpha^n}{\sqrt{n!}} \psi_n,$$

show that its number statistics are Poisson with mean  $|\alpha|^2$ , and derive its sinusoidal position expectation. Analytically continue Mehler's identity with the usual  $t - i0$  prescription to obtain the oscillator propagator for  $t \notin (\pi/\omega)\mathbb{Z}$ ,

$$K_t(x, y) = \sqrt{\frac{M\omega}{2\pi i \hbar \sin(\omega t)}} \exp\left[\frac{iM\omega}{2\hbar \sin(\omega t)} ((x^2 + y^2) \cos(\omega t) - 2xy)\right].$$

Determine the Maslov phase obtained by continuation across each caustic. Prove

$$e^{-2\pi i \hat{H}/(\hbar\omega)} = -I, \quad e^{-4\pi i \hat{H}/(\hbar\omega)} = I,$$

and distinguish the projective and observable revival period  $2\pi/\omega$  from the exact operator period  $4\pi/\omega$ . Use these formulas to predict the spectral spacing, coherent oscillation frequency, occupation-number distribution, and revival period measured in an oscillator experiment.

**Definition 1.18** (Airy function). The Airy function  $\text{Ai}$  is the unique real solution of

$$\text{Ai}''(x) = x \text{Ai}(x)$$

normalized by

$$\text{Ai}(x) \sim \frac{x^{-1/4}}{2\sqrt{\pi}} e^{-2x^{3/2}/3} \quad (x \rightarrow +\infty).$$

Let  $a_n > 0$  be determined by  $\text{Ai}(-a_n) = 0$ , ordered increasingly.

**Exercise 1.28** (Airy spectrum of the quantum bouncer). On  $L^2((0, \infty), dz)$  consider the self-adjoint Dirichlet realization

$$\hat{H} = -\frac{\hbar^2}{2M} \frac{d^2}{dz^2} + Mgz, \quad \psi(0) = 0.$$

With

$$\ell_g = \left( \frac{\hbar^2}{2M^2g} \right)^{1/3},$$

derive

$$E_n = Mg\ell_g a_n, \quad \psi_n(z) = C_n \text{Ai}\left(\frac{z}{\ell_g} - a_n\right).$$

Use the spectral projections of  $\hat{H}$  to prove that a periodic perturbation  $B \cos(\Omega t)$  can produce a transition from the  $n$ th to the  $k$ th level only through the block  $E_{\hat{H}}(\{E_k\}) B E_{\hat{H}}(\{E_n\})$ , at the resonant frequency

$$\Omega_{kn} = \frac{Mg\ell_g}{\hbar} |a_k - a_n|.$$

Deduce the classical turning heights  $z_n = \ell_g a_n$ . Model an absorber with lower face at height  $h$ , width  $w$ , and loss strength  $\Gamma > 0$  by the potential  $-i\Gamma \mathbf{1}_{[h, h+w]}$ . In first-order decay theory, derive

$$\gamma_n(h) = \frac{2\Gamma}{\hbar} \int_h^{h+w} |\psi_n(z)|^2 dz$$

and the transmission factor  $e^{-T\gamma_n(h)}$  after flight time  $T$ . Use the Airy asymptotic to prove that the onset occurs across a height interval of order  $\ell_g$  near  $z_n$ , rather than as a sharp step. State how measured onset heights and resonance frequencies determine  $g$ , and separate the predicted Airy scales from absorber roughness, finite width, and loss.

## 2 Fourier Analysis

**Definition 2.1** (Smooth manifolds and tangent spaces). A *smooth manifold* of dimension  $n$  is a Hausdorff, second-countable space with a maximal atlas of homeomorphisms to open subsets of  $\mathbb{R}^n$  whose transition maps are smooth. A map of smooth manifolds is *smooth* when its chart representatives are smooth; it is a *diffeomorphism* when it has a smooth inverse.

The *tangent space*  $T_p M$  is the vector space of derivations  $v : C^\infty(M) \rightarrow \mathbb{R}$  at  $p$ , satisfying

$$v(fg) = f(p)v(g) + g(p)v(f).$$

The *cotangent space* is  $T_p^* M = (T_p M)^*$ . The differential of a smooth map  $F : M \rightarrow N$  is

$$dF_p : T_p M \longrightarrow T_{F(p)} N, \quad dF_p(v)(f) = v(f \circ F).$$

**Definition 2.2** (Tangent and cotangent bundles). Define

$$TM = \coprod_{p \in M} T_p M, \quad T^*M = \coprod_{p \in M} T_p^* M.$$

A smooth vector bundle is a smooth map  $\pi : E \rightarrow M$  locally isomorphic to  $U \times W \rightarrow U$  for a finite-dimensional real vector space  $W$ , by fiberwise linear maps. A section is a smooth map  $s : M \rightarrow E$  satisfying  $\pi s = \text{id}_M$ . The bundles  $TM$  and  $T^*M$  have the bundle structures induced by the differentials of chart maps and their duals.

**Definition 2.3** (Vector fields and flows). A *vector field* is a smooth section  $X : M \rightarrow TM$ . Its local flow is a family of local diffeomorphisms  $\Phi_t$  satisfying

$$\Phi_0 = \text{id}_M, \quad \Phi_{s+t} = \Phi_s \circ \Phi_t, \quad \frac{d}{dt} f(\Phi_t(p)) = X_{\Phi_t(p)}(f)$$

where both sides are defined. For vector fields  $X, Y$ , define

$$[X, Y](f) = X(Yf) - Y(Xf).$$

**Definition 2.4** (Groups, homomorphisms, and quotients). A *group* is a set  $G$  with an associative multiplication, an identity, and inverses. A *homomorphism*  $\phi : G \rightarrow H$  preserves multiplication. A subgroup  $N \leq G$  is *normal* if  $gNg^{-1} = N$  for every  $g \in G$ ; the *quotient group*  $G/N$  consists of cosets with induced multiplication.

**Definition 2.5** (Actions, orbits, and stabilizers). A *left action* of  $G$  on  $X$  is a homomorphism  $G \rightarrow \text{Bij}(X)$ . The *orbit* and *stabilizer* of  $x \in X$  are  $Gx = \{gx : g \in G\}$  and  $G_x = \{g : gx = x\}$ .

**Definition 2.6** (Representations and intertwiners). A *representation* of  $G$  on  $V$  is a homomorphism  $\rho : G \rightarrow \text{GL}(V)$ . An *intertwiner*  $T : (V, \rho) \rightarrow (W, \sigma)$  satisfies  $T\rho(g) = \sigma(g)T$ . A subspace is *invariant* if it is preserved by every  $\rho(g)$ ; a nonzero representation is *irreducible* if it has no proper nonzero invariant subspace. Direct sums and tensor products carry the actions  $\rho \oplus \sigma$  and  $g \mapsto \rho(g) \otimes \sigma(g)$ .

**Definition 2.7** (Lie groups). A *Lie group* is a topological group with a smooth manifold structure for which multiplication and inversion are smooth. Its *Lie algebra* is  $\mathfrak{g} = T_e G$  with bracket induced by left-invariant vector fields. A *one-parameter subgroup* is a smooth homomorphism  $\mathbb{R} \rightarrow G$ . The *exponential map*  $\exp : \mathfrak{g} \rightarrow G$  sends  $X$  to the value at 1 of the one-parameter subgroup with initial velocity  $X$ .

**Definition 2.8** (Riemannian metrics and Laplacians). A *Riemannian metric* on an  $n$ -manifold  $M$  is a smooth family of positive-definite inner products  $g_p$  on  $T_p M$ . Its Riemannian density is characterized by

$$|\text{vol}_g|_p(v_1 \wedge \cdots \wedge v_n) = \det(g_p(v_i, v_j))^{1/2},$$

and determines the Riemannian measure  $d\text{vol}_g$ . The gradient is defined by

$$g(\text{grad}_g f, X) = Xf.$$

The divergence is determined by

$$\int_M Xf d\text{vol}_g = - \int_M f \text{div}_g X d\text{vol}_g$$

for compactly supported  $f$ , and the Laplace–Beltrami operator is  $\Delta_g = \text{div}_g \text{grad}_g$ .

**Exercise 2.1** (Riemannian invariance). Prove that an isometry preserves the Riemannian density, gradient, divergence, and Laplace–Beltrami operator. Prove that a left-invariant Riemannian metric on a Lie group has left-invariant Riemannian measure. For the Euclidean metric, recover Lebesgue measure and  $\Delta = \sum_j \partial_j^2$ .

**Definition 2.9** (Haar normalization). Every locally compact group below is equipped with a chosen nonzero left Haar measure  $\mu$ . Haar measure on a compact group is normalized by  $\mu(G) = 1$ .

**Definition 2.10** (Regular representation). Let  $G$  be a locally compact group with left Haar measure  $\mu$ . For  $f, g \in C_c(G)$ , define

$$(f * g)(x) = \int_G f(y)g(y^{-1}x) d\mu(y).$$

The *left regular representation* on  $L^2(G)$  is

$$(\lambda_s \xi)(t) = \xi(s^{-1}t).$$

**Exercise 2.2** (Haar regular representation). Identify Haar measure on  $\mathbb{R}^n$ ,  $\mathbb{Z}^n$ , and  $\mathbb{T}^n$ . Construct normalized Haar measure on a compact Lie group from a left-invariant metric. Prove associativity of convolution, the  $L^1$  convolution inequality, and  $\|f * g\|_2 \leq \|f\|_1 \|g\|_2$ . Prove that  $g \mapsto \lambda_g$  is a strongly continuous unitary representation. For a finite-dimensional representation of a compact group, average an inner product over  $G$  to obtain an invariant one.

**Definition 2.11** (Pontryagin dual). For a locally compact abelian group  $G$ , its *Pontryagin dual*  $\widehat{G}$  is the group of continuous characters  $\chi : G \rightarrow U(1)$ , with pointwise multiplication and the compact-open topology. For  $f \in L^1(G)$ , define

$$\widehat{f}(\chi) = \int_G f(x) \overline{\chi(x)} d\mu(x).$$

The *double-dual map* sends  $x \in G$  to the character  $\chi \mapsto \chi(x)$  of  $\widehat{G}$ .

**Exercise 2.3** (Abelian Fourier duality). Identify

$$\widehat{\mathbb{R}^n} \cong \mathbb{R}^n, \quad \widehat{\mathbb{Z}^n} \cong \mathbb{T}^n, \quad \widehat{\mathbb{T}^n} \cong \mathbb{Z}^n,$$

and verify the double-dual map in each case. For a linear subspace  $W \subseteq \mathbb{R}^n$  and a full-rank lattice  $\Gamma \subseteq \mathbb{R}^n$ , compute their annihilators and identify the duals of  $\mathbb{R}^n/W$  and  $\mathbb{R}^n/\Gamma$ .

**Definition 2.12** (Fourier series). On  $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ , write

$$dm_{\mathbb{T}}(x) = \frac{dx}{2\pi}$$

for normalized Haar measure and define

$$\widehat{f}(k) = \int_{\mathbb{T}} f(x) e^{-ikx} dm_{\mathbb{T}}(x), \quad k \in \mathbb{Z}.$$

The *Fourier series* of  $f$  is  $\sum_{k \in \mathbb{Z}} \widehat{f}(k) e^{ikx}$ .

**Definition 2.13** (Schwartz space and convolution). The *Schwartz space*  $\mathcal{S}(\mathbb{R}^n)$  consists of the smooth functions  $f$  satisfying

$$\sup_{x \in \mathbb{R}^n} |x^\alpha \partial^\beta f(x)| < \infty$$

for all multi-indices  $\alpha, \beta$ . For integrable functions, define

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y)g(x - y) dy.$$

**Definition 2.14** (Fourier transform). Let  $V$  be an  $n$ -dimensional real Euclidean space with Lebesgue measure  $dq$ . Equip  $V^*$  with its dual Euclidean structure and the resulting Lebesgue measure  $dp$ . For  $f \in \mathcal{S}(V)$ , the semiclassical Fourier transform is

$$(\mathcal{F}_\hbar f)(p) = (2\pi\hbar)^{-n/2} \int_V e^{-ip(q)/\hbar} f(q) dq.$$

Write  $\mathcal{F} = \mathcal{F}_1$ .

**Exercise 2.4** (Fourier series and Plancherel). Prove that  $(e^{ikx})_{k \in \mathbb{Z}}$  is a complete orthonormal family in  $L^2(\mathbb{T})$  and derive

$$f = \sum_{k \in \mathbb{Z}} \hat{f}(k) e^{ikx}, \quad \|f\|_2^2 = \sum_{k \in \mathbb{Z}} |\hat{f}(k)|^2.$$

**Exercise 2.5** (Euclidean Fourier inversion). Prove Fourier inversion on  $\mathcal{S}(V)$  and prove that  $\mathcal{F}_\hbar$  extends uniquely to a unitary operator on  $L^2(V)$ . For  $v \in V$ , prove

$$\mathcal{F}_\hbar(-i\hbar d_v) \mathcal{F}_\hbar^{-1} \hat{f}(p) = p(v) \hat{f}(p).$$

**Exercise 2.6** (Free-particle dispersion). Substitute the plane wave  $\psi(q, t) = e^{i(k(q) - \omega t)}$ , with  $k \in V^*$ , into the free Schrödinger equation and prove

$$\hbar\omega = \frac{\hbar^2 \|k\|^2}{2m}.$$

Set  $p = \hbar k$  and  $E = \hbar\omega$ . Derive

$$E(p) = \frac{\|p\|^2}{2m}, \quad v_{\text{group}} = \frac{p^\sharp}{m},$$

where  $p^\sharp \in V$  is the Euclidean-dual vector to  $p$ . Deduce the time-of-flight displacement of a wave packet concentrated near  $p_0$  and the de Broglie relation between its measured wavelength and momentum.

**Definition 2.15** (Bessel functions). For  $\alpha > -1$ , the Bessel function of the first kind is

$$J_\alpha(z) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m + \alpha + 1)} \left(\frac{z}{2}\right)^{2m + \alpha},$$

with a branch of  $z^\alpha$  fixed when  $\alpha \notin \mathbb{Z}$ .

**Exercise 2.7** (Radial Bessel transform). Prove

$$z^2 J_\alpha''(z) + z J_\alpha'(z) + (z^2 - \alpha^2) J_\alpha(z) = 0$$

and the recurrence identities

$$J_{\alpha-1}(z) + J_{\alpha+1}(z) = \frac{2\alpha}{z} J_\alpha(z), \quad J_{\alpha-1}(z) - J_{\alpha+1}(z) = 2J_\alpha'(z).$$

Average the Euclidean Fourier kernel over spheres and express the Fourier transform of a radial Schwartz function on  $\mathbb{R}^n$  as a Hankel transform with kernel  $J_{n/2-1}$ . Derive the corresponding inversion identity.

**Exercise 2.8** (Bessel diffraction). Let  $j_{\nu,k}$  be the  $k$ th positive zero of  $J_\nu$ . Separate the wave equation on a circular membrane of radius  $R$  with Dirichlet boundary and prove that its normal-mode frequencies are

$$\omega_{m,k} = \frac{c}{R} j_{|m|,k},$$

with radial factor  $J_{|m|}(j_{|m|,k}r/R)$ . Deduce the nodal circles and angular nodal lines and state how measured resonance ratios determine the ratios of Bessel zeros without knowledge of  $c/R$ .

For a uniformly illuminated circular aperture of radius  $a$ , average the Fraunhofer kernel over the aperture, put  $k = 2\pi/\lambda$ , and derive

$$\frac{I(\theta)}{I(0)} = \left[ \frac{2J_1(ka \sin \theta)}{ka \sin \theta} \right]^2.$$

Prove that the dark-ring angles satisfy  $ka \sin \theta_k = j_{1,k}$  and obtain the small-angle prediction  $\theta_1 \sim j_{1,1}\lambda/(2\pi a)$ . Identify which membrane resonances and diffraction minima measure zeros of the same Bessel functions.

**Definition 2.16** (Lattices and reciprocal lattices). A *lattice* in  $\mathbb{R}^n$  is a subgroup  $\Gamma = A\mathbb{Z}^n$  for some invertible linear map  $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ . Its covolume is  $\text{vol}(\Gamma) = |\det A|$ , and a fundamental cell is  $F = A[0, 1)^n$ . Its *reciprocal lattice* is the annihilator

$$\Gamma^* = \{\xi \in \mathbb{R}^n : \xi \cdot \gamma \in 2\pi\mathbb{Z} \text{ for every } \gamma \in \Gamma\}.$$

**Definition 2.17** (Periodic summation and structure factors). For  $f \in \mathcal{S}(\mathbb{R}^n)$ , its  $\Gamma$ -*periodization* is

$$P_\Gamma f(x) = \sum_{\gamma \in \Gamma} f(x + \gamma).$$

For a finite motif  $B \subset \mathbb{R}^n$  with scattering weights  $c_b$ , its *structure factor* is

$$S_B(q) = \sum_{b \in B} c_b e^{-iq \cdot b}.$$

**Exercise 2.9** (Poisson summation). Prove that  $P_\Gamma f$  converges with all derivatives and compute its Fourier coefficients. With the Fourier-transform normalization above, prove

$$P_\Gamma f(x) = \frac{(2\pi)^{n/2}}{\text{vol}(\Gamma)} \sum_{\xi \in \Gamma^*} \mathcal{F}f(\xi) e^{i\xi \cdot x}.$$

Set  $x = 0$  to obtain the Poisson summation formula. Derive the Jacobi theta transformation by applying it to a Gaussian.

**Exercise 2.10** (Crystal diffraction and Bragg peaks). Let  $\Gamma_N$  be a finite rectangular portion of  $\Gamma$ . In the single-scattering approximation, prove that the amplitude at momentum transfer  $q$  factors as

$$\mathcal{A}_N(q) = S_B(q) \sum_{\gamma \in \Gamma_N} e^{-iq \cdot \gamma}.$$

Use geometric sums and Poisson summation to show that the normalized intensity concentrates at  $q \in \Gamma^*$  as the crystal grows. For elastic scattering with incident and outgoing wavevectors of magnitude  $2\pi/\lambda$ , derive the Laue condition and, for planes spaced by  $d$ , the Bragg relation

$$2d \sin \theta = m\lambda.$$

Show how zeros of  $S_B$  produce missing reflections. Compare the peak condition with W. H. Bragg and W. L. Bragg, *Proceedings of the Royal Society A* **88** (1913).

### 3 Peter–Weyl Theory

**Exercise 3.1** (Maschke’s theorem). Let  $G$  be finite and let the characteristic of  $k$  not divide  $|G|$ . Average an arbitrary projection over  $G$  to construct an invariant complement to every invariant subspace. Deduce that every finite-dimensional representation is a direct sum of irreducible representations.

**Exercise 3.2** (Schur’s lemma). Prove that a nonzero intertwiner between irreducible complex finite-dimensional representations is an isomorphism and that every endomorphism of an irreducible representation is scalar. Deduce uniqueness of irreducible multiplicities in a decomposition.

**Definition 3.1** (Characters). The *character* of a finite-dimensional representation  $V$  is the class function  $\chi_V(g) = \text{Tr}(\rho_V(g))$ . For class functions on a finite group, define

$$\langle f, h \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{f(g)} h(g).$$

**Exercise 3.3** (Character orthogonality and decomposition). Use averaging of linear maps and Schur’s lemma to prove orthonormality of the irreducible characters. Prove that they form a complete orthonormal family in the class functions and that  $\langle \chi_U, \chi_V \rangle_G = \dim \text{Hom}_G(U, V)$ . Deduce the multiplicity formula and the character table constraints.

**Definition 3.2** (Matrix coefficients). A *unitary representation* of a topological group  $G$  on a Hilbert space  $H_\pi$  is a homomorphism  $\pi : G \rightarrow U(H_\pi)$  such that  $g \mapsto \pi(g)x$  is continuous for every  $x \in H_\pi$ . Its *matrix coefficients* are

$$g \longmapsto \langle y, \pi(g)x \rangle, \quad x, y \in H_\pi.$$

It is *irreducible* when it has no proper nonzero closed invariant subspace. The *unitary dual*  $\widehat{G}$  is the set of unitary-equivalence classes of irreducible unitary representations.

**Exercise 3.4** (Exponential map). Prove existence and uniqueness of the one-parameter subgroup generated by  $X \in \mathfrak{g}$ . For a closed matrix group, prove that it is  $t \mapsto e^{tX}$ , identify the commutator bracket, and compute the first nontrivial terms of the Baker–Campbell–Hausdorff formula.

**Definition 3.3** (Infinitesimal representations). For a finite-dimensional smooth representation  $\rho : G \rightarrow \text{GL}(V)$ , its *infinitesimal representation* is

$$d\rho : \mathfrak{g} \rightarrow \text{End}(V), \quad d\rho(X) = \left. \frac{d}{dt} \right|_0 \rho(\exp tX).$$

**Exercise 3.5** (Compact convolution operators). Let  $G$  be compact with normalized Haar measure and  $k \in C(G)$ . Prove that

$$(T_k f)(x) = \int_G k(y^{-1}x) f(y) dy$$

defines a compact operator on  $L^2(G)$  by uniformly approximating its continuous kernel by finite sums of product functions. Prove that  $T_k$  commutes with left translations and that  $k(g^{-1}) = \overline{k(g)}$  makes  $T_k$  self-adjoint. Deduce that every nonzero eigenspace of such a self-adjoint convolution operator is a finite-dimensional invariant subspace of the left regular representation.

**Exercise 3.6** (Density of matrix coefficients). Construct continuous convolution kernels that approximate the identity on  $C(G)$ . Apply the compact spectral theorem to their self-adjoint symmetrizations and prove that the linear span of matrix coefficients of finite-dimensional unitary representations is uniformly dense in  $C(G)$  and dense in  $L^2(G)$ .

**Exercise 3.7** (Schur orthogonality). For irreducible finite-dimensional unitary representations  $\pi$  and  $\sigma$ , prove that

$$A \mapsto \int_G \pi(g) A \sigma(g)^* dg$$

is zero when  $\pi \not\cong \sigma$  and is  $A \mapsto \text{tr}(A) I_{H_\pi} / \dim H_\pi$  when  $\pi = \sigma$ . Deduce that the coefficient map identifies  $H_\pi \otimes H_\pi^\vee$ , after multiplication by  $\sqrt{\dim H_\pi}$ , with an isometric subspace of  $L^2(G)$ , and that distinct irreducible coefficient spaces are orthogonal, where

$$C_\pi : H_\pi \otimes H_\pi^\vee \longrightarrow L^2(G), \quad C_\pi(x \otimes \varphi)(g) = \varphi(\pi(g)x).$$

**Exercise 3.8** (Peter–Weyl decomposition). Use density and Schur orthogonality to prove

$$L^2(G) \cong \widehat{\bigoplus_{\pi \in \widehat{G}} H_\pi \otimes H_\pi^\vee},$$

with the left regular representation acting as  $\pi \otimes I_{H_\pi^\vee}$  on the  $\pi$ -summand. For  $f \in L^2(G) \subseteq L^1(G)$ , define  $\widehat{f}(\pi) = \int_G f(g) \pi(g)^* dg$  and prove

$$f = \sum_{\pi \in \widehat{G}} (\dim H_\pi) \text{tr}(\widehat{f}(\pi) \pi(\cdot)), \quad \|f\|_2^2 = \sum_{\pi \in \widehat{G}} (\dim H_\pi) \text{tr}(\widehat{f}(\pi)^* \widehat{f}(\pi)),$$

where the first sum converges in  $L^2(G)$ . Prove uniform convergence on  $C(G)$  for the summability means supplied by the approximate identities in the density exercise.

**Exercise 3.9** (Character inversion). Let  $G$  be compact with normalized Haar measure, and let  $f$  be a finite linear combination of irreducible characters. For an irreducible unitary representation  $\pi$ , set

$$\widehat{f}(\pi) = \int_G f(s) \pi(s^{-1}) ds.$$

If  $f$  is central, use Schur's lemma to write  $\widehat{f}(\pi) = c_\pi I_{d_\pi}$ . Derive

$$f(e) = \sum_{\pi \in \widehat{G}} d_\pi \text{Tr}(\widehat{f}(\pi)) = \sum_{\pi \in \widehat{G}} d_\pi^2 c_\pi$$

and prove that convolution by  $f$  acts by the scalar  $c_\pi$  on every copy of  $\pi$  in  $L^2(G)$ .

**Definition 3.4** (Rotations and spin). Define

$$\begin{aligned} SO(3) &= \{R \in \text{End}(\mathbb{R}^3) : R^* R = \text{id}, \det R = 1\}, \\ SU(2) &= \{U \in \text{End}(\mathbb{C}^2) : U^* U = \text{id}, \det U = 1\}. \end{aligned}$$

An *angular-momentum representation* is a finite-dimensional smooth unitary representation of  $SU(2)$ ; with Planck constant  $\hbar$ , its infinitesimal generators  $J_1, J_2, J_3$  are normalized by  $[J_i, J_j] = i\hbar \varepsilon_{ijk} J_k$ .

**Exercise 3.10** ( $SU(2)$  double-covers  $SO(3)$ ). Let  $SU(2)$  act by conjugation on the real space of traceless Hermitian  $2 \times 2$  operators. Prove that the resulting homomorphism  $SU(2) \rightarrow SO(3)$  is surjective with kernel  $\{\pm \text{id}\}$ . Deduce the effect of a  $2\pi$  rotation in integer- and half-integer-spin representations and the  $4\pi$  restoration observed by neutron interferometry.



**Exercise 3.11** (Irreducible representations of  $SU(2)$ ). Using raising and lowering operators, prove that every finite-dimensional irreducible smooth complex representation is indexed by  $j \in \{0, \frac{1}{2}, 1, \dots\}$ , has dimension  $2j + 1$ , and has  $J_3$ -eigenvalues  $m\hbar$  for  $m = -j, -j + 1, \dots, j$ . Compute the spectrum of  $J^2 = J_1^2 + J_2^2 + J_3^2$ . For  $j = \frac{1}{2}$ , relate the two  $J_3$  outcomes to the two Stern–Gerlach beams.

**Exercise 3.12** (Clebsch–Gordan decomposition). Prove

$$V_{j_1} \otimes V_{j_2} \cong \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V_j.$$

Construct coupled angular-momentum eigenstates from highest-weight vectors and compute the singlet and triplet decomposition of  $V_{1/2} \otimes V_{1/2}$ .

**Definition 3.5** (Gauss hypergeometric function). Using the rising factorial defined in Spectral Theory, for  $c \notin \mathbb{Z}_{\leq 0}$  the *Gauss hypergeometric function* is

$${}_2F_1\left(\begin{matrix} a, b \\ c \end{matrix}; z\right) = \sum_{m=0}^{\infty} \frac{(a)_m (b)_m}{(c)_m} \frac{z^m}{m!}.$$

It is interpreted as a finite sum when a numerator parameter is a nonpositive integer and otherwise converges for  $|z| < 1$ . The Gauss hypergeometric operator is

$$\mathcal{L}_{a,b,c} = z(1-z) \frac{d^2}{dz^2} + (c - (a+b+1)z) \frac{d}{dz} - ab.$$

**Exercise 3.13** (Gauss equation and integral). Prove that  ${}_2F_1(a, b; c; z)$  is the solution of  $\mathcal{L}_{a,b,c}f = 0$  analytic at  $z = 0$  with  $f(0) = 1$ . Derive Euler’s integral

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1} (1-u)^{c-b-1} (1-zu)^{-a} du,$$

and Gauss’s summation

$${}_2F_1(a, b; c; 1) = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}.$$

State the parameter conditions for the integral and the summation.

**Definition 3.6** (Kummer and Whittaker functions). For  $b \notin \mathbb{Z}_{\leq 0}$ , define the Kummer function

$$M(a, b, z) = {}_1F_1(a; b; z) = \sum_{m=0}^{\infty} \frac{(a)_m}{(b)_m} \frac{z^m}{m!}.$$

For  $2\mu + 1 \notin \mathbb{Z}_{\leq 0}$ , define

$$M_{\lambda,\mu}(z) = e^{-z/2} z^{\mu+1/2} M\left(\mu - \lambda + \frac{1}{2}, 2\mu + 1, z\right),$$

with a branch of  $z^{\mu+1/2}$  fixed. The Whittaker function  $W_{\lambda,\mu}$  is the solution of

$$w'' + \left(-\frac{1}{4} + \frac{\lambda}{z} + \frac{1/4 - \mu^2}{z^2}\right) w = 0$$

normalized by  $W_{\lambda,\mu}(z) \sim e^{-z/2} z^{\lambda}$  as  $z \rightarrow +\infty$ .

**Definition 3.7** (Generalized Laguerre polynomials). For  $\alpha > -1$  and  $n \in \mathbb{Z}_{\geq 0}$ , define

$$L_n^{(\alpha)}(x) = \frac{x^{-\alpha} e^x}{n!} \frac{d^n}{dx^n} \left( e^{-x} x^{n+\alpha} \right) = \frac{(\alpha+1)_n}{n!} M(-n, \alpha+1, x).$$

Write  $L_n = L_n^{(0)}$ .

**Exercise 3.14** (Confluence and Laguerre functions). Prove locally uniformly in  $z$  that

$$\lim_{B \rightarrow +\infty} {}_2F_1 \left( \begin{matrix} a, B \\ c \end{matrix}; \frac{z}{B} \right) = M(a, c, z),$$

and derive Kummer's equation

$$zy'' + (b-z)y' - ay = 0.$$

Apply the substitution  $w(z) = e^{-z/2} z^{\mu+1/2} y(z)$  to obtain the Whittaker equation with  $a = \mu - \lambda + \frac{1}{2}$  and  $b = 2\mu + 1$ .

Derive the equality of the two displayed definitions of  $L_n^{(\alpha)}$ . Use the Rodrigues formula and integration by parts to prove

$$\int_0^\infty L_m^{(\alpha)}(x) L_n^{(\alpha)}(x) x^\alpha e^{-x} dx = \frac{\Gamma(n+\alpha+1)}{n!} \delta_{mn},$$

and derive

$$(n+1)L_{n+1}^{(\alpha)}(x) = (2n+\alpha+1-x)L_n^{(\alpha)}(x) - (n+\alpha)L_{n-1}^{(\alpha)}(x).$$

Prove

$$\sum_{n \geq 0} L_n^{(\alpha)}(x) t^n = (1-t)^{-\alpha-1} \exp\left(-\frac{xt}{1-t}\right), \quad |t| < 1,$$

and obtain the corresponding coefficient contour integral around  $t = 0$ .

**Definition 3.8** (Jacobi polynomials). For  $\alpha, \beta > -1$ , the *Jacobi polynomial* is

$$P_n^{(\alpha, \beta)}(x) = \frac{(\alpha+1)_n}{n!} {}_2F_1 \left( \begin{matrix} -n, n+\alpha+\beta+1 \\ \alpha+1 \end{matrix}; \frac{1-x}{2} \right).$$

**Exercise 3.15** (Jacobi spectral equation). Prove

$$\left( (1-x^2) \frac{d^2}{dx^2} + (\beta - \alpha - (\alpha + \beta + 2)x) \frac{d}{dx} \right) P_n^{(\alpha, \beta)} = -n(n + \alpha + \beta + 1) P_n^{(\alpha, \beta)}.$$

Derive Jacobi orthogonality on  $[-1, 1]$  with weight  $(1-x)^\alpha(1+x)^\beta$  and derive the three-term recurrence.

**Exercise 3.16** (Spherical harmonics and atomic orbitals). Identify  $S^2$  with  $SO(3)/SO(2)$  and  $L^2(S^2)$  with the right  $SO(2)$ -invariant subspace of  $L^2(SO(3))$ . Use Peter-Weyl to obtain

$$L^2(S^2) \cong \widehat{\bigoplus_{\ell=0}^{\infty} V_\ell}$$

For normalized  $J_3$ -eigenstates  $|\ell m\rangle$ , define the Wigner matrix coefficients

$$D_{mn}^\ell(g) = \langle \ell m | \pi_\ell(g) | \ell n \rangle.$$

Prove that the functions  $Y_{\ell m} \propto D_{m0}^\ell$ ,  $-\ell \leq m \leq \ell$ , form a complete orthogonal family in  $V_\ell$ . For the unit round sphere, set

$$\Delta_{S^2} = \frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{\sin^2 \theta} \partial_\phi^2.$$

Derive  $\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1)Y_{\ell m}$  and completeness of the spherical harmonics. Derive the Jacobi realization

$$Y_{\ell m}(\theta, \phi) = C_{\ell m} e^{im\phi} (\sin \theta)^{|m|} P_{\ell-|m|}^{(|m|, |m|)}(\cos \theta)$$

with  $C_{\ell m}$  fixed by unit norm. On

$$L^2(\mathbb{R}^3, d^3x) \cong L^2((0, \infty), r^2 dr) \otimes L^2(S^2, d\Omega),$$

consider the Coulomb Hamiltonian

$$\hat{H}_C = -\frac{\hbar^2}{2M} \Delta_{\mathbb{R}^3} - \frac{\kappa}{r}, \quad M, \kappa > 0.$$

Separate  $\hat{H}_C$  in spherical coordinates and prove that its radial operator on the  $\ell$ th angular sector is

$$-\frac{\hbar^2}{2M} \left( \frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - \frac{\ell(\ell+1)}{r^2} \right) - \frac{\kappa}{r}$$

on  $L^2((0, \infty), r^2 dr)$ . Recover  $\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$ . Put

$$a_0 = \frac{\hbar^2}{M\kappa}, \quad \beta = \frac{\sqrt{-2ME}}{\hbar}, \quad \rho = 2\beta r$$

for  $E < 0$ , and set  $u(r) = rR(r)$ . Reduce the radial eigenvalue equation to the Whittaker equation with

$$\lambda = \frac{1}{a_0\beta}, \quad \mu = \ell + \frac{1}{2}.$$

Use regularity at the origin and square-integrability at infinity to prove that

$$\ell + 1 - \lambda = -n_r, \quad n_r \in \mathbb{Z}_{\geq 0}.$$

Writing  $n = n_r + \ell + 1$ , deduce

$$E_n = -\frac{M\kappa^2}{2\hbar^2 n^2}, \quad R_{n\ell}(r) \propto e^{-r/(na_0)} \left( \frac{2r}{na_0} \right)^\ell L_{n-\ell-1}^{(2\ell+1)} \left( \frac{2r}{na_0} \right).$$

Prove that the spatial multiplicity of  $E_n$  is  $n^2$  and derive the spectral-line frequencies

$$\omega_{n_i \rightarrow n_f} = \frac{M\kappa^2}{2\hbar^3} \left| \frac{1}{n_f^2} - \frac{1}{n_i^2} \right|.$$

Determine the angular nodes of the  $s$ ,  $p$ , and  $d$  orbitals and relate them to symmetry-enforced zeros in photoelectron angular distributions in the plane-wave final-state approximation. Compare the predicted momentum-space nodal suppression with Weiß et al., *Nature Communications* **6** (2015), and distinguish the orbital-symmetry prediction, plane-wave approximation, measured intensity, and final-state scattering corrections.

**Definition 3.9** (Clebsch–Gordan and Wigner coefficients). Choose the normalized  $J_3$ -eigenstates in each irreducible angular-momentum representation by the Condon–Shortley ladder convention

$$J_-|jm\rangle = \hbar\sqrt{(j+m)(j-m+1)}|j, m-1\rangle.$$

Define

$$|jm\rangle = \sum_{m_1+m_2=m} C_{j_1m_1, j_2m_2}^{jm} |j_1m_1\rangle \otimes |j_2m_2\rangle,$$

with all coefficients real and with the coefficient of  $|j_1, j_1\rangle \otimes |j_2, j-j_1\rangle$  in  $|jj\rangle$  positive, and set

$$\begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & m_3 \end{pmatrix} = \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j+1}} C_{j_1m_1, j_2m_2}^{j, -m_3}.$$

Define the Wigner  $6j$  symbol by

$$|(j_1j_2)j_{12}, j_3; jm\rangle = \sum_{j_{23}} (-1)^{j_1+j_2+j_3+j} \sqrt{(2j_{12}+1)(2j_{23}+1)} \begin{Bmatrix} j_1 & j_2 & j_{12} \\ j_3 & j & j_{23} \end{Bmatrix} |j_1, (j_2j_3)j_{23}; jm\rangle.$$

**Exercise 3.17** (Wigner and Jacobi functions). For a rotation through angle  $\beta$  about the second axis, set

$$d_{m'm}^j(\beta) = \langle jm' | e^{-i\beta J_2/\hbar} | jm \rangle.$$

Use the  $SU(2)$  differential equation and the value at  $\beta = 0$  to express  $d_{m'm}^j(\beta)$  as a normalized product of powers of  $\sin(\beta/2)$  and  $\cos(\beta/2)$  with a Jacobi polynomial in  $\cos\beta$ . Deduce its orthogonality from Peter–Weyl. Use the Clebsch–Gordan decomposition to derive the linearization formula for a product of two Wigner coefficients and identify its coefficients as products of Clebsch–Gordan coefficients.

**Exercise 3.18** (Angular amplitudes). Derive the orthogonality relations for the Clebsch–Gordan coefficients and the triangle and magnetic selection rules for the  $3j$  symbols. Prove the Gaunt identity

$$\int_{S^2} \overline{Y_{\ell_f m_f}} Y_{1q} Y_{\ell_i m_i} d\Omega = (-1)^{m_f} \sqrt{\frac{3(2\ell_i+1)(2\ell_f+1)}{4\pi}} \begin{pmatrix} \ell_i & 1 & \ell_f \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell_i & 1 & \ell_f \\ m_i & q & -m_f \end{pmatrix}.$$

Deduce the electric-dipole rules  $\ell_f = \ell_i \pm 1$  and  $m_f = m_i + q$ , and prove that the relative Zeeman-resolved line intensities are the squared absolute values of the second  $3j$  symbol after a common reduced matrix element is removed. Use the  $6j$  identity to derive the corresponding relative intensities when orbital and spin angular momenta are coupled in different orders. For an oriented initial state, express the dipole radiation pattern as the squared absolute value of the resulting spherical-harmonic amplitude and identify its symmetry-enforced dark directions and absent spectral lines.

## 4 Harish–Chandra Theory

**Definition 4.1** (The rank-one symmetric-space model). Let

$$G = SL_2(\mathbb{R}), \quad K = SO(2),$$

and let  $G$  act on the upper half-plane by fractional linear transformations. Define

$$a_r = \begin{pmatrix} e^{r/2} & 0 \\ 0 & e^{-r/2} \end{pmatrix}, \quad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix},$$

with  $r, x \in \mathbb{R}$ . Give the upper half-plane the metric  $ds^2 = (dx^2 + dy^2)/y^2$  and fix its curvature radius to be one.

**Exercise 4.1** (Rank-one Iwasawa and Cartan decompositions). Use the polar and QR decompositions of an element of  $SL_2(\mathbb{R})$  to prove

$$K \times \mathbb{R} \times \mathbb{R} \longrightarrow G, \quad (k, r, x) \longmapsto ka_r n_x$$

is a diffeomorphism and

$$G = K\{a_r : r \geq 0\}K.$$

Prove that  $K$  is the stabilizer of  $i$  and that the orbit map identifies  $G/K$  isometrically with the hyperbolic plane. Identify the two-point Weyl group and prove that a  $K$ -bi-invariant function is determined by its value on  $a_r$ ,  $r \geq 0$ .

**Definition 4.2** (Rank-one spherical transform). For  $\lambda \in \mathbb{R}$ , let  $\varphi_\lambda$  be the normalized smooth  $K$ -invariant Laplace–Beltrami eigenfunction on  $G/K = \mathbb{H}^2$  satisfying

$$\Delta_{\mathbb{H}^2} \varphi_\lambda = -(\lambda^2 + \tfrac{1}{4})\varphi_\lambda, \quad \varphi_\lambda(eK) = 1.$$

Regarded as a  $K$ -bi-invariant function on  $G$  and written in the radial coordinate  $a_r K$ , it is determined by

$$\varphi_\lambda''(r) + \coth r \varphi_\lambda'(r) = -(\lambda^2 + \tfrac{1}{4})\varphi_\lambda(r), \quad \varphi_\lambda(0) = 1, \quad \varphi_\lambda'(0) = 0.$$

For a compactly supported smooth radial function, define

$$\tilde{f}(\lambda) = 2\pi \int_0^\infty f(r) \varphi_\lambda(r) \sinh r \, dr.$$

Define  $c(\lambda)$  by the asymptotic normalization

$$\varphi_\lambda(r) = c(\lambda)e^{(i\lambda-1/2)r} + c(-\lambda)e^{(-i\lambda-1/2)r} + o(e^{-r/2})$$

for real  $\lambda \neq 0$ .

**Exercise 4.2** (Conical eigenfunctions on  $\mathbb{H}^2$ ). For the sign convention in which the Laplace–Beltrami operator is

$$\Delta_{\mathbb{H}^2} = y^2(\partial_x^2 + \partial_y^2),$$

derive its radial part

$$\Delta_{\text{rad}} = \frac{d^2}{dr^2} + \coth r \frac{d}{dr}.$$

Show that the normalized radial eigenfunction satisfying

$$\Delta \varphi_\lambda = -(\lambda^2 + \tfrac{1}{4})\varphi_\lambda, \quad \varphi_\lambda(0) = 1,$$

is the conical Legendre function

$$\varphi_\lambda(r) = P_{-1/2+i\lambda}(\cosh r).$$

For real  $\lambda, \mu$ , derive the Wronskian identity

$$(\mu^2 - \lambda^2) \int_0^R \varphi_\lambda(r) \varphi_\mu(r) \sinh r \, dr = \left[ \sinh r (\varphi_\mu \varphi'_\lambda - \varphi_\lambda \varphi'_\mu) \right]_0^R.$$

Fix one self-adjoint boundary condition  $a\varphi(R) + b\varphi'(R) = 0$ , with  $(a, b) \neq (0, 0)$ . Prove orthogonality only for the discrete parameters  $\lambda_j$  satisfying this condition. Compare the large- $r$  equation with a constant-coefficient equation and show how the discrete parameters become continuous as  $R \rightarrow \infty$ .

**Exercise 4.3** (Harish–Chandra normalization). Use the large- $r$  asymptotics of the conical Legendre function and the Wronskian identity to derive

$$c(\lambda) = \frac{\Gamma(i\lambda)}{\sqrt{\pi} \Gamma(\frac{1}{2} + i\lambda)}, \quad |c(\lambda)|^{-2} = \pi \lambda \tanh(\pi \lambda).$$

**Exercise 4.4** (Mehler–Fock inversion). For  $\lambda, \mu > 0$ , prove in the distributional sense that

$$2\pi \int_0^\infty \varphi_\lambda(r) \varphi_\mu(r) \sinh r \, dr = \frac{2\pi}{\lambda \tanh(\pi \lambda)} \delta(\lambda - \mu).$$

Deduce the Mehler–Fock inversion and Plancherel formulas

$$f(r) = \frac{1}{2\pi^2} \int_0^\infty \tilde{f}(\lambda) P_{-1/2+i\lambda}(\cosh r) |c(\lambda)|^{-2} d\lambda,$$

$$\|f\|_2^2 = \frac{1}{2\pi^2} \int_0^\infty |\tilde{f}(\lambda)|^2 |c(\lambda)|^{-2} d\lambda.$$

**Exercise 4.5** (Hyperbolic wave propagation). Let  $u_{tt} = \Delta_{\mathbb{H}^2} u$  have radial initial data  $u(0, r) = f(r)$  and  $u_t(0, r) = g(r)$  in  $C_c^\infty$ . Put  $\omega_\lambda = (\lambda^2 + 1/4)^{1/2}$ . Use Mehler–Fock inversion to prove

$$u(t, r) = \frac{1}{2\pi^2} \int_0^\infty \left( \cos(t\omega_\lambda) \tilde{f}(\lambda) + \frac{\sin(t\omega_\lambda)}{\omega_\lambda} \tilde{g}(\lambda) \right) P_{-1/2+i\lambda}(\cosh r) |c(\lambda)|^{-2} d\lambda.$$

Deduce the radial arrival profile of a localized pulse and its high-frequency spectral density from  $|c(\lambda)|^{-2} = \pi \lambda \tanh(\pi \lambda)$ .

**Exercise 4.6** (Hyperbolic-drum resonances). Let  $D_R \subset \mathbb{H}^2$  be the geodesic disk of radius  $R$  in curvature  $-1$ , impose Dirichlet boundary conditions, and let the wave speed be  $v$ . Prove that the radial resonances are indexed by the positive solutions of

$$P_{-1/2+i\lambda_j}(\cosh R) = 0$$

and have angular frequencies

$$\omega_j = v \sqrt{\lambda_j^2 + \frac{1}{4}}.$$

Use the  $c$ -function asymptotics to derive the large- $R$  quantization law

$$\lambda_j R + \arg c(\lambda_j) = \left(j + \frac{1}{2}\right) \pi + o(1),$$

with the indexing convention stated explicitly, and deduce the predicted radial frequency spacings.

For the  $\{3, 7\}$  tessellation used in Lenggenhager et al., *Nature Communications* **13** (2022), let  $Q_R$  be the graph Laplacian in their sign convention, put  $L_R = -Q_R \geq 0$ , and write  $\Lambda_j(L_R)$  for its

eigenvalues. The experiment uses curvature  $-4$  and lattice parameter  $h = \tanh(0.545275)$ . Prove the long-wavelength calibration

$$\Lambda_j(L_R) = \frac{7}{4}h^2\nu_j^{(-4)} + O(h^3), \quad \nu_j^{(-4)} = 4\nu_j^{(-1)},$$

where  $\nu_j^{(-1)} = \lambda_j^2 + \frac{1}{4}$  is the curvature- $-1$  continuum eigenvalue. Prove that the metric and geodesic-radius normalizations satisfy  $g_{-1} = 4g_{-4}$  and  $R_{-1} = 2R_{-4}$ .

For circuit capacitance  $C$  and grounding inductance  $L$ , derive from the grounded circuit Laplacian

$$J(\omega) = -i\omega C Q_R + \frac{1}{i\omega L} I$$

that the measured circuit resonance satisfies

$$\omega_j^{\text{circ}} = \frac{1}{\sqrt{LC \Lambda_j(L_R)}}, \quad \nu_j^{(-1)} \sim \frac{\Lambda_j(L_R)}{7h^2} = \frac{1}{28\pi^2 h^2 LC (f_j^{\text{circ}})^2}.$$

Use this calibration to compare the inferred conical parameters, measured voltage eigenmodes, and radial phase fronts with the conical-function zeros and Mehler–Fock wave kernel. State separately the curvature rescaling, continuum prediction, circuit resonance law, measured quantities, and the errors caused by boundary shape, loss, and discretization.

## 5 Automorphic Forms

**Definition 5.1** (Integral lattices and theta series). A *Euclidean lattice*  $L \subset \mathbb{R}^d$  is a discrete subgroup for which  $\mathbb{R}^d/L$  is compact. Its dual is

$$L^* = \{y \in \mathbb{R}^d : \langle x, y \rangle \in \mathbb{Z} \text{ for every } x \in L\}.$$

It is *integral* if  $L \subseteq L^*$ , *unimodular* if  $L = L^*$ , and *even* if  $\|x\|^2 \in 2\mathbb{Z}$  for every  $x \in L$ . For  $\tau$  in the upper half-plane and  $q = e^{2\pi i\tau}$ , its theta series is

$$\Theta_L(\tau) = \sum_{x \in L} q^{\|x\|^2/2}.$$

Its coefficient of  $q^n$  counts lattice vectors of squared length  $2n$ . If the same lattice is used as a crystal lattice, its physical reciprocal lattice is

$$L_{\text{physical}}^* = 2\pi L^*.$$

**Exercise 5.1** (Poisson duality for lattice theta series). Apply Exercise 2.9 to a Gaussian and prove

$$\Theta_L(-1/\tau) = \frac{(-i\tau)^{d/2}}{\text{vol}(L)} \Theta_{L^*}(\tau),$$

using the branch positive on the positive imaginary axis. Show that evenness gives  $\Theta_L(\tau+1) = \Theta_L(\tau)$ . Interpret the first identity as momentum–winding duality for a compact free boson and as the reciprocal-lattice relation behind diffraction. Explain why a general lattice theta series transforms with its dual rather than being a scalar modular form by itself.

**Definition 5.2** (The modular group and modular forms). The modular group  $SL_2(\mathbb{Z})$  acts on the upper half-plane by

$$\gamma\tau = \frac{a\tau + b}{c\tau + d}, \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

and is generated by  $S : \tau \mapsto -1/\tau$  and  $T : \tau \mapsto \tau + 1$ . A *modular form of integral weight  $k$*  for  $SL_2(\mathbb{Z})$  is a holomorphic function  $f$  on the upper half-plane satisfying

$$f(\gamma\tau) = (c\tau + d)^k f(\tau)$$

and holomorphic at the cusp, meaning that its Fourier expansion  $f(\tau) = \sum_{n \geq 0} a_n q^n$  has no negative powers. It is a *cusp form* if  $a_0 = 0$ .

**Definition 5.3** (Zeta and Eisenstein functions). For  $\text{Re } s > 1$ , define the Riemann zeta function by

$$\zeta(s) = \sum_{n \geq 1} n^{-s}.$$

For a  $d$ -dimensional Euclidean lattice  $L$ , with  $d \geq 1$ , and  $\text{Re } s > d/2$ , define its Epstein zeta function by

$$Z_L(s) = \sum_{x \in L \setminus \{0\}} \|x\|^{-2s}.$$

For  $m \geq 2$  and  $\tau$  in the upper half-plane, define

$$G_{2m}(\tau) = \sum_{(c,d) \in \mathbb{Z}^2 \setminus \{(0,0)\}} \frac{1}{(c\tau + d)^{2m}}, \quad E_{2m}(\tau) = \frac{G_{2m}(\tau)}{2\zeta(2m)}.$$

**Exercise 5.2** (Theta Mellin transform and zeta functional equation). Put  $\theta_L(t) = \Theta_L(it)$ . For  $\text{Re } s > d/2$ , prove

$$\pi^{-s} \Gamma(s) Z_L(s) = \int_0^\infty (\theta_L(t) - 1) t^{s-1} dt.$$

Split the integral at 1 and use Poisson duality to prove the meromorphic continuation

$$\begin{aligned} \pi^{-s} \Gamma(s) Z_L(s) &= \int_1^\infty (\theta_L(t) - 1) t^{s-1} dt \\ &+ \frac{1}{\text{vol}(L)} \int_1^\infty (\theta_{L^*}(t) - 1) t^{d/2-s-1} dt + \frac{1}{\text{vol}(L)(s - d/2)} - \frac{1}{s}. \end{aligned}$$

Deduce

$$\pi^{-s} \Gamma(s) Z_L(s) = \frac{1}{\text{vol}(L)} \pi^{-(d/2-s)} \Gamma(d/2 - s) Z_{L^*}(d/2 - s),$$

$Z_L(0) = -1$ , and

$$\text{res}_{s=d/2} Z_L(s) = \frac{\pi^{d/2}}{\text{vol}(L) \Gamma(d/2)}.$$

For  $L = \mathbb{Z}$ , use  $Z_{\mathbb{Z}}(s) = 2\zeta(2s)$  to derive the functional equation of  $\zeta$ . Use Fourier series and Parseval to prove

$$\zeta(2) = \frac{\pi^2}{6}, \quad \zeta(4) = \frac{\pi^4}{90},$$

and then deduce

$$\zeta(-1) = -\frac{1}{12}, \quad \zeta(-3) = \frac{1}{120}.$$



**Exercise 5.3** (Holomorphic Eisenstein series). Prove absolute local uniform convergence of  $G_{2m}$  and prove

$$G_{2m}\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^{2m} G_{2m}(\tau)$$

for every  $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ . Using Poisson summation in one lattice direction, derive

$$G_{2m}(\tau) = 2\zeta(2m) + \frac{2(2\pi i)^{2m}}{(2m-1)!} \sum_{n \geq 1} \sigma_{2m-1}(n) q^n, \quad \sigma_r(n) = \sum_{d|n} d^r.$$

Deduce that  $E_{2m}$  is a modular form of weight  $2m$  and that

$$E_4(\tau) = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n.$$

**Definition 5.4** (Theta and eta functions). Set

$$\begin{aligned} \vartheta_2(\tau) &= \sum_{n \in \mathbb{Z}} q^{(n+1/2)^2/2}, \\ \vartheta_3(\tau) &= \sum_{n \in \mathbb{Z}} q^{n^2/2}, & \eta(\tau) &= q^{1/24} \prod_{m \geq 1} (1 - q^m). \\ \vartheta_4(\tau) &= \sum_{n \in \mathbb{Z}} (-1)^n q^{n^2/2}, \end{aligned}$$

**Exercise 5.4** (Theta transformations). Use Poisson summation on shifted and phase-twisted Gaussians to derive

$$\vartheta_2(-1/\tau) = (-i\tau)^{1/2} \vartheta_4(\tau), \quad \vartheta_3(-1/\tau) = (-i\tau)^{1/2} \vartheta_3(\tau), \quad \vartheta_4(-1/\tau) = (-i\tau)^{1/2} \vartheta_2(\tau).$$

Use the branch positive for  $\tau$  on the positive imaginary axis. Prove directly from the series that

$$\vartheta_2(\tau + 1) = e^{\pi i/4} \vartheta_2(\tau), \quad \vartheta_3(\tau + 1) = \vartheta_4(\tau), \quad \vartheta_4(\tau + 1) = \vartheta_3(\tau).$$

Write these  $S$ - and  $T$ -laws as a vector-valued transformation law of weight  $\frac{1}{2}$  for  $(\vartheta_2, \vartheta_3, \vartheta_4)^T$  and prove that it is not an instance of the preceding scalar integral-weight definition.

**Exercise 5.5** (Jacobi triple product and eta transformations). Establish the Jacobi triple product, first as an identity of formal Laurent series,

$$\sum_{n \in \mathbb{Z}} z^n q^{n^2/2} = \prod_{m \geq 1} (1 - q^m)(1 + zq^{m-1/2})(1 + z^{-1}q^{m-1/2}),$$

and use specializations to recover product forms for the theta functions. Deduce

$$2\eta(\tau)^3 = \vartheta_2(\tau)\vartheta_3(\tau)\vartheta_4(\tau).$$

Use the theta transformations and the branch positive on the positive imaginary axis to derive

$$\eta(\tau + 1) = e^{\pi i/12} \eta(\tau), \quad \eta(-1/\tau) = (-i\tau)^{1/2} \eta(\tau).$$

Prove that these laws give  $\eta$  weight  $\frac{1}{2}$  with a nontrivial scalar multiplier, rather than the integral-weight trivial-multiplier law defined above. Explain how the two sides encode charge sectors and independent oscillator modes, respectively.

**Exercise 5.6** (Theta diffraction). Let  $T_L = \mathbb{R}^d/L$  carry its flat metric, and normalize  $L^*$  by  $\langle x, \xi \rangle \in \mathbb{Z}$ . Let  $K_t(x, y)$  be the heat kernel of the flat Laplacian and prove

$$\int_{T_L} K_t(x, x) dx = \sum_{\xi \in L^*} e^{-4\pi^2 t \|\xi\|^2} = \Theta_{L^*}(4\pi i t).$$

Apply Poisson duality to derive the short-time image sum and its leading term  $\text{vol}(T_L)(4\pi t)^{-d/2}$ . For a periodic array with motif  $B$ , prove that the elastic intensity at reciprocal vector  $2\pi\xi$  is proportional to

$$|S_B(2\pi\xi)|^2.$$

Prove that the heat-trace exponents and diffraction-peak radii determine the same reciprocal squared lengths  $\|\xi\|^2$ . Deduce the predicted peak locations, systematic absences, and thermal decay rates from the theta coefficients of  $L^*$ .

**Exercise 5.7** (Zeta-regularized Casimir pressure). For a massless scalar field between parallel Dirichlet plates of area  $A$  and separation  $a$ , neglect edge effects and take the normal-mode frequencies to be

$$\omega_{n,k} = c \sqrt{|k|^2 + \left(\frac{\pi n}{a}\right)^2}, \quad n \geq 1, \quad k \in \mathbb{R}^2.$$

For  $\mu > 0$  and  $\text{Re } s$  sufficiently large, define the regulated vacuum energy density by

$$\frac{\mathcal{E}_D(s)}{A} = \frac{\hbar c}{2} \mu^{2s} \sum_{n \geq 1} \int_{\mathbb{R}^2} \frac{d^2 k}{(2\pi)^2} \left( |k|^2 + \frac{\pi^2 n^2}{a^2} \right)^{1/2-s}.$$

Use the gamma integral to continue this expression as

$$\frac{\mathcal{E}_D(s)}{A} = \frac{\hbar c \mu^{2s}}{8\pi} \frac{\Gamma(s-3/2)}{\Gamma(s-1/2)} \left(\frac{\pi}{a}\right)^{3-2s} \zeta(2s-3).$$

After subtracting the infinite-separation and plate-local terms, evaluate at  $s = 0$  and prove

$$\frac{\mathcal{E}_D}{A} = -\frac{\pi^2 \hbar c}{1440 a^3}.$$

For the ideal electromagnetic field, account for the two polarizations and deduce

$$\frac{\mathcal{E}_{\text{EM}}}{A} = -\frac{\pi^2 \hbar c}{720 a^3}, \quad P = -\frac{\partial(\mathcal{E}_{\text{EM}}/A)}{\partial a} = -\frac{\pi^2 \hbar c}{240 a^4}.$$

State how a force measurement tests the  $a^{-4}$  law and its coefficient, separating the ideal prediction from finite conductivity, temperature, roughness, alignment, and edge corrections.

## Part II

# Counting

## 6 Graded Characters

**Definition 6.1** (Probability laws). A probability space is a measure space  $(\Omega, \mathcal{F}, \mathbb{P})$  with  $\mathbb{P}(\Omega) = 1$ . A random variable with values in a metric space  $(S, d)$  is a measurable map  $X : \Omega \rightarrow S$ , and its

law is the pushforward  $X_*\mathbb{P}$ . For an integrable complex random variable define  $\mathbb{E}X = \int_{\Omega} X d\mathbb{P}$ . For square-integrable complex random variables define

$$\text{Var}(X) = \mathbb{E}|X - \mathbb{E}X|^2, \quad \text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(\overline{Y - \mathbb{E}Y})].$$

Random variables  $X_i : \Omega \rightarrow S_i$ ,  $1 \leq i \leq r$ , are independent when their joint law is the product of their marginal laws. For random variables  $X_N, X : \Omega \rightarrow S$ , define convergence in probability by

$$\mathbb{P}(d(X_N, X) > \varepsilon) \longrightarrow 0 \quad (\varepsilon > 0),$$

and convergence in distribution by convergence of their laws against bounded continuous functions. The standard normal law  $N(0, 1)$  on  $\mathbb{R}$  has density  $(2\pi)^{-1/2}e^{-x^2/2}$  with respect to Lebesgue measure.

**Definition 6.2** (Symmetric powers and graded traces). For a finite-dimensional complex vector space  $V$ , define

$$\text{Sym}^m V = V^{\otimes m} / \langle \xi - \sigma\xi : \xi \in V^{\otimes m}, \sigma \in S_m \rangle.$$

An endomorphism  $T$  induces  $\text{Sym}^m T$ . Using the exterior powers and trace defined in Spectral Theory, define

$$Z_B(T; z) = \sum_{m \geq 0} \text{tr}(\text{Sym}^m T) z^m, \quad Z_F(T; z) = \sum_{m \geq 0} \text{tr}(\Lambda^m T) z^m.$$

**Definition 6.3** (Partitions). An *integer partition* of  $n \geq 0$  is a finite nonincreasing sequence  $\lambda = (\lambda_1, \dots, \lambda_r)$  of positive integers with  $|\lambda| = \sum_i \lambda_i = n$ . Let  $p(n)$  be the number of partitions, with  $p(0) = 1$ . For partitions of equal size, write  $\mu < \lambda$  in dominance order when

$$\sum_{i=1}^k \mu_i \leq \sum_{i=1}^k \lambda_i$$

for every  $k \geq 1$ , with missing parts interpreted as zero, and at least one inequality strict.

**Definition 6.4** (Symmetric polynomials). For  $N \geq 1$ , let

$$\Lambda_N = \mathbb{C}[x_1, \dots, x_N]^{S_N}.$$

For a partition  $\lambda$  of length at most  $N$ , let  $m_\lambda$  be the sum of the distinct permutations of  $x_1^{\lambda_1} \cdots x_N^{\lambda_N}$ . Define

$$e_r = \sum_{1 \leq i_1 < \dots < i_r \leq N} x_{i_1} \cdots x_{i_r}, \quad h_r = \sum_{1 \leq i_1 \leq \dots \leq i_r \leq N} x_{i_1} \cdots x_{i_r},$$

$$p_r = \sum_{i=1}^N x_i^r, \quad e_0 = h_0 = 1.$$

Set  $e_r = 0$  for  $r > N$  and  $h_r = 0$  for  $r < 0$ .

**Exercise 6.1** (Fundamental symmetric identities). Prove that every element of  $\Lambda_N$  is uniquely a polynomial in  $e_1, \dots, e_N$ . Prove

$$\sum_{r=0}^N e_r t^r = \prod_{i=1}^N (1 + x_i t), \quad \sum_{r \geq 0} h_r t^r = \prod_{i=1}^N \frac{1}{1 - x_i t},$$

and deduce

$$\left( \sum_{r \geq 0} h_r t^r \right) \left( \sum_{r=0}^N (-1)^r e_r t^r \right) = 1, \quad r h_r = \sum_{j=1}^r p_j h_{r-j}.$$

**Exercise 6.2** (Bosonic and fermionic trace generating functions). Let  $V$  be a finite-dimensional complex vector space and  $T \in \text{End}(V)$ . Using the intrinsic determinant defined by the action on the highest exterior power, prove

$$Z_B(T; z) = \frac{1}{\det(I - zT)}, \quad Z_F(T; z) = \det(I + zT).$$

The first identity holds for  $|z|$  below the reciprocal spectral radius; the second is a polynomial identity. Prove them first for a direct sum of generalized spectral subspaces and then by polynomial continuation, without choosing vectors in those subspaces. If  $\lambda_1, \dots, \lambda_N$  are the eigenvalues counted with algebraic multiplicity, prove

$$\text{tr}(\text{Sym}^m T) = h_m(\lambda_1, \dots, \lambda_N), \quad \text{tr}(\Lambda^m T) = e_m(\lambda_1, \dots, \lambda_N).$$

Let the one-particle Hamiltonian have one-dimensional spectral subspaces at energies  $j\hbar\omega$ ,  $j \geq 1$ . Apply the identities to the finite spectral truncation and then pass coefficientwise to the limit to derive

$$\prod_{j \geq 1} \frac{1}{1 - q^j} = \sum_{n \geq 0} p(n) q^n, \quad \prod_{j \geq 1} (1 + q^j) = \sum_{n \geq 0} p_{\text{dist}}(n) q^n,$$

where  $p_{\text{dist}}(n)$  counts partitions into distinct parts. Construct the multiplicity map separating powers of two and prove

$$\prod_{j \geq 1} (1 + q^j) = \prod_{j \geq 1} \frac{1}{1 - q^{2j-1}}.$$

Interpret the coefficient of  $q^n$  as the exact degeneracy at excitation energy  $n\hbar\omega$  for the corresponding bosonic or fermionic system.

For the bosonic system put  $q = e^{-\beta_{\text{th}}\hbar\omega}$ , where  $\beta_{\text{th}} = (k_B T)^{-1}$ , and define

$$Z(\beta_{\text{th}}) = \prod_{j \geq 1} \frac{1}{1 - e^{-\beta_{\text{th}} j \hbar \omega}}, \quad \Psi(\beta_{\text{th}}) = \log Z(\beta_{\text{th}}).$$

Define

$$U = -\frac{d\Psi}{d\beta_{\text{th}}}.$$

For

$$\mathcal{S} = k_B(\Psi + \beta_{\text{th}} U), \quad C = k_B \beta_{\text{th}}^2 \frac{d^2 \Psi}{d\beta_{\text{th}}^2},$$

prove  $d\mathcal{S} = k_B \beta_{\text{th}} dU$  as  $\beta_{\text{th}}$  varies and express  $d^2 \Psi / d\beta_{\text{th}}^2$  as the energy variance.

## 7 Modular Characters

**Definition 7.1** (Virasoro characters). The Virasoro algebra has generators  $L_n$ ,  $n \in \mathbb{Z}$ , and a central generator  $C$ , with

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{1}{12}(m^3 - m)\delta_{m+n,0}C, \quad [C, L_n] = 0.$$

A module has central charge  $c$  when  $C$  acts by  $cI$ . A highest-weight vector  $|h\rangle$  satisfies  $L_0|h\rangle = h|h\rangle$  and  $L_n|h\rangle = 0$  for  $n > 0$ . For  $\tau$  in the upper half-plane put  $q = e^{2\pi i \tau}$ . If  $H_a$  is a positive-energy module with finite-dimensional  $L_0$  spectral subspaces, define

$$\chi_a(\tau) = \text{Tr}_{H_a} q^{L_0 - c/24}$$

when the displayed operator is trace class; the same expression denotes the formal graded trace otherwise.

**Exercise 7.1** (Ordered Virasoro descendants). Order the negative Virasoro modes by their indices. Use the commutation relations to rewrite every word as an ordered word. Check every three-letter overlap and use the Jacobi identity to prove that the rewriting is confluent. Filter by word length and deduce that the associated graded algebra is the symmetric algebra on the negative modes. Conclude that the ordered monomials in the negative modes are linearly independent in the universal highest-weight module.

**Exercise 7.2** (Verma and vacuum characters). Use the ordered descendants to prove that the Verma-module character is

$$\chi_{c,h}^{\text{Verma}}(\tau) = q^{h-c/24} \prod_{m \geq 1} (1 - q^m)^{-1}.$$

For the vacuum impose  $L_{-1}|0\rangle = 0$  and derive the corresponding generic vacuum product.

**Exercise 7.3** (Null-vector quotients). Let a singular vector of degree  $r$  in a Verma module generate a submodule isomorphic to the Verma module of highest weight  $h + r$ . Prove that the quotient character is

$$\chi_{c,h}^{\text{Verma}} - \chi_{c,h+r}^{\text{Verma}}.$$

For several singular vectors, prove that their submodules must be combined by inclusion–exclusion, and show that independent subtraction is valid only when the intersections of their descendant submodules have been accounted for.

**Definition 7.2** (Unitary modular character systems). A finite character vector  $\chi$  is a *unitary modular character system* when

$$\chi(-1/\tau) = S\chi(\tau), \quad \chi(\tau + 1) = T\chi(\tau)$$

for unitary matrices  $S, T$ . For a matrix  $M$  with nonnegative integral entries define

$$Z_M(\tau, \bar{\tau}) = \sum_{a,b} \overline{\chi_a(\tau)} M_{ab} \chi_b(\tau).$$

It is *modular invariant* when

$$S^* M S = M, \quad T^* M T = M.$$

**Exercise 7.4** (Ising characters and modular invariance). For the Ising character system with  $c = 1/2$  and weights  $0, 1/2, 1/16$ , choose the square roots whose  $q$ -expansions have the displayed positive leading coefficients and put

$$\begin{aligned} \chi_0 &= \frac{1}{2} \left( \sqrt{\frac{\vartheta_3}{\eta}} + \sqrt{\frac{\vartheta_4}{\eta}} \right), \\ \chi_{1/2} &= \frac{1}{2} \left( \sqrt{\frac{\vartheta_3}{\eta}} - \sqrt{\frac{\vartheta_4}{\eta}} \right), \\ \chi_{1/16} &= \frac{1}{\sqrt{2}} \sqrt{\frac{\vartheta_2}{\eta}}. \end{aligned}$$

Use the theta and eta transformations developed in Automorphic Forms to derive

$$S = \frac{1}{2} \begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{pmatrix}, \quad T_{aa} = e^{2\pi i(h_a - c/24)}.$$

Expand each character as  $q^{h-c/24}$  times a power series, identify its coefficients with exact descendant counts, and verify the modular invariance of  $\sum_a |\chi_a|^2$ .

**Exercise 7.5** (Eta asymptotics and bosonic thermodynamics). Use Exercise 5.5 to prove

$$\prod_{j \geq 1} (1 - e^{-tj})^{-1} \sim \sqrt{\frac{t}{2\pi}} \exp\left(\frac{\pi^2}{6t}\right) \quad (t \downarrow 0).$$

For  $t = \beta_{\text{th}} \hbar \omega$ , differentiate the transformed eta-product identity to strengthen this to

$$\Psi(\beta_{\text{th}}) = \frac{\pi^2}{6t} + \frac{1}{2} \log \frac{t}{2\pi} - \frac{t}{24} + O(e^{-4\pi^2/t})$$

and derive

$$U = \hbar \omega \left( \frac{\pi^2}{6t^2} - \frac{1}{2t} + \frac{1}{24} + o(1) \right),$$

$$C = k_B \left( \frac{\pi^2}{3t} - \frac{1}{2} + o(1) \right), \quad S = k_B \left( \frac{\pi^2}{3t} + \frac{1}{2} \log \frac{t}{2\pi} - \frac{1}{2} + o(1) \right).$$

**Exercise 7.6** (Partition saddle). Insert this estimate into Cauchy's coefficient integral, locate its positive real saddle, prove that its equation is

$$n = \frac{U}{\hbar \omega},$$

and identify the leading saddle exponent with the Legendre transform  $S/k_B$ . Expand the phase through quadratic order and derive the major-arc contribution

$$\frac{1}{4n\sqrt{3}} \exp\left(\pi \sqrt{\frac{2n}{3}}\right).$$

**Exercise 7.7** (Partition minor arcs). On the coefficient circle through the positive saddle, use

$$|1 - re^{i\theta}|^2 = (1 - r)^2 + 4r \sin^2(\theta/2)$$

together with the modular transformation at each rational cusp met by the circle dissection to bound the complementary arcs by an exponentially smaller quantity. Deduce

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi \sqrt{\frac{2n}{3}}\right).$$

**Exercise 7.8** (Ising state counts). For the diagonal Ising system, define nonnegative integers  $d_a(N)$  by

$$\chi_a(\tau) = q^{h_a - c/24} \sum_{N \geq 0} d_a(N) q^N$$

and define the resolved two-sided degeneracy

$$\rho_{\text{Ising}}(N, \bar{N}) = \sum_{a \in \{0, 1/2, 1/16\}} d_a(N) d_a(\bar{N}).$$

Use the explicit theta-product formulas and the  $S$ -transformation to control each character in a complex neighborhood of the positive real saddle. Bound the complementary arcs as in the partition calculation and prove, as  $N, \bar{N} \rightarrow \infty$  with  $N/\bar{N}$  in a fixed compact subset of  $(0, \infty)$ ,

$$\log \rho_{\text{Ising}}(N, \bar{N}) \sim 2\pi \sqrt{\frac{cN}{6}} + 2\pi \sqrt{\frac{c\bar{N}}{6}}, \quad c = \frac{1}{2}.$$

Derive the resulting asymptotic number of states below a specified high energy on a circle and the corresponding entropy. State the circumference, propagation speed, energy window, vacuum assumption, sector resolution, saddle-point regime, and finite-size limitations required when comparing these predictions with measured or transfer-matrix spectra.

## 8 Transfer Matrices

**Definition 8.1** (Transfer-matrix counts). Let  $\mathcal{R}_N$  be a finite set of allowed width- $N$  configurations and let  $T_N(z)$  be an operator on the configuration space whose matrix element between  $a, b \in \mathcal{R}_N$  is the weight of adjoining  $b$  to  $a$ . For a height- $M$  periodic lattice define

$$Z_{N,M}(z) = \text{Tr}(T_N(z)^M).$$

When  $T_N(z)$  has nonnegative entries, write  $\lambda_N(z)$  for its spectral radius.

**Definition 8.2** (Hard-hexagon rows). Let

$$\mathcal{R}_N = \{a \in \{0, 1\}^{\mathbb{Z}/N\mathbb{Z}} : a_i a_{i+1} = 0 \text{ for every } i\}.$$

Two rows  $a, b \in \mathcal{R}_N$  are compatible when

$$a_i b_i = 0, \quad a_i b_{i+1} = 0 \quad (i \in \mathbb{Z}/N\mathbb{Z}).$$

For  $z \geq 0$ , with  $|a| = \sum_i a_i$ , define

$$T_N(z)_{ab} = \mathbf{1}_{\{a, b \text{ compatible}\}} z^{(|a|+|b|)/2}.$$

**Exercise 8.1** (Hard-hexagon trace counts). Identify the horizontal, vertical, and diagonal nearest-neighbor edges encoded by the two compatibility conditions. Prove

$$\text{Tr}(T_N(z)^M) = \sum_I z^{|I|},$$

where the sum is over independent vertex sets of the specified triangular  $N \times M$  torus. Prove that every coefficient is an exact state count. Let 0 denote the empty row. For  $z > 0$ , prove

$$(T_N(z)^2)_{ab} \geq T_N(z)_{a0} T_N(z)_{0b} > 0.$$

Apply the Perron–Frobenius theorem to show that  $\lambda_N(z)$  is simple, that every other spectral value has smaller modulus, and that

$$\lim_{M \rightarrow \infty} \frac{1}{NM} \log Z_{N,M}(z) = \frac{1}{N} \log \lambda_N(z).$$

Put  $\eta = \log z = \beta_{\text{th}}\mu$  and

$$\Psi_{N,M}(\eta) = \log Z_{N,M}(e^\eta).$$

Set  $A = \partial_\eta \Psi_{N,M}$  and prove

$$A = \mathbb{E}_\eta |I|, \quad \partial_\eta A = \text{Var}_\eta(|I|),$$

and derive the mean occupied fraction and isothermal finite-size susceptibility

$$\rho_{N,M} = \frac{A}{NM}, \quad \chi_{N,M} = \frac{\partial \rho_{N,M}}{\partial \mu} = \frac{\beta_{\text{th}}}{NM} \text{Var}_\eta(|I|).$$

State the lattice geometry, boundary conditions, and finite  $N, M$  limitations required when these quantities are compared with adsorption or exclusion-process data.

**Definition 8.3** (Critical Ising transfer matrix). Let  $\mathcal{S}_N = \{-1, 1\}^{\mathbb{Z}/N\mathbb{Z}}$  and  $K_c = \frac{1}{2} \log(1 + \sqrt{2})$ . For row configurations  $s, t \in \mathcal{S}_N$  define

$$T_N^{\text{Is}}(s, t) = \exp \left[ \frac{K_c}{2} \sum_i (s_i s_{i+1} + t_i t_{i+1}) + K_c \sum_i s_i t_i \right].$$

Let  $R_N$  be cyclic row translation. Then  $R_N T_N^{\text{Is}} = T_N^{\text{Is}} R_N$ . Write  $a_x$  and  $a_t$  for the spatial and transfer-step lattice spacings and  $L = Na_x$  for the circumference.

**Exercise 8.2** (Theta-character transfer spectra). Let  $\Lambda_0(L) > 0$  be the Perron eigenvalue of  $T_N^{\text{Is}}$  and let  $\Lambda_{a,r,\bar{r}}(L)$  denote its low-lying symmetry-resolved eigenvalues. Define

$$E_{a,r,\bar{r}}(L) = -\frac{1}{a_t} \log |\Lambda_{a,r,\bar{r}}(L)|.$$

Assume the critical finite-size asymptotic

$$E_{a,r,\bar{r}}(L) = e_\infty L + \frac{2\pi v}{L} \left( h_a + \bar{h}_a + r + \bar{r} - \frac{c + \bar{c}}{24} \right) + o(L^{-1}).$$

If  $R_N$  acts on the same state by  $e^{iPa_x}$ , assume

$$P = \frac{2\pi}{L} (h_a - \bar{h}_a + r - \bar{r}) \pmod{2\pi/a_x}.$$

Use the Ising character system from the preceding section to prove that the coefficient of  $q^{h_a+r-c/24} \bar{q}^{\bar{h}_a+\bar{r}-\bar{c}/24}$  gives the predicted multiplicity at these energy and momentum values. Express every excitation gap directly as

$$E_j(L) - E_0(L) = -\frac{1}{a_t} \log \left| \frac{\Lambda_j(L)}{\Lambda_0(L)} \right|.$$

Compute the first spectral gaps and degeneracies in the three Ising sectors. Deduce the universal gap ratios and the  $L^{-1}$  Casimir-energy correction. State how finite-size transfer eigenvalues determine  $c$ , the primary weights, and  $v$ , including the symmetry-sector resolution, anisotropy corrections, and irrelevant-operator corrections required for the comparison.



## 9 Determinantal Ensembles

**Exercise 9.1** (Hermite Gaussian moments). For  $X \sim N(0, 1)$ , prove

$$\mathbb{E}[e^{tX}] = e^{t^2/2}.$$

Define the probabilists' Hermite polynomials by

$$e^{xt-t^2/2} = \sum_{n \geq 0} \text{He}_n(x) \frac{t^n}{n!}$$

and prove

$$\mathbb{E}[\text{He}_m(X) \text{He}_n(X)] = n! \delta_{mn}, \quad \mathbb{E}[X^{2n}] = \frac{(2n)!}{2^n n!}.$$

Relate  $\text{He}_n$  to the oscillator Hermite functions developed in Spectral Theory.

**Definition 9.1** (Dyson classes). For  $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{H}$ , put  $\beta_{\mathbb{D}} = \dim_{\mathbb{R}} \mathbb{F} \in \{1, 2, 4\}$ . The Gaussian Dyson ensemble on the self-adjoint operators on  $\mathbb{F}^N$  has density

$$d\mathbb{P}_{N, \beta_{\mathbb{D}}}(H) = C_{N, \beta_{\mathbb{D}}} \exp\left[-\frac{\beta_{\mathbb{D}} N}{4\sigma^2} \text{Tr}(H^2)\right] dH$$

for the invariant Hilbert–Schmidt volume. For  $\lambda = (\lambda_1, \dots, \lambda_N)$  define

$$\Delta(\lambda) = \det[\lambda_i^{j-1}]_{i,j=1}^N.$$

**Exercise 9.2** (Unitary orbit Jacobian). Use the exterior-power determinant to prove

$$\Delta(\lambda) = \prod_{i < j} (\lambda_j - \lambda_i).$$

Compute the Jacobian of the unitary conjugation-orbit map on the regular spectral locus, including the stabilizer and permutation factors, and derive

$$p_{N,2}(\lambda) = \frac{1}{Z_{N,2}} e^{-N \sum_i \lambda_i^2 / (2\sigma^2)} |\Delta(\lambda)|^2.$$

**Exercise 9.3** (Dyson level repulsion). For the real and quaternionic classes, show that each off-diagonal spectral direction has real multiplicity  $\beta_{\mathbb{D}}$  and deduce

$$p_{N, \beta_{\mathbb{D}}}(\lambda) = \frac{1}{Z_{N, \beta_{\mathbb{D}}}} e^{-\beta_{\mathbb{D}} N \sum_i \lambda_i^2 / (4\sigma^2)} |\Delta(\lambda)|^{\beta_{\mathbb{D}}}.$$

For  $2 \times 2$  GOE and GUE, separate the trace from the traceless operator, rescale the gap to unit mean, and obtain

$$P_1(s) = \frac{\pi}{2} s e^{-\pi s^2/4}, \quad P_2(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}.$$

Identify the exponent  $\beta_{\mathbb{D}}$  as the small-gap repulsion prediction and state the symmetry-sector separation required in spectral data.

**Definition 9.2** (Fredholm determinants). Let  $A$  be a positive compact operator on a Hilbert space. It is *trace class* when its nonzero spectral values, repeated with multiplicity, satisfy  $\sum_j \lambda_j(A) < \infty$ . Define

$$\det(I + zA) = \prod_j (1 + z\lambda_j(A)), \quad \text{Tr}(A^m) = \sum_j \lambda_j(A)^m.$$

**Exercise 9.4** (Fredholm trace identity). Prove absolute local uniform convergence of the defining product and

$$\det(I - zA) = \exp\left(-\sum_{m \geq 1} \frac{z^m}{m} \operatorname{Tr}(A^m)\right) \quad (|z| < \|A\|^{-1}).$$

Prove that the determinant is entire, identify its zeros, and show that its finite-rank specialization agrees with the exterior-power identity from the opening section.

**Definition 9.3** (Point processes and correlation densities). Let  $X$  be a locally compact second-countable Hausdorff space and let  $\mu$  be a Radon measure on  $X$ . A simple point process is a random locally finite counting measure  $\Xi = \sum_i \delta_{x_i}$  with distinct points almost surely. When its  $k$ th factorial moment measure is absolutely continuous with respect to  $\mu^{\otimes k}$ , its density  $\rho_k$  is defined, for every nonnegative compactly supported measurable function  $f$ , by

$$\mathbb{E} \sum_{i_1, \dots, i_k}^{\neq} f(x_{i_1}, \dots, x_{i_k}) = \int_{X^k} f \rho_k d\mu^{\otimes k}.$$

A *determinantal point process* with measurable Hermitian kernel  $K$  is one for which

$$\rho_k(x_1, \dots, x_k) = \det[K(x_i, x_j)]_{i,j=1}^k.$$

For the existence and counting formulas below, the integral operator defined by  $K$  is a locally trace-class positive contraction on  $L^2(X, \mu)$ .

**Exercise 9.5** (Determinantal counting statistics). For a relatively compact measurable set  $B$ , put  $N_B = \Xi(B)$  and  $K_B = \mathbf{1}_B K \mathbf{1}_B$ . Use factorial moments and the Fredholm series to prove

$$\mathbb{E}[z^{N_B}] = \det(I + (z - 1)K_B), \quad \mathbb{P}(N_B = 0) = \det(I - K_B).$$

Prove that  $N_B$  is distributed as a sum of independent Bernoulli variables whose parameters are the eigenvalues of  $K_B$ .

**Definition 9.4** (Unitary polynomial ensembles). Let  $\mu$  be a Radon measure on  $\mathbb{R}$  such that  $w(x)d\mu(x)$  has infinite support and finite moments, and let  $P_j$  be its monic orthogonal polynomials, normalized by

$$\int P_j(x) P_k(x) w(x) d\mu(x) = H_j \delta_{jk}, \quad H_j > 0.$$

The associated unitary polynomial ensemble has density, with respect to  $\mu^{\otimes N}$ ,

$$\mathfrak{p}_N(x_1, \dots, x_N) = \frac{1}{N! \prod_{j=0}^{N-1} H_j} \Delta(x)^2 \prod_{i=1}^N w(x_i)$$

and kernel

$$K_N(x, y) = \sqrt{w(x)w(y)} \sum_{j=0}^{N-1} \frac{P_j(x)P_j(y)}{H_j}.$$

**Exercise 9.6** (Andréief and Christoffel–Darboux identities). Prove Andréief's identity

$$\int_{X^N} \det[f_i(x_j)] \det[g_i(x_j)] \prod_j d\mu(x_j) = N! \det\left[\int_X f_i g_j d\mu\right].$$

Deduce that the unitary polynomial ensemble is determinantal with kernel  $K_N$ . Derive the Christoffel–Darboux formula from the three-term recurrence and prove

$$\int K_N(x, z) K_N(z, y) d\mu(z) = K_N(x, y), \quad \int K_N(x, x) d\mu(x) = N.$$

**Definition 9.5** (GUE and its limiting kernels). For the GUE weight  $w_N(x) = e^{-Nx^2/(2\sigma^2)}$ , write  $\psi_j^H$  for the normalized oscillator Hermite functions developed in Spectral Theory and define

$$\phi_{j,N}(x) = \left(\frac{N}{2\sigma^2}\right)^{1/4} \psi_j^H\left(\sqrt{\frac{N}{2\sigma^2}}x\right), \quad K_N^{\text{GUE}}(x, y) = \sum_{j=0}^{N-1} \phi_{j,N}(x)\phi_{j,N}(y).$$

Define

$$\begin{aligned} \rho_{\text{sc}}(x) &= \frac{1}{2\pi\sigma^2} \sqrt{4\sigma^2 - x^2} \mathbf{1}_{[-2\sigma, 2\sigma]}(x), \\ K_{\sin}(u, v) &= \frac{\sin \pi(u - v)}{\pi(u - v)}, \\ K_{\text{Ai}}(u, v) &= \frac{\text{Ai}(u) \text{Ai}'(v) - \text{Ai}'(u) \text{Ai}(v)}{u - v}, \quad F_2(s) = \det(I - K_{\text{Ai}})_{L^2(s, \infty)}. \end{aligned}$$

Diagonal values are defined by continuity.

**Definition 9.6** (Hastings–McLeod function). The *Hastings–McLeod function* is the real pole-free solution  $q$  of the second Painlevé equation

$$q''(s) = sq(s) + 2q(s)^3$$

with

$$q(s) \sim \text{Ai}(s) \quad (s \rightarrow +\infty).$$

**Exercise 9.7** (Hermite steepest descent). Derive a contour integral for the normalized Hermite functions from their generating function. For  $E \in (-2\sigma, 2\sigma)$ , perform steepest descent at the two simple saddles; at  $2\sigma$ , rescale the coalescing saddles to the Airy integral. Establish uniform asymptotic formulas on compact bulk windows and uniform exponentially weighted bounds on edge half-lines.

**Exercise 9.8** (GUE bulk statistics). Insert the bulk Hermite asymptotics into the Christoffel–Darboux identity and prove

$$\frac{1}{N\rho_{\text{sc}}(E)} K_N^{\text{GUE}}\left(E + \frac{u}{N\rho_{\text{sc}}(E)}, E + \frac{v}{N\rho_{\text{sc}}(E)}\right) \longrightarrow K_{\sin}(u, v)$$

locally uniformly. For the limiting sine process prove

$$\text{Var } N_{[0, L]} = \int_0^L K_{\sin}(u, u) du - \int_0^L \int_0^L |K_{\sin}(u, v)|^2 du dv = \frac{1}{\pi^2} \log L + O(1).$$

State the unfolding, symmetry-sector separation, and finite-window assumptions required when testing the spacing and number-variance predictions with resonance or Hamiltonian spectra.

**Exercise 9.9** (GUE edge statistics). Insert the turning-point asymptotics into the Christoffel–Darboux identity and prove

$$\frac{\sigma}{N^{2/3}} K_N^{\text{GUE}}\left(2\sigma + \frac{\sigma u}{N^{2/3}}, 2\sigma + \frac{\sigma v}{N^{2/3}}\right) \longrightarrow K_{\text{Ai}}(u, v).$$

Use the exponentially weighted bounds to pass to the Fredholm series and deduce the GUE largest-eigenvalue law

$$\mathbb{P}\left(\frac{N^{2/3}}{\sigma}(\lambda_{\max} - 2\sigma) \leq s\right) \longrightarrow F_2(s).$$

State the centering, scale calibration, symmetry class, and finite-size assumptions required when testing this prediction with extreme resonance or Hamiltonian levels.

**Exercise 9.10** (Painlevé form of the Tracy–Widom law). Let  $K_s$  be the Airy-kernel operator on  $L^2((s, \infty))$  and define

$$Q_s = (I - K_s)^{-1} \text{Ai}, \quad P_s = (I - K_s)^{-1} \text{Ai}'.$$

With the inner product on  $L^2((s, \infty))$ , put

$$q(s) = Q_s(s), \quad p(s) = P_s(s), \quad u(s) = \langle Q_s, \text{Ai} \rangle, \quad v(s) = \langle P_s, \text{Ai} \rangle.$$

Use the integrable form of  $K_{\text{Ai}}$ , the Airy equation, and differentiation of the moving endpoint to prove

$$q' = p - qu, \quad p' = sq + pu - 2qv, \quad u' = -q^2, \quad v' = -pq.$$

Prove that  $u^2 - 2v - q^2$  is constant and vanishes as  $s \rightarrow +\infty$ . Deduce

$$q'' = sq + 2q^3, \quad q(s) \sim \text{Ai}(s) \quad (s \rightarrow +\infty).$$

Use the positive integral representation

$$K_{\text{Ai}}(x, y) = \int_0^\infty \text{Ai}(x+t) \text{Ai}(y+t) dt$$

to prove that  $I - K_s$  is invertible for every real  $s$ . Conclude that the resolvent construction gives the real pole-free Hastings–McLeod solution. For the resolvent  $R_s = K_s(I - K_s)^{-1}$ , prove

$$\frac{d}{ds} \log \det(I - K_s) = R_s(s, s) = u(s) = \int_s^\infty q(x)^2 dx.$$

Conclude that the Fredholm and Painlevé definitions agree:

$$F_2(s) = \exp \left[ - \int_s^\infty (x-s) q(x)^2 dx \right].$$

Use the Airy asymptotic of the Hastings–McLeod solution to derive

$$1 - F_2(s) \sim \frac{e^{-4s^{3/2}/3}}{16\pi s^{3/2}} \quad (s \rightarrow +\infty).$$

State how exceedance counts for centered and edge-scaled extreme eigenvalues or resonance levels test this right-tail prediction, including the finite-size range over which the asymptotic is used.

## 10 Schur Measures

**Definition 10.1** (Schur polynomials). For a partition  $\lambda$  of length at most  $N$ , define

$$s_\lambda(x_1, \dots, x_N) = \det[h_{\lambda_i - i + j}]_{i,j=1}^N.$$

**Exercise 10.1** (Schur identities). Derive the alternant formula

$$s_\lambda(x) = \frac{\det[x_i^{\lambda_j + N - j}]_{i,j=1}^N}{\det[x_i^{N-j}]_{i,j=1}^N}$$

from the fundamental symmetric identities. Prove  $s_{(r)} = h_r$  and  $s_{(1^r)} = e_r$ .

**Definition 10.2** (Row insertion and RSK). A semistandard Young tableau has weakly increasing rows and strictly increasing columns. To insert an integer  $x$ , replace the first entry strictly larger than  $x$  in the first row and insert the replaced entry in the next row; append when no such entry exists. For  $W = (W_{ij}) \in M_{m,n}(\mathbb{Z}_{\geq 0})$ , form the lexicographically ordered two-line word containing  $W_{ij}$  copies of  $(i, j)$ , insert the lower letters, and record the upper letters in the created boxes. The resulting pair  $(P, Q)$  of tableaux of common shape  $\lambda$  is the RSK image of  $W$ .

**Definition 10.3** (Last-passage values). For nonnegative weights  $W = (W_{ij})$ , define

$$G(m, n) = \max_{\pi: (1,1) \nearrow (m,n)} \sum_{(i,j) \in \pi} W_{ij}.$$

For parameters  $\alpha_i, \gamma_j \geq 0$  with  $\alpha_i \gamma_j < 1$ , the geometric model has independent weights

$$\mathbb{P}(W_{ij} = k) = (1 - \alpha_i \gamma_j)(\alpha_i \gamma_j)^k, \quad k \in \mathbb{Z}_{\geq 0}.$$

**Exercise 10.2** (RSK and the finite Schur measure). Prove that row insertion is reversible and hence that RSK is a bijection. Show that the length of the first row of its shape is the maximal weight of an up-right path:

$$\lambda_1 = G(m, n).$$

For variables  $\alpha = (\alpha_1, \dots, \alpha_m)$ , prove the tableau formula

$$s_\lambda(\alpha) = \sum_T \prod_i \alpha_i^{m_i(T)}.$$

Use the contents of the two RSK tableaux and the geometric weight law to derive

$$\mathbb{P}(\lambda) = \prod_{i,j} (1 - \alpha_i \gamma_j) s_\lambda(\alpha) s_\lambda(\gamma)$$

and the Cauchy identity

$$\sum_\lambda s_\lambda(\alpha) s_\lambda(\gamma) = \prod_{i,j} (1 - \alpha_i \gamma_j)^{-1}.$$

**Definition 10.4** (Interlacing partitions). For partitions  $\mu$  and  $\lambda$ , write  $\mu \prec \lambda$  when

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \dots.$$

**Exercise 10.3** (Nonintersecting Schur paths). Prove the branching formula

$$s_\lambda(\alpha_1, \dots, \alpha_m) = \sum_{\emptyset = \lambda^{(0)} \prec \dots \prec \lambda^{(m)} = \lambda} \prod_{k=1}^m \alpha_k^{|\lambda^{(k)}| - |\lambda^{(k-1)}|}.$$

Encode each interlacing chain by nonintersecting paths on the integer lattice. For a finite directed acyclic graph with compatible sources  $u_i$  and sinks  $v_i$ , let  $Z(u_i, v_j)$  be the weighted sum of paths from  $u_i$  to  $v_j$ . Use the first intersection and tail exchange as a sign-reversing involution to prove

$$\sum_{\substack{(\pi_1, \dots, \pi_r) \\ \text{vertex-disjoint}}} \prod_i \text{wt}(\pi_i) = \det[Z(u_i, v_j)]_{i,j=1}^r.$$

Apply this identity to the interlacing-chain encoding of  $s_\lambda$ .

**Definition 10.5** (Finite Schur kernel). For the preceding finite alphabets, set

$$\Phi(z) = \frac{\prod_j (1 - \gamma_j/z)}{\prod_i (1 - \alpha_i z)}.$$

Choose concentric contours with

$$\max_j \gamma_j < |w| < |z| < \min_i \alpha_i^{-1}.$$

For  $r, s \in \mathbb{Z} + \frac{1}{2}$ , define

$$K_{\alpha, \gamma}(r, s) = \frac{1}{(2\pi i)^2} \oint \oint \frac{\Phi(z)}{\Phi(w)} \frac{1}{z - w} z^{-r-1/2} w^{s-1/2} dz dw.$$

**Exercise 10.4** (Schur correlation kernel). Encode  $\lambda$  by

$$\mathcal{X}(\lambda) = \{\lambda_i - i + \frac{1}{2} : i \geq 1\}.$$

Glue the two nonintersecting path ensembles associated with  $s_\lambda(\alpha)$  and  $s_\lambda(\gamma)$  along their common boundary  $\lambda$ . Apply the Andréief identity successively to prove that the occupation correlations of the glued ensemble are determinants. Evaluate the resulting one-particle propagators by coefficient extraction and derive the kernel  $K_{\alpha, \gamma}$ . Deduce the exact finite Fredholm formula

$$\mathbb{P}(G(m, n) \leq L) = \det(I - K_{\alpha, \gamma})_{\ell^2(\{L+\frac{1}{2}, L+\frac{3}{2}, \dots\})}.$$

State how repeated last-passage or growth experiments with the specified geometric weights test this distribution without invoking an asymptotic limit.

**Definition 10.6** (Exponential last-passage model). Let  $E_{ij}$  be independent rate-one exponential random variables and define

$$G_{\text{exp}}(m, n) = \max_{\pi: (1,1) \nearrow (m,n)} \sum_{(i,j) \in \pi} E_{ij}.$$

**Exercise 10.5** (Exponential last-passage kernel). Set  $\alpha_i = \gamma_j = e^{-\varepsilon/2}$  in the  $n \times n$  geometric model and prove that  $\varepsilon W_{ij}$  converges in distribution to a rate-one exponential variable as  $\varepsilon \downarrow 0$ . Take this limit in the finite Schur Fredholm formula and identify the resulting finite- $n$  kernel with the Laguerre Christoffel–Darboux kernel.

**Exercise 10.6** (Laguerre edge limit). Use the Laguerre contour integral to locate the double saddle at the upper spectral edge. Perform the cubic rescaling, prove uniform exponential bounds on the deformed contours, and derive convergence of the rescaled Laguerre kernel to  $K_{\text{Ai}}$ . Use the bounds to pass to the Fredholm series.

**Exercise 10.7** (Last-passage fluctuations). Combine the exponential last-passage determinant with the Laguerre edge limit to prove

$$\frac{G_{\text{exp}}(n, n) - 4n}{2^{4/3} n^{1/3}} \implies \chi_2, \quad \mathbb{P}(\chi_2 \leq s) = F_2(s).$$

Deduce the  $n^{1/3}$  height-fluctuation scale. Balance this scale against the quadratic curvature of the deterministic limit shape to predict the transverse  $n^{2/3}$  scale. State the independent-weight, boundary, centering, scale-calibration, and finite-size assumptions required when the Tracy–Widom prediction is compared with interface-height or directed transport data.

## Part III

# Quantization

## 11 Geometric Quantization

**Definition 11.1** (Differential forms). A *differential  $k$ -form* on  $M$  is a smooth section of  $\Lambda^k T^*M$ ; write  $\Omega^k(M)$  for the space of differential  $k$ -forms. The exterior products on the cotangent spaces define

$$\wedge : \Omega^r(M) \times \Omega^s(M) \longrightarrow \Omega^{r+s}(M).$$

For a smooth map  $F : M \rightarrow N$ , define

$$(F^*\alpha)_p(v_1, \dots, v_k) = \alpha_{F(p)}(dF_p v_1, \dots, dF_p v_k).$$

For a vector field  $X$ , define contraction by

$$(\iota_X \alpha)_p(v_2, \dots, v_k) = \alpha_p(X_p, v_2, \dots, v_k).$$

The *exterior derivative* is the family  $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$  satisfying  $df(X) = Xf$ , the graded Leibniz rule,  $d^2 = 0$ , and  $F^*d = dF^*$ . For a vector field  $X$  with local flow  $\Phi_t$ , define

$$\mathcal{L}_X \alpha = \left. \frac{d}{dt} \right|_{t=0} \Phi_t^* \alpha.$$

**Exercise 11.1** (Cartan calculus). Construct  $d$  locally and prove its uniqueness. For contraction  $\iota_X$ , prove Cartan's identity

$$\mathcal{L}_X = d\iota_X + \iota_X d.$$

**Exercise 11.2** (Local Poincaré lemma). Let  $U$  be a star-shaped open neighborhood of 0 in a finite-dimensional real vector space. For  $\alpha \in \Omega^k(U)$  with  $k > 0$ , define

$$(K\alpha)_x(v_1, \dots, v_{k-1}) = \int_0^1 t^{k-1} \alpha_{tx}(x, v_1, \dots, v_{k-1}) dt.$$

Use Cartan's identity for the radial flow to prove

$$dK + Kd = \text{id}$$

on positive-degree forms. Deduce that every closed form of positive degree on  $U$  is exact, with a primitive vanishing at the origin.

**Definition 11.2** (Cotangent phase space). For the projection  $\pi : T^*Q \rightarrow Q$ , define the tautological one-form by

$$\theta_\alpha(v) = \alpha(d\pi(v)), \quad \alpha \in T^*Q, \quad v \in T_\alpha(T^*Q).$$

The canonical two-form on  $T^*Q$  is  $\omega = d\theta$ .

**Exercise 11.3** (From linear to cotangent phase space). For a finite-dimensional real vector space  $V$ , construct the canonical identifications

$$TV \cong V \times V, \quad T^*V \cong V \times V^*.$$

Prove that a phase  $S \in C^\infty(Q)$  determines the momentum  $p = dS_q \in T_q^*Q$  and wave covector  $p/\hbar$ . For a diffeomorphism  $F : Q \rightarrow Q$ , construct its cotangent lift and prove that it preserves  $\theta$  and  $d\theta$ . Under  $T^*V \cong V \times V^*$ , prove

$$\omega((q, p), (q', p')) = p(q') - p'(q).$$

**Definition 11.3** (Symplectic manifolds). A *symplectic form* on a smooth manifold  $M$  is a closed nondegenerate two-form  $\omega$ . A *symplectic vector space* is a finite-dimensional real vector space with a nondegenerate alternating bilinear form, regarded as a constant symplectic form. For  $f \in C^\infty(M)$ , define

$$\iota_{X_f}\omega = -df, \quad \{f, g\} = -\omega(X_f, X_g).$$

The local flow of  $X_f$  is the *Hamiltonian flow* of  $f$ . A diffeomorphism  $F : (M, \omega_M) \rightarrow (N, \omega_N)$  is a *canonical transformation* when  $F^*\omega_N = \omega_M$ .

**Exercise 11.4** (Hamiltonian identities). Prove that the canonical two-form  $d\theta$  on  $T^*Q$  is nondegenerate and hence symplectic. Prove that the Poisson bracket is antisymmetric, satisfies the Leibniz rule and Jacobi identity, and is preserved by canonical transformations. For the Hamiltonian flow  $\Phi_t$  of  $H$ , prove

$$\frac{d}{dt}f(\Phi_t(x)) = \{f, H\}(\Phi_t(x)).$$

**Exercise 11.5** (Liouville's theorem). On a  $2n$ -dimensional symplectic manifold, prove that Hamiltonian flow preserves  $\omega$ , the volume form  $\omega^n/n!$ , and every density depending only on the Hamiltonian. On a symplectic vector space, prove that the infinitesimal generator of a linear Hamiltonian flow has trace zero and that the flow has determinant one.

**Definition 11.4** (Contact manifolds). Let  $Y$  be a smooth manifold of dimension  $2n+1$ . A one-form  $\alpha$  is a *contact form* when

$$\alpha \wedge (d\alpha)^n$$

is nowhere zero. Its kernel  $\xi = \ker \alpha$  is the associated cooriented *contact structure*; contact forms defining the same cooriented contact structure differ by multiplication by a positive smooth function. The *Reeb vector field*  $R_\alpha$  is defined by

$$\alpha(R_\alpha) = 1, \quad \iota_{R_\alpha}d\alpha = 0.$$

An immersed  $n$ -manifold  $\iota : L \rightarrow Y$  is *Legendrian* when  $\iota^*\alpha = 0$ . A diffeomorphism  $F : (Y, \xi) \rightarrow (Y', \xi')$  is a *contact transformation* when  $F^*\alpha' = h\alpha$  for a positive smooth function  $h$ .

**Exercise 11.6** (Contactification and symplectization). Let  $(M, \omega = d\lambda)$  be an exact symplectic manifold. On  $M \times \mathbb{R}$ , with coordinate function  $u$  on the second factor, prove that

$$\alpha = du - \lambda$$

is a contact form and that  $R_\alpha$  is the infinitesimal generator of translation in the  $\mathbb{R}$  factor. For a manifold  $X$ , identify  $J^1X = T^*X \times \mathbb{R}$  with this contactification and prove that

$$j^1f : X \longrightarrow J^1X, \quad j^1f(x) = (df_x, f(x)),$$

is Legendrian for every  $f \in C^\infty(X)$ . Prove that every section of  $J^1X \rightarrow X$  with Legendrian image is of this form.

For a contact manifold  $(Y, \alpha)$  and the positive coordinate  $r$  on  $\mathbb{R}_{>0}$ , prove that

$$\Omega = d(r\alpha)$$

is symplectic on  $\mathbb{R}_{>0} \times Y$ . Determine the vector field  $Z$  satisfying  $\iota_Z\Omega = r\alpha$  and prove that multiplication of  $\alpha$  by a positive function gives a symplectomorphic symplectization.



**Definition 11.5** (Integration of forms). An *orientation* of an  $n$ -manifold is a continuous choice of component of the nonzero elements in  $\Lambda^n T_p^* M$ . The integral  $\int_M \omega$  of a compactly supported top form is defined using orientation-preserving charts and a partition of unity.

**Exercise 11.7** (Stokes' theorem). Prove that the integral is independent of charts and partition of unity. For an oriented manifold with boundary, prove

$$\int_M d\omega = \int_{\partial M} \omega.$$

Deduce the fundamental theorem of calculus, Green's theorem, the divergence theorem, and the classical curl theorem. For time-dependent magnetic two-form  $B$  and electric one-form  $E$ , prove the equivalence of

$$\int_{\partial S} E = -\frac{d}{dt} \int_S B \quad \text{and} \quad dE = -\partial_t B,$$

recovering the observed flux–emf law.

**Definition 11.6** (de Rham cohomology). A differential form  $\omega$  is *closed* if  $d\omega = 0$  and *exact* if  $\omega = d\eta$  for some form  $\eta$ . Since  $d^2 = 0$ , the *de Rham cohomology* of  $M$  is

$$H_{\text{dR}}^k(M) = \frac{\ker(d : \Omega^k(M) \rightarrow \Omega^{k+1}(M))}{\text{im}(d : \Omega^{k-1}(M) \rightarrow \Omega^k(M))}.$$

Pullback induces maps on de Rham cohomology, and the wedge product makes  $H_{\text{dR}}^*(M)$  a graded-commutative algebra.

**Exercise 11.8** (de Rham classes and periods). Use the local Poincaré lemma and Stokes' theorem to prove that an exact  $k$ -form has zero integral over every closed oriented  $k$ -manifold mapped smoothly into  $M$ . Prove  $H_{\text{dR}}^1(S^1) \cong \mathbb{R}$ , with the isomorphism given by integration.

On  $\mathbb{R}^2 \setminus \{0\}$ , show that

$$\alpha = \frac{-y dx + x dy}{x^2 + y^2}$$

is closed and has integral  $2\pi$  around the unit circle, hence is not exact. Compute  $H_{\text{dR}}^1(\mathbb{R}^2 \setminus \{0\}) = \mathbb{R} \cdot [\alpha]$ . Deduce that a closed electromagnetic field strength has local potentials, while it has a global potential  $F = dA$  exactly when  $[F] = 0$  in de Rham cohomology.

**Definition 11.7** (Connections and parallel transport). A *connection* on a vector bundle  $E \rightarrow M$  is an  $\mathbb{R}$ -bilinear map  $\nabla : \Gamma(TM) \times \Gamma(E) \rightarrow \Gamma(E)$ , linear over smooth functions in the first variable and satisfying  $\nabla_X(fs) = X(f)s + f\nabla_X s$ . Along a curve  $\gamma$  it induces a covariant derivative  $D/dt$ ; its solutions  $Ds/dt = 0$  define *parallel transport* between endpoint fibers.

**Definition 11.8** (Curvature and holonomy). The *curvature* of  $\nabla$  is the  $\text{End}(E)$ -valued two-form

$$R^\nabla(X, Y)s = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X, Y]} s.$$

The *holonomy group* at  $p$  consists of the automorphisms of  $E_p$  obtained by parallel transport around loops based at  $p$ .

**Exercise 11.9** (Parallel transport and curvature). Prove existence, uniqueness, composition, and inverse properties of parallel transport. Prove that curvature is tensorial and gives the leading failure of transport around an infinitesimal parallelogram to return a vector to itself. Deduce that a flat connection has path-independent transport on every simply connected coordinate domain.

**Definition 11.9** (Principal connections). For a Lie group  $G$ , a *principal  $G$ -bundle* is a smooth map  $P \rightarrow M$  with a free right  $G$ -action, transitive on each fiber, and local  $G$ -equivariant trivializations. For a representation  $G \rightarrow \mathrm{GL}(W)$ , its associated vector bundle is  $P \times_G W = (P \times W)/((pg, w) \sim (p, gw))$ . A *principal connection* is an equivariant  $\mathfrak{g}$ -valued one-form  $A$  on  $P$  reproducing fundamental vertical fields. It induces connections on every associated bundle, and its *curvature* is  $F_A = dA + \frac{1}{2}[A \wedge A]$ . A *gauge transformation* is a  $G$ -equivariant bundle automorphism covering  $\mathrm{id}_M$ .

**Exercise 11.10** (Gauge covariance and Bianchi identity). Prove  $d_A F_A = 0$ . For a gauge transformation  $\Phi$ , prove  $F_{\Phi^* A} = \Phi^* F_A$  and that holonomy changes by conjugacy. Prove that a flat connection has trivial holonomy on contractible loops and is gauge-equivalent to a product connection on a connected base exactly when its holonomy representation is trivial. For  $G = U(1)$ , compute the holonomy around a generator of the fundamental group and exhibit the obstruction to removing a flat connection globally.

**Definition 11.10** (Prequantum line bundles). A *prequantum line bundle* over  $(M, \omega)$  is a Hermitian line bundle  $L \rightarrow M$  with a unitary connection  $\nabla$  satisfying

$$F_\nabla = -\frac{i}{\hbar}\omega.$$

For  $f \in C^\infty(M)$ , its *Kostant–Souriau operator* is

$$\mathcal{Q}_{\mathrm{pre}}(f) = -i\hbar\nabla_{X_f} + f.$$

**Definition 11.11** (Metaplectic correction). A *complex polarization* is an involutive Lagrangian subbundle  $P \subset T_{\mathbb{C}}M$ . Its annihilator and canonical line are

$$P^\circ = \{\alpha \in T_{\mathbb{C}}^*M : \alpha|_P = 0\}, \quad K_P = \Lambda^{\mathrm{top}} P^\circ.$$

A *half-form bundle* is a line bundle  $\delta_P$  with  $\delta_P^{\otimes 2} \cong K_P$ , equipped with the partial connection along  $P$  induced by Lie derivatives. A corrected polarized section of  $L \otimes \delta_P$  is covariantly constant along  $P$  and square-integrable for the density determined by the half-form pairing. The corrected quantum space is its Hilbert completion when this pairing is positive.

**Exercise 11.11** (Prequantum integrality). Prove that a prequantum line bundle exists if and only if the period condition

$$\frac{1}{2\pi\hbar} \int_\Sigma f^* \omega \in \mathbb{Z}$$

holds for every smooth map from a closed oriented surface  $f : \Sigma \rightarrow M$ . Construct the line-bundle transition functions from local symplectic potentials and prove that the period condition is exactly their cocycle condition.

**Exercise 11.12** (Kostant–Souriau commutators). Prove

$$[\mathcal{Q}_{\mathrm{pre}}(f), \mathcal{Q}_{\mathrm{pre}}(g)] = i\hbar \mathcal{Q}_{\mathrm{pre}}(\{f, g\}).$$

**Exercise 11.13** (Contactification). Let  $\pi : S(L) \rightarrow M$  be the unit-circle bundle of  $L$ , let  $A \in \Omega^1(S(L); i\mathbb{R})$  be its connection form normalized by  $A(\xi) = i$  on the circle generator  $\xi$ , and put  $\alpha = -iA$ . Prove

$$d\alpha = -\frac{1}{\hbar}\pi^*\omega,$$

that  $\alpha$  is a contact form with Reeb vector field  $\xi$ , and that sections of  $L$  correspond to functions  $F : S(L) \rightarrow \mathbb{C}$  satisfying  $F(ue^{i\theta}) = e^{-i\theta}F(u)$ .

**Exercise 11.14** (Polarized quantum spaces). For  $T^*V$  with the vertical polarization, recover  $L^2(V)$  and the operators

$$\psi(x) \mapsto \ell(x)\psi(x), \quad \psi(x) \mapsto -i\hbar d\psi_x(v).$$

For a compact Kähler manifold with  $P = T^{0,1}M$ , prove  $P^\circ = T^{*1,0}M$  and  $K_P = K_M$ . For a positive prequantum line bundle, identify the corrected quantum space with  $H^0(M, L \otimes K_M^{1/2})$ .

**Definition 11.12** (Theta functions with characteristics). For  $a, b \in \mathbb{R}$ ,  $z \in \mathbb{C}$ , and  $\tau$  in the upper half-plane, define

$$\vartheta \begin{bmatrix} a \\ b \end{bmatrix} (z \mid \tau) = \sum_{n \in \mathbb{Z}} \exp\left(\pi i(n+a)^2\tau + 2\pi i(n+a)(z+b)\right).$$

**Exercise 11.15** (Landau theta functions). Let  $Q = \mathbb{R}^2/(L_x\mathbb{Z} \oplus L_y\mathbb{Z})$  have area  $A = L_xL_y$ , let a particle of charge  $e$  experience the constant magnetic field  $B dx \wedge dy$ , and assume  $eB > 0$ . Put

$$N = \frac{eBA}{2\pi\hbar}, \quad \tau = i\frac{L_y}{L_x}, \quad z = \frac{x}{L_x} + \tau \frac{y}{L_y}.$$

Use the prequantization condition to prove that a magnetic line bundle  $L \rightarrow Q$  exists exactly when  $N \in \mathbb{Z}$ . For  $N > 0$ , use the holomorphic-gauge automorphy laws

$$f(z+1) = f(z), \quad f(z+\tau) = e^{-\pi i N \tau - 2\pi i N z} f(z)$$

to prove the magnetic-translation decomposition

$$H^0(Q, L) = \bigoplus_{r \in \mathbb{Z}/N\mathbb{Z}} \mathbb{C} \vartheta \begin{bmatrix} r/N \\ 0 \end{bmatrix} (Nz \mid N\tau).$$

Derive the automorphy and magnetic-translation eigenvalue laws directly from the defining series. Relate the holomorphic-gauge sections to physical wavefunctions by their common unitary-gauge factor.

For magnetic translations by  $L_x/N$  and  $L_y/N$ , use holonomy around a fundamental magnetic cell to prove

$$UV = e^{2\pi i/N} VU.$$

For the covariant Landau Hamiltonian, construct the raising and lowering operators and prove

$$E_n = \hbar \frac{eB}{m} \left(n + \frac{1}{2}\right), \quad \dim E_{\widehat{H}}(\{E_n\})H = N.$$

Deduce the cyclotron resonance spacing, the degeneracy density  $eB/(2\pi\hbar)$ , and the magnetic-translation interference phase. State how measurements of these three quantities separately test the energy, flux-integrality, and theta-function predictions.

## 12 Deformation Quantization

**Definition 12.1** (Coadjoint action and stabilizers). For a Lie group  $G$  with Lie algebra  $\mathfrak{g}$ , define

$$\text{Ad}_g = d(h \mapsto ghg^{-1})_e, \quad (\text{Ad}_g^* \ell)(X) = \ell(\text{Ad}_{g^{-1}} X).$$

For  $\ell \in \mathfrak{g}^*$ , define

$$\mathcal{O}_\ell = \text{Ad}^*(G)\ell, \quad G_\ell = \{g \in G : \text{Ad}_g^* \ell = \ell\},$$

and

$$\mathfrak{g}_\ell = \{X \in \mathfrak{g} : \ell([X, Y]) = 0 \text{ for every } Y \in \mathfrak{g}\}.$$

For  $X \in \mathfrak{g}$ , set

$$(\text{ad}_X^* \ell)(Y) = -\ell([X, Y]).$$

**Definition 12.2** (Kirillov form). On  $T_\ell \mathcal{O}_\ell$ , define

$$\omega_\ell(\text{ad}_X^* \ell, \text{ad}_Y^* \ell) = \ell([X, Y]).$$

For  $\eta = \text{Ad}_g^* \ell$ , transport  $\omega_\ell$  by the coadjoint action to define  $\omega_\eta$ .

**Exercise 12.1** (Geometry of coadjoint orbits). Prove

$$T_\ell \mathcal{O}_\ell = \{\text{ad}_X^* \ell : X \in \mathfrak{g}\} \cong \mathfrak{g}/\mathfrak{g}_\ell.$$

Prove that the Kirillov form is well-defined, alternating, nondegenerate, and  $G$ -invariant. Prove the cyclic identity

$$\ell([X, [Y, Z]]) + \ell([Y, [Z, X]]) + \ell([Z, [X, Y]]) = 0$$

and identify it with the closedness identity for the invariant two-form  $\omega$  on  $\mathcal{O}_\ell$ .

**Definition 12.3** (Projective representations). A *normalized multiplier* on a locally compact group  $G$  is a continuous map  $\sigma : G \times G \rightarrow \mathbb{T}$  satisfying

$$\sigma(x, e) = \sigma(e, x) = 1, \quad \sigma(x, y)\sigma(xy, z) = \sigma(y, z)\sigma(x, yz).$$

A  $\sigma$ -projective unitary representation is a strongly continuous map  $U : G \rightarrow U(H)$  satisfying

$$U_x U_y = \sigma(x, y) U_{xy}, \quad U_e = I.$$

Multipliers  $\sigma$  and  $\sigma'$  are *cohomologous* when there is a continuous map  $b : G \rightarrow \mathbb{T}$ ,  $b(e) = 1$ , such that

$$\sigma'(x, y) = b(x)b(y)\overline{b(xy)}\sigma(x, y).$$

**Definition 12.4** (Weyl multipliers). Let  $(E, \omega)$  be a symplectic vector space. For  $\lambda \in \mathbb{R}$ , define

$$\sigma_{\lambda\omega}(z, z') = e^{i\lambda\omega(z, z')/2}.$$

For physical momentum variables and  $\hbar > 0$ , the *Weyl multiplier* is  $\sigma_{\omega/\hbar}$ .

**Exercise 12.2** (The Weyl multiplier). Prove that  $\sigma_{\lambda\omega}$  is a normalized multiplier for every  $\lambda \in \mathbb{R}$ .

**Definition 12.5** (Commutator bicharacter). For a multiplier  $\sigma$  on an abelian group  $A$ , define

$$c_\sigma(x, y) = \sigma(x, y)\overline{\sigma(y, x)}.$$

**Exercise 12.3** (Commutator bicharacter). Prove that  $c_\sigma$  is an alternating bicharacter and depends only on the cohomology class of  $\sigma$ . For the Weyl multiplier on  $V \oplus V^*$ , prove

$$c_{\sigma_{\omega/\hbar}}((q, p), (q', p')) = e^{i(p(q') - p'(q))/\hbar}.$$

Prove

$$c_{\sigma_{\lambda\omega}}(z, z') = e^{i\lambda\omega(z, z')}.$$

**Definition 12.6** (Heisenberg group). For a symplectic vector space  $(E, \omega)$ , its *Heisenberg group* is

$$\mathbb{H}(E, \omega) = E \times \mathbb{R}$$

with multiplication

$$(z, s)(z', s') = (z + z', s + s' + \frac{1}{2}\omega(z, z')).$$

Write  $\mathbb{H}_n = \mathbb{H}(\mathbb{R}^n \oplus (\mathbb{R}^n)^*, \omega)$  for the canonical symplectic form on  $\mathbb{R}^n \oplus (\mathbb{R}^n)^*$ . Its Lie algebra is  $\mathfrak{h}(E, \omega) = E \oplus \mathbb{R}Z$  with

$$[(z, sZ), (z', s'Z)] = \omega(z, z')Z.$$

For  $E = V \oplus V^*$  and  $\lambda \neq 0$ , the *Schrödinger representation* on  $L^2(V)$  is

$$(\pi_\lambda(q, p, s)\xi)(x) = e^{i\lambda(s+p(x-q/2))}\xi(x - q).$$

For  $\hbar > 0$ ,  $\ell \in V^*$ , and  $v \in V$ , define on  $\mathcal{S}(V)$

$$Q_\ell \psi(x) = \ell(x)\psi(x), \quad P_v \psi(x) = -i\hbar d\psi_x(v).$$

**Definition 12.7** (Polarization). For a connected simply connected nilpotent group  $G$  and  $\ell \in \mathfrak{g}^*$ , a *polarization* is a Lie subalgebra  $\mathfrak{p} \subseteq \mathfrak{g}$  satisfying

$$\ell([\mathfrak{p}, \mathfrak{p}]) = 0, \quad \dim \mathfrak{p} = \frac{\dim \mathfrak{g} + \dim \mathfrak{g}_\ell}{2}.$$

For  $P = \exp(\mathfrak{p})$ , set  $\chi_\ell(\exp X) = e^{i\ell(X)}$ . The induced representation  $\text{Ind}_P^G \chi_\ell$  acts by left translation on the  $L^2$ -sections over  $G/P$  satisfying  $F(gp) = \chi_\ell(p)^{-1}F(g)$ .

**Exercise 12.4** (Central extensions). Prove that  $\mathbb{H}(E, \omega)$  is a locally compact group with center  $\{0\} \times \mathbb{R}$  and that

$$\pi_\lambda(0, s) = e^{i\lambda s}I.$$

Prove that projective representations of  $E$  with multiplier  $e^{i\lambda\omega/2}$  correspond to representations of  $\mathbb{H}(E, \omega)$  with central character  $e^{i\lambda s}$ . Let  $p : \mathbb{H}(E, \omega) \rightarrow E$  be the quotient map and define

$$\alpha_{(z,s)}(v, r) = r - \frac{1}{2}\omega(z, v).$$

Prove that  $\alpha$  is left-invariant,

$$d\alpha = -p^*\omega,$$

and that  $\alpha$  is a contact form whose Reeb flow is translation in the central direction.

**Exercise 12.5** (Canonical commutation and Weyl relations). Prove

$$[Q_\ell, P_v] = i\hbar\ell(v)I.$$

For

$$(T_q\psi)(x) = \psi(x - q), \quad (M_p^\hbar\psi)(x) = e^{ip(x)/\hbar}\psi(x),$$

prove

$$M_p^\hbar T_q = e^{ip(q)/\hbar} T_q M_p^\hbar.$$

For

$$W_\hbar(q, p) = e^{-ip(q)/(2\hbar)} M_p^\hbar T_q,$$

prove

$$W_\hbar(z)W_\hbar(z') = e^{i\omega(z, z')/(2\hbar)} W_\hbar(z + z')$$

and identify  $W_\hbar(z)$  with  $\pi_{1/\hbar}(z, 0)$  under the canonical set-theoretic section  $z \mapsto (z, 0)$ . Prove that this section is not a homomorphism and that its failure to preserve multiplication is exactly the displayed multiplier.

**Exercise 12.6** (Schrödinger model). Identify

$$\mathfrak{h}(E, \omega)^* = E^* \oplus \mathbb{R}Z^*.$$

For  $(\xi, \lambda) \in E^* \oplus \mathbb{R}Z^*$ , prove

$$\text{Ad}_{(z,s)}^*(\xi, \lambda) = (\xi - \lambda\omega(z, \cdot), \lambda).$$

Deduce that the orbits with  $\lambda = 0$  are points and that, for  $\lambda \neq 0$ ,

$$\mathcal{O}_\lambda = \{(\xi, \lambda) : \xi \in E^*\}.$$

Prove that

$$E \longrightarrow \mathcal{O}_\lambda, \quad z \longmapsto (-\lambda\omega(z, \cdot), \lambda),$$

pulls the Kirillov form back to  $\lambda\omega$ .

For  $E = V \oplus V^*$  and  $\mathfrak{p} = V^* \oplus \mathbb{R}Z$ , prove that  $\mathfrak{p}$  is a polarization at  $(0, \lambda)$ . Identify  $\mathbb{H}(E, \omega)/\exp(\mathfrak{p})$  with  $V$  and compute the induced action to obtain

$$(\pi_\lambda(q, p, s)\psi)(x) = e^{i\lambda(s+p(x-q/2))}\psi(x-q).$$

For  $\lambda = 0$ , recover every character of  $E$  from a point orbit.

**Exercise 12.7** (Irreducibility of the Schrödinger representation). Prove that an operator on  $L^2(V)$  commuting with every translation and every modulation is scalar: first use the Fourier transform to identify the commutant of translations, and then impose commutation with modulations. Deduce that  $\pi_\lambda$  is irreducible for  $\lambda \neq 0$ .

**Exercise 12.8** (Stone–von Neumann theorem). Let  $U$  be a strongly continuous irreducible unitary representation of  $\mathbb{H}(V \oplus V^*, \omega)$  with central character  $U(0, s) = e^{i\lambda s}I$ , where  $\lambda \neq 0$ . Integrate Schwartz functions on  $V \oplus V^*$  against the canonical section of  $U$ . Use Fourier inversion and a Gaussian approximate identity to show that this integrated representation contains the same nonzero finite-rank operators as the Schrödinger model. Construct a nonzero intertwiner and use irreducibility to prove  $U \simeq \pi_\lambda$ .

**Definition 12.8** (Trace and Hilbert–Schmidt classes). For  $x, y \in H$ , let

$$\theta_{x,y}(\xi) = \langle y, \xi \rangle x.$$

The *trace class*  $S_1(H)$  consists of operators admitting a representation

$$A = \sum_{j=1}^{\infty} \theta_{x_j, y_j}, \quad \sum_{j=1}^{\infty} \|x_j\| \|y_j\| < \infty,$$

where the series converges in operator norm, with norm the infimum of the displayed sums. Define

$$\text{Tr}(A) = \sum_{j=1}^{\infty} \langle y_j, x_j \rangle.$$

The *Hilbert–Schmidt class* is

$$S_2(H) = \{A \in B(H) : A^*A \in S_1(H)\}, \quad \|A\|_2^2 = \text{Tr}(A^*A).$$

**Exercise 12.9** (Trace ideals). Prove that  $S_1(H)$ ,  $\text{Tr}$ , and  $S_2(H)$  are well-defined. Prove that  $\langle A, B \rangle_{S_2} = \text{Tr}(A^*B)$  makes  $S_2(H)$  a Hilbert space and that the pairing  $(A, T) \mapsto \text{Tr}(AT)$  identifies  $S_1(H)^*$  isometrically with  $B(H)$ .

**Definition 12.9** (Coefficient transforms). Let  $(X, \mu)$  be a measure space and let  $U : X \rightarrow U(H)$  be a weakly measurable map. The *coefficient transform* of the unitary system  $U$  is

$$\mathcal{V}_U : S_1(H) \longrightarrow L^\infty(X), \quad \mathcal{V}_U(A)(x) = \text{Tr}(AU_x).$$

**Definition 12.10** (Fourier–Wigner transform). Let  $V$  be an  $n$ -dimensional real vector space with Lebesgue measure  $dq$ , let  $\hbar > 0$ , and equip  $V^*$  with the dual measure  $dp$  for which  $(2\pi\hbar)^{-n/2} \int_V e^{-ip(q)/\hbar} f(q) dq$  is unitary on  $L^2$ . For  $q \in V$  and  $p \in V^*$ , define

$$(T_q \xi)(x) = \xi(x - q), \quad (M_p^\hbar \xi)(x) = e^{ip(x)/\hbar} \xi(x).$$

For  $z = (q, p) \in V \oplus V^*$ , define

$$W_\hbar(z) = e^{-ip(q)/(2\hbar)} M_p^\hbar T_q.$$

The *Fourier–Wigner transform* is

$$\mathcal{W}_\hbar : S_1(L^2(V)) \longrightarrow C_b(V \oplus V^*), \quad \mathcal{W}_\hbar(A)(z) = (2\pi\hbar)^{-n/2} \text{Tr}(AW_\hbar(z)).$$

**Exercise 12.10** (Fourier–Wigner Plancherel). Prove

$$W_\hbar(z)^* = W_\hbar(-z).$$

For  $A, B \in S_1(L^2(V)) \cap S_2(L^2(V))$ , prove the Fourier–Wigner orthogonality identity

$$\int_{V \oplus V^*} \overline{\mathcal{W}_\hbar(A)(z)} \mathcal{W}_\hbar(B)(z) dz = \text{Tr}(A^*B).$$

Deduce that  $\mathcal{W}_\hbar$  extends uniquely to a unitary map

$$S_2(L^2(V)) \longrightarrow L^2(V \oplus V^*)$$

and, whenever  $\mathcal{W}_\hbar(A) \in L^1(V \oplus V^*)$ , prove

$$A = (2\pi\hbar)^{-n/2} \int_{V \oplus V^*} \mathcal{W}_\hbar(A)(z) W_\hbar(z)^* dz$$

in the weak operator topology.

**Exercise 12.11** (Schrödinger character). For  $V = \mathbb{R}^n$ , let  $f \in C^\infty(\mathbb{H}_n)$  satisfy

$$(q, p, s) \longmapsto f(\exp(q, p, s)) \in \mathcal{S}(\mathbb{R}^{2n+1})$$

under the canonical identification  $\mathfrak{h}_n = \mathbb{R}^n \oplus (\mathbb{R}^n)^* \oplus \mathbb{R}$ . With Haar measure  $dq dp ds$ , put

$$\pi_\lambda(f) = \int_{\mathbb{H}_n} f(g) \pi_\lambda(g) dg.$$

Prove

$$\text{Tr } \pi_\lambda(f) = (2\pi)^n |\lambda|^{-n} \int_{\mathbb{R}} f(0, 0, s) e^{i\lambda s} ds.$$

Derive this identity directly from the Schrödinger kernel and identify the factor  $|\lambda|^{-n}$  with the symplectic-volume scaling of  $\mathcal{O}_\lambda$ .

**Exercise 12.12** (Heisenberg Plancherel and inversion). For  $f \in L^1(\mathbb{H}_n) \cap L^2(\mathbb{H}_n)$ , use Haar measure  $dq dp ds$  and the scalar Fourier transform  $\hat{h}(\lambda) = \int_{\mathbb{R}} h(s) e^{-i\lambda s} ds$ , and define

$$\hat{f}(\lambda) = \int_{\mathbb{H}_n} f(g) \pi_\lambda(g)^* dg.$$

Compute the integral kernel of  $\hat{f}(\lambda)$  and apply Euclidean Plancherel in the central and phase-space variables. Deduce the Plancherel measure  $(2\pi)^{-(n+1)} |\lambda|^n d\lambda$  and prove

$$\|f\|_2^2 = (2\pi)^{-(n+1)} \int_{\mathbb{R} \setminus \{0\}} \|\hat{f}(\lambda)\|_{\text{HS}}^2 |\lambda|^n d\lambda.$$

Prove the inversion formula and the distributional orthogonality of Weyl operators. Prove that representations with trivial central character have zero Plancherel measure.

**Definition 12.11** (Weyl quantization). For  $a \in \mathcal{S}(V \oplus V^*)$ , define  $\text{Op}_h^W(a) : \mathcal{S}(V) \rightarrow \mathcal{S}(V)$  by

$$(\text{Op}_h^W(a)\psi)(x) = \frac{1}{(2\pi\hbar)^n} \int_{V \times V^*} e^{ip(x-y)/\hbar} a\left(\frac{x+y}{2}, p\right) \psi(y) dy dp.$$

For polynomial symbols, use the same formula as an oscillatory integral on  $\mathcal{S}(V)$ .

**Definition 12.12** (Wigner distributions). For  $\psi, \varphi \in \mathcal{S}(V)$ , define

$$W_h(\psi, \varphi)(q, p) = \frac{1}{(2\pi\hbar)^n} \int_V e^{-ip(y)/\hbar} \overline{\psi(q - y/2)} \varphi(q + y/2) dy.$$

Write  $W_h[\psi] = W_h(\psi, \psi)$ .

**Exercise 12.13** (Weyl calculus and the Moyal product). For  $\ell \in V^*$  and  $v \in V$ , prove

$$\text{Op}_h^W((q, p) \mapsto \ell(q)) = Q_\ell, \quad \text{Op}_h^W((q, p) \mapsto p(v)) = P_v.$$

Prove

$$\text{Op}_h^W(a)^* = \text{Op}_h^W(\bar{a}), \quad \|\text{Op}_h^W(a)\|_{\text{HS}} = (2\pi\hbar)^{-n/2} \|a\|_2,$$

and

$$\langle \psi, \text{Op}_h^W(a)\varphi \rangle = \int_{V \oplus V^*} a(q, p) W_h(\psi, \varphi)(q, p) dq dp.$$

Prove that for  $a, b \in \mathcal{S}(V \oplus V^*)$  there is a unique  $a \star_h b \in \mathcal{S}(V \oplus V^*)$  satisfying

$$\text{Op}_h^W(a \star_h b) = \text{Op}_h^W(a) \text{Op}_h^W(b).$$

Derive the oscillatory-integral formula for  $\star_h$  and prove

$$a \star_h b = ab + \frac{i\hbar}{2} \{a, b\} + O(\hbar^2), \quad \frac{a \star_h b - b \star_h a}{i\hbar} = \{a, b\} + O(\hbar^2)$$

in the Schwartz-symbol topology.

**Exercise 12.14** (Groenewold–van Hove obstruction). Specialize to  $V = \mathbb{R}$  and write  $Q = Q_{\text{id}_{\mathbb{R}}}$  and  $P = P_1$ . Prove that the polynomials of degree at most two form a Lie subalgebra for the Poisson bracket and that Weyl quantization is an exact Lie-algebra quantization on this subalgebra. Quantize  $q^2 p^2$  using

$$q^2 p^2 = \frac{1}{9} \{q^3, p^3\} = \frac{1}{3} \{q^2 p, qp^2\}$$



and prove that there is no linear quantization of the polynomial Poisson algebra satisfying

$$\mathcal{Q}(1) = I, \quad \mathcal{Q}(q) = Q, \quad \mathcal{Q}(p) = P, \quad \mathcal{Q}(q^m) = Q^m, \quad \mathcal{Q}(p^m) = P^m,$$

together with

$$\mathcal{Q}(\{f, g\}) = (i\hbar)^{-1}[\mathcal{Q}(f), \mathcal{Q}(g)]$$

on a common invariant domain.

**Definition 12.13** (Classical and quantum evolution). Let  $H \in C^\infty(V \oplus V^*, \mathbb{R})$  have bounded derivatives of order at least two, let  $\Phi_t$  be its Hamiltonian flow, and let  $\text{Op}_h^W(H)$  denote its self-adjoint realization. Define

$$U_h(t) = e^{-it \text{Op}_h^W(H)/\hbar}.$$

**Exercise 12.18** (Laguerre tomography). For the dimensionless quadrature

$$X_\theta = \sqrt{\frac{m\omega}{\hbar}} Q \cos \theta + \frac{P}{\sqrt{m\hbar\omega}} \sin \theta,$$

prove that its probability density in a state  $\psi$  is the Radon transform of  $W_\hbar[\psi]$  along the corresponding phase-space lines. Derive the Fourier-slice inversion formula reconstructing the Wigner distribution from the family of quadrature densities. For the oscillator state  $\psi_n$ , use the Wigner–Laguerre identity to prove that the reconstructed radial profile is

$$\frac{(-1)^n}{\pi\hbar} e^{-2H_{\text{cl}}/(\hbar\omega)} L_n\left(\frac{4H_{\text{cl}}}{\hbar\omega}\right),$$

while every quadrature density is a scaled  $e^{-x^2} H_n(x)^2$ . Deduce the predicted number of quadrature nodes, the radial zero circles determined by  $L_n$ , and the sign at the phase-space origin. State how homodyne quadrature data distinguish adjacent number states and detect Wigner negativity after inverse Radon reconstruction.

Prove associativity directly by checking the multiplier identity for the phase. Differentiate in  $k$  and  $\ell$  to recover the canonical commutation relations. Prove that every affine symplectic transformation of  $E$  preserves  $\star$ . For  $E = V \oplus V^*$ , prove that the restriction to Schwartz functions is the formal  $\hbar$ -expansion of the Weyl product defined by operator composition above.

**Definition 12.16** (Magnetic lattice Hamiltonian). Let  $U, V$  be unitary translations satisfying

$$UV = e^{2\pi i \alpha} VU,$$

where  $\alpha$  is the magnetic flux through one lattice cell in units of the flux quantum. The magnetic nearest-neighbor Hamiltonian is

$$H_\alpha = U + U^* + V + V^*.$$

In the position realization on  $\ell^2(\mathbb{Z})$ , with transverse phase  $\theta$ , it is the Harper operator

**Exercise 13.1** (Infinitesimal Poisson–Lie structure). Differentiate a Poisson–Lie structure at the identity and prove that it defines a Lie bialgebra. For a coboundary Lie bialgebra, define

$$\text{CYB}(r) = [r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}].$$

Prove that the co-Jacobi identity is equivalent to  $\text{CYB}(r)$  being invariant under the diagonal adjoint action. Under the additional triangular hypothesis  $\text{CYB}(r) = 0$ , recover the classical Yang–Baxter equation; state the modified classical Yang–Baxter equation as  $\text{CYB}(r) = \Omega$  for an invariant tensor  $\Omega$  and distinguish the case  $\Omega \neq 0$ .

**Definition 13.2** (Bialgebras and Hopf algebras). A *bialgebra* is a unital associative algebra  $H$  with unital algebra homomorphisms

$$\Delta : H \rightarrow H \otimes H, \quad \varepsilon : H \rightarrow \mathbb{C}$$

satisfying co

**Exercise 13.3** (Gaussian and quantum-group binomial identities). Derive

$$\begin{bmatrix} n \\ r \end{bmatrix}_u = \begin{bmatrix} n-1 \\ r \end{bmatrix}_u + u^{n-r} \begin{bmatrix} n-1 \\ r-1 \end{bmatrix}_u$$

and prove

$$\prod_{j=0}^{n-1} (1 + tu^j) = \sum_{r=0}^n u^{r(r-1)/2} \begin{bmatrix} n \\ r \end{bmatrix}_u t^r.$$

For  $YX = uXY$ , prove the noncommutative identity

$$(X + Y)^n = \sum_{r=0}^n \begin{bmatrix} n \\ r \end{bmatrix}_u X^r Y^{n-r}.$$

Set  $u = v^2$  and apply it to  $X = E \otimes 1$  and  $Y = K \otimes E$

**Definition 13.8** ( $SU_v(2)$ ). Let  $\text{Pol}(SU_v(2))$  be the universal unital  $*$ -algebra generated by  $\alpha, \gamma$  so that

$$u = \begin{pmatrix} \alpha & -v\gamma^* \\ \gamma & \alpha^* \end{pmatrix}$$

is unitary. Equivalently, its relations are

$$\begin{aligned} \alpha^* \alpha + \gamma^* \gamma &= 1, & \alpha \alpha^* + v^2 \gamma \gamma^* &= 1, & \gamma^* \gamma &= \gamma \gamma^*, \\ \alpha \gamma &= v \gamma \alpha, & \alpha \gamma^* &= v \gamma^* \alpha, \end{aligned}$$

Define  $\Delta(u_{ij}) = \sum_k u_{ik} \otimes u_{kj}$ , and let  $C(SU_v(2))$  be the universal  $C^*$ -completion.

**Exercise 13.5** (<

**Definition 13.10** (Conjugates and quantum dimension). For a finite-dimensional  $H$ -module  $V$ , its *left dual*  $V^\vee$  has action

$$(x\varphi)(v) = \varphi(S(x)v),$$

evaluation  $\text{ev}_V(\varphi \otimes v) = \varphi(v)$ , and coevaluation  $\text{coev}_V$ . In a rigid  $C^*$ -tensor category, a *conjugate* of  $V$  is an object  $\bar{V}$  with morphisms

$$R_V : \mathbf{1} \longrightarrow \bar{V} \otimes V, \quad \bar{R}_V : \mathbf{1} \longrightarrow V \otimes \bar{V}$$

satisfying

$$(\bar{R}_V^* \otimes \text{id}_V)(\text{id}_V \otimes R_V) = \text{id}_V, \quad (R_V^* \otimes \text{id}_{\bar{V}})(\text{id}_{\bar{V}} \otimes \bar{R}_V) = \text{id}_{$$

**Exercise 13.9** (Basic hypergeometric summations). Prove the  $Q$ -Gauss identity

$${}_2\phi_1\left(\begin{matrix} a, b \\ c \end{matrix}; Q, \frac{c}{ab}\right) = \frac{(c/a, c/b; Q)_\infty}{(c, c/(ab); Q)_\infty}, \quad \left|\frac{c}{ab}\right| < 1,$$

and the terminating  $Q$ -Pfaff–Saalschütz identity

$${}_3\phi_2\left(\begin{matrix} a, b, Q^{-n} \\ c, abQ^{1-n}/c \end{matrix}; Q, Q\right) = \frac{(c/a, c/b; Q)_n}{(c, c/(ab); Q)_n}$$

by comparing coefficients after multiplication by the denominator  $Q$ -products in the first identity, and by showing that the two sides of the second identity satisfy the same recurrence in  $n$  and have the same initial value.

**Definition 13.12** (Quantum



**Definition 13.13** (Universal  $R$ -matrix and braiding). A *universal  $R$ -matrix* for a Hopf algebra  $H$  is an invertible element  $\mathcal{R}$  in a completion of  $H \otimes H$  on which the finite-dimensional tensor-product representations under consideration extend, satisfying

$$\mathcal{R}\Delta(x)\mathcal{R}^{-1} = \Delta^{\text{op}}(x), \quad (\Delta \otimes \text{id})(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{23}, \quad (\text{id} \otimes \Delta)(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{12}.$$

For finite-dimensional modules  $V, W$ , define

$$c_{V,W} = \tau \circ (\pi_V \otimes \pi_W)(\mathcal{R}) : V \otimes W \longrightarrow W \otimes V.$$

The pair  $(H, \mathcal{R})$  is *quasitriangular*.

**Definition 13.14** (Spectral exchange operators). Let  $V$  be finite-dimensional. A meromorphic family  $R(u) \in \text{End}(V \otimes V)$  satisfies the spectral Yang–Baxter equation when

$$R_{12}(u/v)R_{13}(u/w)R_{23}(v/w) = R_{23}(v/w)R_{13}(u/w)R_{12}(u/v).$$

It is regular when  $R(1)$  is a nonzero scalar multiple of the flip and unitary up to scalar when  $R_{12}(u)R_{21}(u^{-1})$  is scalar.

**Exercise 13.13** (Hexagon identities). Prove that  $c_{V,W}$  is a natural module intertwiner and that the two coproduct identities for  $\mathcal{R}$  give the two hexagon identities. Derive

$$\mathcal{R}_{12}\mathcal{R}_{13}\mathcal{R}_{23} = \mathcal{R}_{23}\mathcal{R}_{13}\mathcal{R}_{12}$$

and, on  $V^{\otimes 3}$ ,

$$c_{12}c_{23}c_{12} = c_{23}c_{12}c_{23}.$$

Prove that the operators  $c_{j,j+1}$  define braid-group representations on tensor powers of every finite-dimensional type-1 module.

**Definition 13.15** (Ribbon categories). Let  $\mathcal{C}$  be a braided rigid  $\mathbb{C}$ -linear monoidal category. A *twist* is a natural isomorphism  $\theta : \text{id}_{\mathcal{C}} \rightarrow \text{id}_{\mathcal{C}}$  satisfying

$$\theta_1 = \text{id}_1, \quad \theta_{X \otimes Y} = c_{Y,X}c_{X,Y}(\theta_X \otimes \theta_Y).$$

The category is *ribbon* when

$$\theta_{X^\vee} = (\theta_X)^\vee$$

for every object  $X$ .

**Exercise 13.14** (Ribbon elements). For a quasitriangular Hopf algebra  $(H, \mathcal{R})$ , set

$$u = m(S \otimes \text{id})(\mathcal{R}_{21}).$$

Let  $\nu_R \in H$  be central and invertible and satisfy

$$\nu_R^2 = uS(u), \quad S(\nu_R) = \nu_R, \quad \varepsilon(\nu_R) = 1, \quad \Delta(\nu_R) = (\mathcal{R}_{21}\mathcal{R})^{-1}(\nu_R \otimes \nu_R).$$

Prove that

$$\theta_V = \pi_V(\nu_R^{-1})$$

defines a ribbon twist on the category of finite-dimensional  $H$ -modules. Derive invariance of the quantum trace under framed ribbon isotopy.

**Exercise 13.15** (Chebyshev quantum dimensions). For the irreducible  $SU_v(2)$  objects  $V_n$ , use  $V_1 \otimes V_n \cong V_{n+1} \oplus V_{n-1}$  to prove

$$d_{n+1} = (v + v^{-1})d_n - d_{n-1}, \quad d_0 = 1, \quad d_1 = v + v^{-1},$$

where  $d_n = \dim_v(V_n)$ . If  $U_n$  is the Chebyshev polynomial determined by  $U_0(x) = 1$ ,  $U_1(x) = 2x$ , and  $U_{n+1}(x) = 2xU_n(x) - U_{n-1}(x)$ , deduce

$$\dim_v(V_n) = U_n\left(\frac{v + v^{-1}}{2}\right) = \frac{v^{n+1} - v^{-(n+1)}}{v - v^{-1}}.$$

**Definition 13.16** (Quantum recoupling coefficients). The matrix coefficients of the recoupling map

$$(V_a \otimes V_b) \otimes V_c \longrightarrow V_a \otimes (V_b \otimes V_c)$$

between fixed intermediate irreducible summands are the  $v$ -6j coefficients.

**Exercise 13.16** (Quantum Racah recoupling). Compute the change between the two coupling orders of  $V_a \otimes V_b \otimes V_c$  and express its entries as terminating  ${}_4\phi_3$  series with base  $Q = v^2$ . Identify them with  $Q$ -Racah polynomials and derive their orthogonality from unitarity of the recoupling map for the compact real form. Prove the pentagon identity by comparing the five coupling orders of four factors. For three resolved XXZ spin sectors, derive the recoupling probabilities as squared  $v$ -6j coefficients and state the unitarity and spectral-resolution assumptions required for this probability interpretation.

**Definition 13.17** (Type-A Hecke actions). For  $v \neq 0$ , the Hecke algebra  $\mathcal{H}_v(S_r)$  is generated by  $T_1, \dots, T_{r-1}$  with

$$\begin{aligned} T_i T_{i+1} T_i &= T_{i+1} T_i T_{i+1}, & T_i T_j &= T_j T_i \quad (|i - j| > 1), \\ (T_i - v)(T_i + v^{-1}) &= 0. \end{aligned}$$

Let  $V_1$  be the fundamental  $U_v(\mathfrak{sl}_2)$ -module. On  $V_1 \otimes V_1 \cong V_2 \oplus V_0$ , let  $P_2, P_0$  be the invariant orthogonal projections, and define

$$T = vP_2 - v^{-1}P_0, \quad \rho_r(T_i) = \text{id}^{\otimes(i-1)} \otimes T \otimes \text{id}^{\otimes(r-i-1)}.$$

**Exercise 13.17** (Fundamental Hecke symmetry). Prove that the coproduct action of  $U_v(\mathfrak{sl}_2)$  commutes with  $\rho_r(T_i)$ . Derive the braid relation from the  $Q$ -Racah recoupling identity and the Hecke relation from the two displayed spectral projections. Put  $e_i = (v - T_i)/(v + v^{-1})$  and prove that the action factors through the Temperley-Lieb relations

$$e_i^2 = e_i, \quad e_i e_{i \pm 1} e_i = (v + v^{-1})^{-2} e_i.$$

**Exercise 13.18** (Hecke Poincaré identity). Put  $\tau = v^2$ . For  $w \in S_r$ , let  $\ell(w)$  be the least number of adjacent transpositions in an expression for  $w$ , and let  $T_w = T_{i_1} \cdots T_{i_{\ell(w)}}$  for any reduced expression  $w = s_{i_1} \cdots s_{i_{\ell(w)}}$ . Prove from the braid relations that  $T_w$  is well-defined and that

$$\sum_{w \in S_r} \tau^{\ell(w)} = \prod_{j=1}^r \frac{1 - \tau^j}{1 - \tau}.$$

Prove that

$$e_+ = \frac{\sum_{w \in S_r} v^{\ell(w)} T_w}{\sum_{w \in S_r} v^{2\ell(w)}}, \quad e_- = \frac{\sum_{w \in S_r} (-v^{-1})^{\ell(w)} T_w}{\sum_{w \in S_r} v^{-2\ell(w)}}$$

are idempotents satisfying  $T_i e_+ = v e_+$  and  $T_i e_- = -v^{-1} e_-$ . Under the fundamental Hecke action from the preceding exercise, identify their images with the quantum symmetric and exterior state spaces and interpret the Poincaré polynomial as their graded state count.

**Definition 13.18** (Affine Hecke algebras). The extended affine Hecke algebra  $\mathcal{H}_v^{\text{aff}}(GL_r)$  is generated by  $T_1, \dots, T_{r-1}$  and commuting invertible elements  $X_1, \dots, X_r$ , subject to the finite Hecke relations and

$$T_i X_i T_i = X_{i+1}, \quad T_i X_j = X_j T_i \quad (j \neq i, i+1).$$

Equivalently, it is the Hecke quotient of the extended affine braid group of  $\mathbb{Z}^r \rtimes S_r$ .

**Definition 13.19** (Baxterized Hecke operators). For an invertible spectral parameter  $u$ , define

$$\check{R}_i(u) = uT_i - u^{-1}T_i^{-1}.$$

Its eigenvalues on the  $v$  and  $-v^{-1}$  eigenspaces of  $T_i$  are

$$uv - u^{-1}v^{-1}, \quad -uv^{-1} + u^{-1}v.$$

**Exercise 13.19** (Baxterization). Use the braid and Hecke relations for  $T_i$  to prove

$$\check{R}_i(u)\check{R}_{i+1}(uv)\check{R}_i(v) = \check{R}_{i+1}(v)\check{R}_i(uv)\check{R}_{i+1}(u).$$

Prove that  $\check{R}_i(1)$  is a scalar multiple of the identity and compute  $\check{R}_i(u)\check{R}_i(u^{-1})$ . For  $|u| = 1$ , prove that  $\check{R}_i(u)^* = \check{R}_i(u^{-1})$  and normalize the operators to be unitary wherever the scalar product is nonzero. Prove that their two eigenvalues depend on  $u$  and hence that the spectral family itself does not satisfy one parameter-independent Hecke polynomial.

**Definition 13.20** ( $q$ KZ transport). Let  $v$  be the quantum-group parameter and let  $\kappa \neq 0$  be an independent shift parameter. Let  $D \in \text{End}(V)$  be invertible and satisfy  $[D \otimes D, \check{R}(u)] = 0$ . Let  $D_1$  act on the transported tensor factor and define

$$\mathcal{K}_1(z_1, \dots, z_r) = D_1 \check{R}_{1r}(z_1/z_r) \cdots \check{R}_{12}(z_1/z_2).$$

The first  $q$ KZ equation is

$$\Psi(z_2, \dots, z_r, \kappa z_1) = \mathcal{K}_1(z_1, \dots, z_r) \Psi(z_1, \dots, z_r),$$

and the remaining equations are obtained by cyclic conjugation.

**Exercise 13.20** ( $q$ KZ exchange transport). Use Baxterization to prove consistency of the adjacent exchange rules and the  $q$ KZ difference equations. Identify the constant exchanges and winding operators with the generators of  $\mathcal{H}_v^{\text{aff}}(GL_r)$ , keeping  $v$  and  $\kappa$  independent. For the unitary normalization, prepare and detect three resolved XXZ spin sectors and prove that the two factorizations of a three-spin exchange give the same output probabilities. Express this equality as the  $Q$ -Racah recoupling identity. Derive the phase accumulated in one  $\kappa$ -shift cycle and state the twist, unitarity, preparation, and detection assumptions under which it is an interference observable.

**Definition 13.21** (Temperley–Lieb algebras). For  $d \in \mathbb{C}$ , the Temperley–Lieb algebra  $TL_n(d)$  is generated by  $U_1, \dots, U_{n-1}$  with

$$U_i^2 = dU_i, \quad U_i U_{i\pm 1} U_i = U_i, \quad U_i U_j = U_j U_i \quad (|i - j| > 1).$$

Its diagram trace is normalized by

$$\text{tr}_0(1) = 1, \quad \text{tr}_{n+1}(x) = d \text{tr}_n(x), \quad \text{tr}_{n+1}(x U_n) = \text{tr}_n(x),$$

where  $x \in TL_n(d)$  is embedded in  $TL_{n+1}(d)$ .

**Definition 13.22** (Jones–Wenzl recursion). Put

$$\zeta = e^{\pi i/(k+2)}, \quad d = \zeta + \zeta^{-1}, \quad [n]_\zeta = \frac{\zeta^n - \zeta^{-n}}{\zeta - \zeta^{-1}}.$$

With  $f_0 = f_1 = 1$ , define recursively in  $TL_{n+1}(d)$

$$f_{n+1} = f_n - \frac{[n]_\zeta}{[n+1]_\zeta} f_n U_n f_n \quad (1 \leq n \leq k).$$

**Definition 13.23** (Negligible morphisms). In a pivotal category with trace  $\text{tr}$ , a morphism  $f : X \rightarrow Y$  is *negligible* when

$$\text{tr}(gf) = 0 \quad \text{for every } g : Y \rightarrow X.$$

Let  $\mathcal{N}$  be the tensor ideal of negligible morphisms. Let  $\mathcal{C}_k$  be the idempotent completion of the quotient of the Temperley–Lieb category at  $d = \zeta + \zeta^{-1}$  by  $\mathcal{N}$ , and write  $V_a$  for the image of  $f_a$ ,  $0 \leq a \leq k$ .

**Exercise 13.21** (Jones–Wenzl trace). Use the Temperley–Lieb relations to prove inductively

$$f_n^2 = f_n, \quad U_i f_n = f_n U_i = 0 \quad (1 \leq i < n), \quad \text{tr}(f_n) = [n+1]_\zeta.$$

Deduce  $\text{tr}(f_{k+1}) = [k+2]_\zeta = 0$ .

**Exercise 13.22** (Temperley–Lieb semisimplification). Prove that  $\mathcal{N}$  is the tensor ideal generated by  $f_{k+1}$ , and prove that the quotient removes every trace-zero summand and is finite semisimple.

**Exercise 13.23** (Truncated fusion). Derive

$$V_a \otimes V_b \cong \bigoplus_{\substack{|a-b| \leq c \leq \min(a+b, 2k-a-b) \\ a+b+c \equiv 0 \pmod{2}}} V_c$$

and

$$d_a = [a+1]_\zeta = \frac{\sin((a+1)\pi/(k+2))}{\sin(\pi/(k+2))}.$$

**Definition 13.24** ( $SU(2)_k$  modular data). For  $0 \leq a, b \leq k$ , define

$$S_{ab} = \sqrt{\frac{2}{k+2}} \sin\left(\frac{(a+1)(b+1)\pi}{k+2}\right), \quad \theta_a = \exp\left(\frac{\pi i a(a+2)}{2(k+2)}\right).$$

Let  $N_{ab}^c$  be the multiplicity of  $V_c$  in  $V_a \otimes V_b$ .

**Exercise 13.24** (Verlinde diagonalization). Use sine orthogonality to prove that  $S$  is real, symmetric, and unitary. For each  $a$ , let  $(N_a)_{bc} = N_{ab}^c$ . Use the truncated Chebyshev recursion to prove

$$N_a = S \text{diag}\left(\frac{S_{ax}}{S_{0x}}\right)_{x=0}^k S^{-1}$$

and hence

$$N_{ab}^c = \sum_{x=0}^k \frac{S_{ax} S_{bx} \overline{S_{cx}}}{S_{0x}}.$$

Deduce that the number of fusion channels from  $a_1, \dots, a_n$  to  $c$  is  $(N_{a_1} \cdots N_{a_n})_{0c}$  and express it as a finite sum of products of sine functions. Apply the same formula to the Ising modular matrix derived in Counting and recover the Ising fusion multiplicities.

**Exercise 13.25** (Anyon interference and fusion counting). For simple charges  $a, b$ , prove that the normalized monodromy amplitude is

$$M_{ab} = \frac{S_{ab}S_{00}}{S_{0a}S_{0b}}.$$

In a two-path anyon interferometer with path intensities  $I_1, I_2$ , derive

$$I_b(\phi) = I_1 + I_2 + 2\sqrt{I_1 I_2} \Re(e^{i\phi} M_{ab}).$$

Insert the sine formula and predict the visibility and phase for every enclosed charge. For a collection of prepared charges, use the Verlinde formula to predict the exact number of fusion outcomes in each total-charge sector. Identify indistinguishable charges for one probe and prove that the full family of probes distinguishes them. State the coherence, topological-gap, charge-preparation, and readout assumptions separately from the predicted fringes and integer state counts.

## 14 Integrable Systems

**Exercise 14.1** (Elliptic pendulum period). On  $T^*S^1$  consider

$$H(\theta, p) = \frac{p^2}{2m\ell^2} + mg\ell(1 - \cos \theta).$$

For an oscillation of amplitude  $0 < \theta_0 < \pi$ , set  $k = \sin(\theta_0/2)$  and define

$$K(k) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}.$$

Use energy conservation to prove

$$T(\theta_0) = 4\sqrt{\frac{\ell}{g}} K(k).$$

Apply the generalized binomial theorem with  $a = \frac{1}{2}$ . For

$$I_m = \int_0^{\pi/2} \sin^{2m} \phi \, d\phi,$$

use integration by parts to prove

$$I_0 = \frac{\pi}{2}, \quad I_m = \frac{2m-1}{2m} I_{m-1} \quad (m \geq 1),$$

and obtain

$$K(k) = \frac{\pi}{2} \sum_{m=0}^{\infty} \left( \frac{(1/2)_m}{m!} \right)^2 k^{2m}, \quad |k| < 1.$$

Deduce

$$T(\theta_0) = 2\pi\sqrt{\frac{\ell}{g}} \left( 1 + \frac{\theta_0^2}{16} + O(\theta_0^4) \right).$$

Deduce the experimentally testable increase of period with amplitude and the logarithmic divergence as  $\theta_0 \uparrow \pi$ .

**Definition 14.1** (Sutherland model). For  $N$  identical particles of mass  $m$  on a ring of radius  $R$  and  $g \geq 1$ , let

$$\text{Conf}_N(S^1) = \{(\theta_1, \dots, \theta_N) : \theta_i \neq \theta_j \text{ for } i \neq j\}.$$

On  $C_c^\infty(\text{Conf}_N(S^1))^{S_N}$  define

$$\hat{H}_g = \frac{\hbar^2}{2mR^2} \left[ -\sum_{j=1}^N \partial_{\theta_j}^2 + \frac{g(g-1)}{2} \sum_{i < j} \frac{1}{\sin^2((\theta_i - \theta_j)/2)} \right]$$

and let the same symbol denote its Friedrichs realization in  $L^2((S^1)^N)^{S_N}$ . Define

$$\Psi_0(\theta) = \prod_{i < j} |e^{i\theta_i} - e^{i\theta_j}|^g.$$

For a partition  $\lambda$  of length at most  $N$ , use the monomial symmetric polynomial  $m_\lambda$  and dominance order defined in Counting, evaluated at  $z_j = e^{i\theta_j}$ .

**Definition 14.2** (Jack polynomials). The Jack polynomial  $P_\lambda^{(1/g)}$  is the symmetric polynomial normalized by

$$P_\lambda^{(1/g)} = m_\lambda + \sum_{\mu < \lambda} c_{\lambda\mu}(g) m_\mu$$

and by satisfying

$$\left[ \sum_i (z_i \partial_{z_i})^2 + g \sum_{i < j} \frac{z_i + z_j}{z_i - z_j} (z_i \partial_{z_i} - z_j \partial_{z_j}) \right] P_\lambda^{(1/g)} = \left[ \sum_i (\lambda_i^2 + g(N+1-2i)\lambda_i) \right] P_\lambda^{(1/g)};$$

here  $<$  is dominance order.

**Exercise 14.2** (Sutherland spectrum). Conjugate the quantum Hamiltonian by  $\Psi_0$ , identify its commuting quantum integrals, and prove that

$$\Psi_\lambda = \Psi_0 P_\lambda^{(1/g)}(e^{i\theta_1}, \dots, e^{i\theta_N})$$

is a joint eigenfunction with

$$E_\lambda = \frac{\hbar^2}{2mR^2} \left[ \frac{g^2 N(N^2 - 1)}{12} + \sum_{j=1}^N (\lambda_j^2 + g(N+1-2j)\lambda_j) \right].$$

At  $g = 1$ , prove that  $P_\lambda^{(1)} = s_\lambda$  for the Schur polynomial defined in Counting.

**Exercise 14.3** (Jack transition spectrum). Let  $\rho_r = \sum_{j=1}^N e^{ir\theta_j}$  be a density mode. Derive the Jack Pieri identity in orthonormal normalization for  $\rho_1 P_\lambda^{(1/g)}$  and show that only nonincreasing tuples obtained from  $\lambda$  by increasing one entry by one occur. For a weak periodic drive proportional to  $\rho_1 e^{-i\Omega t} + \rho_{-1} e^{i\Omega t}$ , deduce the allowed resonance frequencies

$$\Omega_{\mu\lambda} = \frac{|E_\mu - E_\lambda|}{\hbar}$$

and express their relative intensities as squared normalized Jack Pieri coefficients. State how spectroscopy of particles confined to a ring determines  $g$  from the line shifts and tests the Jack selection rule from the missing lines.

**Definition 14.3** (Double-affine braid group). For  $r \geq 2$ , the *double-affine braid group*  $\tilde{B}_r$  of type  $GL_r$  is generated by  $T_i$  for  $1 \leq i < r$ , commuting invertible elements  $X_j$ , commuting invertible elements  $Y_j$ , and a central invertible element  $\mathbf{q}$ , subject to the braid relations and

$$\begin{aligned} T_i X_i T_i &= X_{i+1}, & T_i X_j &= X_j T_i \quad (j \neq i, i+1), \\ T_i^{-1} Y_i T_i^{-1} &= Y_{i+1}, & T_i Y_j &= Y_j T_i \quad (j \neq i, i+1), \\ Y_1 X_1 \cdots X_r &= \mathbf{q} X_1 \cdots X_r Y_1, & X_1^{-1} Y_2 &= Y_2 X_1^{-1} T_1^{-2}. \end{aligned}$$

**Exercise 14.4** (From affine to double-affine braids). Let  $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ . Interpret  $T_i$  as an adjacent braid and  $X_j, Y_j$  as motions of the  $j$ th point around the two cycles of  $T^2$ . Derive the displayed relations from these motions and identify  $\tilde{B}_r$  with the central extension of

$$\pi_1(\text{Conf}_r(T^2)/S_r)$$

whose total cycle commutator is  $\mathbf{q}$ . Prove that deleting either the  $X$ -family or the  $Y$ -family leaves an affine braid group.

**Definition 14.4** (Type- $A$  double affine Hecke algebra). For nonzero parameters  $q, t$ , the *double affine Hecke algebra*  $H_{q,t}(GL_r)$  is the quotient of  $\mathbb{C}(q, t)[\tilde{B}_r]$  by

$$\mathbf{q} = q, \quad (T_i - t^{1/2})(T_i + t^{-1/2}) = 0.$$

**Definition 14.5** (Rank-one DAHA). For nonzero  $k_0, k_1, u_0, u_1$  and a choice of  $q^{1/2}$ , the rank-one DAHA  $H_q(C_1^\vee, C_1)$  is generated by  $T_0^{\pm 1}, T_1^{\pm 1}, (T_0^\vee)^{\pm 1}, (T_1^\vee)^{\pm 1}$  subject to

$$\begin{aligned} (T_i - k_i)(T_i + k_i^{-1}) &= 0, & (T_i^\vee - u_i)(T_i^\vee + u_i^{-1}) &= 0 \quad (i = 0, 1), \\ T_1^\vee T_1 T_0 T_0^\vee &= q^{-1/2}. \end{aligned}$$

For the finite Hecke generator  $T_1$ , define

$$e = \frac{T_1 + k_1^{-1}}{k_1 + k_1^{-1}}.$$

**Definition 14.6** (Rank-one polynomial representation). Fix nonzero  $k_0, k_1, u_0, u_1$  and choose  $q^{1/2}$ . On  $\mathcal{L} = \mathbb{C}[z^{\pm 1}]$  put

$$\begin{aligned} (s_1 f)(z) &= f(z^{-1}), & (s_0 f)(z) &= f(qz^{-1}), \\ (T_1 f)(z) &= k_1 f(z^{-1}) + \frac{(k_1 - k_1^{-1}) + (u_1 - u_1^{-1})z}{1 - z^2} (f(z) - f(z^{-1})), \\ (T_0 f)(z) &= k_0 f(qz^{-1}) + \frac{(k_0 - k_0^{-1}) + (u_0 - u_0^{-1})q^{1/2}z^{-1}}{1 - qz^{-2}} (f(z) - f(qz^{-1})). \end{aligned}$$

Let  $X$  act by multiplication by  $z$ . The Askey–Wilson parameters in this convention are

$$a = k_1 u_1, \quad b = -k_1 u_1^{-1}, \quad c = q^{1/2} k_0 u_0, \quad d = -q^{1/2} k_0 u_0^{-1}.$$

Put

$$T_1^\vee = X^{-1} T_1^{-1}, \quad T_0^\vee = q^{-1/2} T_0^{-1} X.$$

**Definition 14.7** (Generalized hypergeometric series). If no denominator parameter  $b_j$  is a nonpositive integer, define

$${}_rF_s \left( \begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; z \right) = \sum_{m=0}^{\infty} \frac{(a_1)_m \cdots (a_r)_m}{(b_1)_m \cdots (b_s)_m} \frac{z^m}{m!}.$$

The series is finite when a numerator parameter is a nonpositive integer. Otherwise it converges for every  $z$  when  $r \leq s$ , for  $|z| < 1$  when  $r = s + 1$ , and only at  $z = 0$  when  $r > s + 1$ .

**Definition 14.8** (Askey–Wilson operator and polynomials). For  $0 < q < 1$  and nonzero  $a, b, c, d$ , define

$$A(z) = \frac{(1-az)(1-bz)(1-cz)(1-dz)}{(1-z^2)(1-qz^2)}$$

and, on  $f(z) = f(z^{-1})$ ,

$$(\mathcal{D}_{AW}f)(z) = A(z)(f(qz) - f(z)) + A(z^{-1})(f(q^{-1}z) - f(z)).$$

For  $x = (z + z^{-1})/2$ , the *Askey–Wilson polynomial* is

$$p_n(x; a, b, c, d \mid q) = a^{-n}(ab, ac, ad; q)_n \times {}_4\phi_3 \left( \begin{matrix} q^{-n}, abcdq^{n-1}, az, az^{-1} \\ ab, ac, ad \end{matrix}; q, q \right).$$

Its weight on the unit circle is

$$w_{AW}(z) = \frac{(z^2, z^{-2}; q)_{\infty}}{(az, a/z, bz, b/z, cz, c/z, dz, d/z; q)_{\infty}}.$$

**Exercise 14.5** (Rank-one spherical operator). Verify that the displayed reflection–difference operators preserve  $\mathcal{L}$ , that  $T_0, T_1, T_0^{\vee}, T_1^{\vee}$  satisfy the four quadratic Hecke relations and the product relation, and that they give the rank-one DAHA polynomial representation with the stated parameter map. Prove

$$e\mathcal{L} = \mathcal{L}^{z \mapsto z^{-1}} = \mathbb{C}[z + z^{-1}],$$

and prove that the spherical action contains multiplication by  $z + z^{-1}$  and  $\mathcal{D}_{AW}$ .

**Exercise 14.6** (Askey–Wilson spectral theorem). Prove

$$\mathcal{D}_{AW}p_n = (q^{-n} - 1)(1 - abcdq^{n-1})p_n.$$

For  $a, b, c, d \in (-1, 1)$ , prove that  $\mathcal{D}_{AW}$  is symmetric for

$$\langle f, g \rangle_{AW} = \int_{|z|=1} f(z) \overline{g(z)} w_{AW}(z) \frac{dz}{2\pi iz},$$

and deduce the orthogonality and three-term recurrence of the Askey–Wilson polynomials.

**Definition 14.9** (Wilson polynomials). For positive  $a, b, c, d$ , define

$$W_n(x^2; a, b, c, d) = (a+b)_n(a+c)_n(a+d)_n \times {}_4F_3 \left( \begin{matrix} -n, n+a+b+c+d-1, a+ix, a-ix \\ a+b, a+c, a+d \end{matrix}; 1 \right).$$



**Exercise 14.7** (Wilson degeneration). Set  $q = e^{-\varepsilon}$  and scale the Askey–Wilson parameters and spectral variable by

$$(a, b, c, d, z) = (q^A, q^B, q^C, q^D, q^{ix}).$$

After the scalar normalization by  $(1 - q)^{-3n}$ , prove that the Askey–Wilson polynomial tends to  $W_n(x^2; A, B, C, D)$  as  $\varepsilon \downarrow 0$ . Derive the Wilson difference equation, weight, orthogonality, and three-term recurrence as limits of the corresponding Askey–Wilson identities.

**Definition 14.10** (Type- $A$  polynomial representation). Let

$$\mathcal{P}_r = \mathbb{C}(q, t^{1/2})[x_1^{\pm 1}, \dots, x_r^{\pm 1}],$$

let  $X_j$  act by multiplication by  $x_j$ , and let  $s_i$  exchange  $x_i$  and  $x_{i+1}$ . Define

$$T_i = t^{1/2}s_i + (t^{1/2} - t^{-1/2}) \frac{s_i - 1}{x_i/x_{i+1} - 1}, \quad 1 \leq i < r,$$

and

$$(\pi f)(x_1, \dots, x_r) = f(x_2, \dots, x_r, qx_1).$$

For generic independent parameters  $q, t$ , call the operator algebra generated by  $X_j^{\pm 1}$ ,  $T_i$ , and  $\pi^{\pm 1}$  the type- $A$  polynomial operator model.

**Exercise 14.8** (DAHA polynomial quotient). Prove that the displayed operators preserve  $\mathcal{P}_r$  and satisfy the braid and Hecke relations. Prove

$$\pi X_i \pi^{-1} = X_{i+1} \quad (1 \leq i < r), \quad \pi X_r \pi^{-1} = qX_1,$$

and identify the two affine-Hecke subalgebras generated respectively by  $T_i, X_j$  and by  $T_i, Y_j$ . Prove that this representation is obtained from the double-affine braid-group algebra by imposing

$$(T_i - t^{1/2})(T_i + t^{-1/2}) = 0$$

and specializing its central parameter to  $q$ . Conclude that the polynomial operator model is a realization of  $\mathbf{H}_{q,t}(GL_r)$ .

**Definition 14.11** (Macdonald operators and polynomials). Let

$$(T_{q,x_i} f)(x_1, \dots, x_r) = f(x_1, \dots, qx_i, \dots, x_r).$$

The first type- $A$  Macdonald operator is

$$D_{q,t} = \sum_{i=1}^r \left( \prod_{j \neq i} \frac{tx_i - x_j}{x_i - x_j} \right) T_{q,x_i}.$$

For generic  $q, t$ , the Macdonald polynomial  $P_\lambda(x; q, t)$  is the symmetric polynomial triangular in the monomial symmetric functions and satisfying

$$D_{q,t} P_\lambda = \left( \sum_{i=1}^r q^{\lambda_i} t^{r-i} \right) P_\lambda, \quad \lambda_1 \geq \dots \geq \lambda_r \geq 0.$$

For  $0 < q, t < 1$ , define

$$\Delta_{q,t}(x) = \prod_{i \neq j} \frac{(x_i/x_j; q)_\infty}{(tx_i/x_j; q)_\infty},$$

and

$$\langle f, g \rangle_{q,t} = \frac{1}{r!} \int_{(S^1)^r} f(x) \overline{g(x)} \Delta_{q,t}(x) \prod_{j=1}^r \frac{dx_j}{2\pi i x_j}.$$

**Exercise 14.9** (Cherednik operators). Define

$$Y_i = T_i \cdots T_{r-1} \pi T_1^{-1} \cdots T_{i-1}^{-1}.$$

Use the displayed polynomial representation and the affine braid relations to prove that the  $Y_i$  commute. Prove that their symmetric Laurent polynomials preserve  $\mathcal{P}_r^{S_r}$  and identify  $t^{-(r-1)/2}(Y_1 + \cdots + Y_r)$  with  $D_{q,t}$ .

**Definition 14.12** (DAHA Fourier assignment). Define the DAHA Fourier assignment on the torus, winding, and Hecke generators by

$$\mathcal{F}(X_j) = Y_j^{-1}, \quad \mathcal{F}(Y_j) = X_j, \quad \mathcal{F}(T_i) = T_i.$$

For  $\rho = (r-1, r-2, \dots, 0)$  and a partition  $\lambda$ , put

$$\xi_\lambda = (q^{\lambda_1} t^{r-1}, q^{\lambda_2} t^{r-2}, \dots, q^{\lambda_r}).$$

**Exercise 14.10** (Macdonald Fourier duality). Verify the DAHA relations after applying  $\mathcal{F}$ , prove that it extends to an automorphism, and construct its intertwining transform on the polynomial representation. Normalize  $P_\lambda$  by its value at  $t^\rho$  and prove the evaluation symmetry

$$\frac{P_\lambda(\xi_\mu; q, t)}{P_\lambda(t^\rho; q, t)} = \frac{P_\mu(\xi_\lambda; q, t)}{P_\mu(t^\rho; q, t)}.$$

Derive the duality between multiplication recurrences in  $\lambda$  and the commuting Macdonald difference equations in the spectral variable.

**Exercise 14.11** (Macdonald spectral theory). Order monomial symmetric functions by dominance and prove that  $D_{q,t}$  is triangular with the displayed diagonal entries. For generic  $q, t$ , deduce existence and uniqueness of  $P_\lambda$ . Prove directly from the torus integral that  $D_{q,t}$  is symmetric when  $0 < q, t < 1$ , and deduce orthogonality and joint diagonalization for symmetric Laurent polynomials in the commuting  $Y_i$ .

**Definition 14.13** (Symmetric-function Macdonald pairing). Let  $\Lambda_{\text{sym}} = \mathbb{C}[p_1, p_2, \dots]$  be graded by  $\deg p_r = r$ . For a partition  $\mu$ , let  $m_r(\mu)$  be the number of its parts equal to  $r$ , and define

$$z_\mu = \prod_{r \geq 1} r^{m_r(\mu)} m_r(\mu)!, \quad p_\mu = \prod_j p_{\mu_j}, \quad p_r(x) = \sum_i x_i^r.$$

Define the symmetric-function Macdonald pairing by

$$\langle p_\mu, p_\nu \rangle_{q,t}^{\text{sym}} = \delta_{\mu\nu} z_\mu \prod_j \frac{1 - q^{\mu_j}}{1 - t^{\mu_j}}.$$

For a cell  $s = (i, j)$  in the diagram of  $\lambda$ , define its arm and leg by

$$a_\lambda(s) = \lambda_i - j, \quad \ell_\lambda(s) = \lambda'_j - i,$$

and put

$$b_\lambda(q, t) = \prod_{s \in \lambda} \frac{1 - q^{a_\lambda(s)} t^{\ell_\lambda(s)+1}}{1 - q^{a_\lambda(s)+1} t^{\ell_\lambda(s)}}.$$

**Exercise 14.12** (Stable Macdonald pairing and Cauchy identity). Prove that the finite-variable Macdonald polynomials are stable under adjoining a zero variable and hence determine elements  $P_\lambda$  of  $\Lambda_{\text{sym}}$ . Put  $Q_\lambda = b_\lambda(q, t)P_\lambda$ . Use the adjointness of the DAHA difference operators and triangularity to prove

$$\langle P_\lambda, P_\mu \rangle_{q, t}^{\text{sym}} = \delta_{\lambda\mu} b_\lambda(q, t)^{-1}.$$

Deduce that  $P_\lambda$  and  $Q_\lambda$  are dual. Expand the reproducing kernel in the power sums and use the  $q$ -binomial identity to prove

$$\sum_{\lambda} P_\lambda(a; q, t) Q_\lambda(b; q, t) = \prod_{i, j} \frac{(ta_i b_j; q)_\infty}{(a_i b_j; q)_\infty}$$

for finite alphabets with  $|a_i b_j| < 1$ .

**Definition 14.14** (Finite Macdonald measures). Let  $0 \leq q, t < 1$ , let  $a = (a_1, \dots, a_N)$  and  $b = (b_1, \dots, b_M)$  be nonnegative with  $a_i b_j < 1$ , and let  $P_\lambda, Q_\lambda$  be the preceding Macdonald functions. Define

$$\Pi(a; b) = \sum_{\lambda} P_\lambda(a; q, t) Q_\lambda(b; q, t), \quad \mathbb{P}_{q, t}(\lambda) = \frac{P_\lambda(a; q, t) Q_\lambda(b; q, t)}{\Pi(a; b)}.$$

**Exercise 14.13** (One Macdonald observable). Apply the first Macdonald difference operator in the  $a$  variables to the Cauchy identity. For distinct  $a_i$ , prove

$$\mathbb{E}_{q, t} \left[ \sum_{i=1}^N q^{\lambda_i} t^{N-i} \right] = \sum_{i=1}^N \prod_{j \neq i} \frac{ta_i - a_j}{a_i - a_j} \prod_{k=1}^M \frac{1 - a_i b_k}{1 - ta_i b_k},$$

and extend the identity to coincident parameters by continuity. Verify the Schur specialization  $q = t$  and the  $q$ -Whittaker specialization  $t = 0$ . For samples from the finite measure, identify the displayed expectation as a directly estimable statistic and state the positivity, truncation, and sampling assumptions required for comparison with the formula.

**Definition 14.15** (Multiplicative theta function). For  $|p| < 1$ , define

$$\theta(z; p) = (z; p)_\infty (p/z; p)_\infty.$$

**Definition 14.16** (Elliptic Ruijsenaars operator). For type  $A_{r-1}$ , define

$$D_{p, q, t}^{\text{ell}} = \sum_{i=1}^r \left( \prod_{j \neq i} \frac{\theta(tx_i/x_j; p)}{\theta(x_i/x_j; p)} \right) T_{q, x_i}.$$

**Exercise 14.14** (Ruijsenaars degeneration). Prove directly from  $\theta(z; 0) = 1 - z$  that

$$D_{p, q, t}^{\text{ell}} \longrightarrow D_{q, t} \quad (p \rightarrow 0).$$

Set  $q = e^{-\varepsilon}$  and  $t = e^{-g\varepsilon}$  and derive the Heckman–Opdam differential operator from the second-order term as  $\varepsilon \rightarrow 0$ . Take the rational scaling limit of its hyperbolic coefficients and obtain the rational Calogero–Moser operator.

**Exercise 14.15** (The Calogero–Sutherland limit). Write  $x_i = e^{z_i}$ , set  $q = e^{-\varepsilon}$  and  $t = e^{-g\varepsilon}$ , and expand  $D_{q, t}$  through order  $\varepsilon^2$ . After subtracting its scalar value on 1 and removing the center-of-mass drift, obtain

$$\mathcal{L}_g = \sum_i \partial_{z_i}^2 + g \sum_{i < j} \coth\left(\frac{z_i - z_j}{2}\right) (\partial_{z_i} - \partial_{z_j}).$$

On the circle, put  $z_i = i\theta_i$  and conjugate by

$$\Psi_0(\theta) = \prod_{i < j} \left| 2 \sin \frac{\theta_i - \theta_j}{2} \right|^g.$$

Recover, up to its ground-state energy, the Sutherland Hamiltonian

$$H_g = - \sum_i \partial_{\theta_i}^2 + \frac{g(g-1)}{2} \sum_{i < j} \frac{1}{\sin^2((\theta_i - \theta_j)/2)}.$$

Compute the eigenfunctions of degrees zero and one and, for  $r = 2$ , degree two. Identify their Sutherland eigenvalues.

**Exercise 14.16** (Hypergeometric degeneration). Set  $q = e^{-\varepsilon}$ . Prove

$$\lim_{\varepsilon \rightarrow 0^+} \frac{(q^a; q)_m}{(1-q)^m} = (a)_m, \quad \lim_{q \rightarrow 1^-} (1-q)^{1-a} \frac{(q; q)_\infty}{(q^a; q)_\infty} = \Gamma(a).$$

Deduce

$$\lim_{q \rightarrow 1^-} {}_r\phi_{r-1} \left( \begin{matrix} q^{a_1}, \dots, q^{a_r} \\ q^{b_1}, \dots, q^{b_{r-1}} \end{matrix}; q, z \right) = {}_rF_{r-1} \left( \begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_{r-1} \end{matrix}; z \right).$$

Derive Gauss's summation from the  $q$ -Gauss summation. Prove the corresponding degeneration of the basic hypergeometric  $q$ -difference equation to the Gauss equation.

**Definition 14.17** ( $BC_r$  root-system weight). Write  $f(z^{\pm m}) = f(z^m)f(z^{-m})$  and

$$f(z_i^{\pm 1} z_j^{\pm 1}) = \prod_{\epsilon, \delta = \pm 1} f(z_i^\epsilon z_j^\delta).$$

For  $|q|, |t|, |a_s| < 1$ , define

$$\begin{aligned} \Delta_{BC_r}(z) &= \prod_{i=1}^r \frac{(z_i^{\pm 2}; q)_\infty}{\prod_{s=1}^4 (a_s z_i^{\pm 1}; q)_\infty} \\ &\quad \times \prod_{1 \leq i < j \leq r} \frac{(z_i^{\pm 1} z_j^{\pm 1}; q)_\infty}{(t z_i^{\pm 1} z_j^{\pm 1}; q)_\infty}. \end{aligned}$$

Use the normalized torus measure

$$\frac{1}{2^r r!} \prod_{i=1}^r \frac{dz_i}{2\pi i z_i}.$$

**Definition 14.18** (Koornwinder operator). For  $z = (z_1, \dots, z_r)$  put

$$A_i(z) = \frac{\prod_{s=1}^4 (1 - a_s z_i)}{(1 - z_i^2)(1 - q z_i^2)} \prod_{j \neq i} \frac{(1 - t z_i z_j)(1 - t z_i / z_j)}{(1 - z_i z_j)(1 - z_i / z_j)}.$$

On Laurent polynomials invariant under permutations and inversions define

$$\mathcal{D}_K = \sum_{i=1}^r \left[ A_i(z)(T_{q, z_i} - 1) + A_i(z^{-1})(T_{q, z_i}^{-1} - 1) \right].$$

The *Koornwinder polynomial*  $K_\lambda$  is the Weyl-invariant Laurent polynomial triangular in orbit sums and satisfying

$$\mathcal{D}_K K_\lambda = \left( \sum_{i=1}^r \left[ a_1 a_2 a_3 a_4 q^{-1} t^{2r-i-1} (q^{\lambda_i} - 1) + t^{i-1} (q^{-\lambda_i} - 1) \right] \right) K_\lambda,$$

with the scalar eigenvalue fixed by the displayed convention.

**Exercise 14.17** ( $BC_r$  weight). Prove that  $\Delta_{BC_r}$  is invariant under permutations and independent inversions of the  $z_i$ . Identify its factors with the roots  $\pm e_i$ ,  $\pm 2e_i$ , and  $\pm e_i \pm e_j$ . For  $r = 1$ , recover the Askey–Wilson weight and its order-two Weyl normalization, and identify  $\mathcal{D}_K$  with  $\mathcal{D}_{AW}$ .

**Exercise 14.18** (Koornwinder spectral theory). Prove that  $\mathcal{D}_K$  is symmetric for the  $\Delta_{BC_r}$  inner product and that its triangular eigenfunctions are mutually orthogonal when the displayed eigenvalues are distinct.

**Exercise 14.19** ( $BC_r$  limit). Set  $t = q^g$  with  $g \in \mathbb{N}$ , reduce the quotients of  $q$ -products to finite products, and compute the limit  $q \rightarrow 1^-$  with  $z_i = e^{i\theta_i}$ .

**Exercise 14.20** (Askey–Wilson spectrum). Fix  $M \geq 4$  and parameters  $a, b, c, d \in (-1, 1)$  for which the Askey–Wilson weight is positive and the first  $M + 1$  eigenvalues are distinct. Let  $\mathcal{H}_M$  be the span of the normalized Askey–Wilson polynomials  $p_0, \dots, p_M$  with their positive Askey–Wilson inner product. Model a finite programmable  $q$ -difference simulator by the self-adjoint Hamiltonian  $\hat{H}_M = \varepsilon \mathcal{D}_{AW}|_{\mathcal{H}_M}$ , where  $\varepsilon > 0$  is a calibrated energy scale. Prove that its spectral values are

$$E_n = \varepsilon(q^{-n} - 1)(1 - abcdq^{n-1}).$$

Write the three-term recurrence in the form

$$(z + z^{-1})p_n = A_n p_{n+1} + B_n p_n + C_n p_{n-1}$$

and compute the corresponding normalized matrix elements. Deduce that a periodic perturbation proportional to the compression of  $z + z^{-1}$  to  $\mathcal{H}_M$  produces only neighboring transitions, with resonance frequencies  $|E_{n+1} - E_n|/\hbar$  and intensities determined by  $A_n$  and  $C_{n+1}$ . Assuming  $\varepsilon$  is known and at least three nondegenerate lines are resolved, derive the equations satisfied by  $q$  and the product  $abcd$ . Compute the Jacobian of two independent gap ratios and state the nonvanishing and parameter-range hypotheses under which these two quantities are locally identifiable.

Form the data map from  $(a, b, c, d)$  to four selected squared normalized neighboring matrix elements. Prove that it is invariant under permutation of the parameters. At a chosen parameter point, compute its differential in the four elementary symmetric functions and state the full-rank hypothesis under which the measured strengths locally determine the unordered set  $\{a, b, c, d\}$ . When this rank condition fails, identify the remaining fiber instead of asserting identifiability; state also that squared strengths cannot distinguish any additional discrete ambiguity in that fiber. State that these are predictions for the engineered simulator.

**Exercise 14.21** (Macdonald–Ruijsenaars spectrum). Let  $0 < q, t < 1$ , fix  $M \geq 0$ , and define

$$\mathcal{H}_{q,t}^{(M)} = \text{span}\{P_\lambda(x; q, t) : |\lambda| \leq M\}$$

with the positive Macdonald inner product. Put

$$d_0 = \sum_{i=1}^r t^{r-i}, \quad \hat{H}_{q,t}^{(M)} = \varepsilon(d_0 - D_{q,t})|_{\mathcal{H}_{q,t}^{(M)}}, \quad \varepsilon > 0.$$

Identify  $D_{q,t}$  with the first symmetric function of the commuting DAHA winding operators and derive its compatibility with the Hecke exchange relations. Prove that  $\mathcal{H}_{q,t}^{(M)}$  is invariant, that  $\hat{H}_{q,t}^{(M)}$  is self-adjoint, and that the normalized Macdonald states have energies

$$E_\lambda = \varepsilon \sum_{i=1}^r t^{r-i} (1 - q^{\lambda_i}).$$

**Exercise 14.22** (Macdonald transition spectrum). Model a finite programmable  $q$ -difference simulator by this operator. Use the Macdonald Pieri identity for  $\rho_1 = x_1 + \cdots + x_r$  and its adjoint to determine every nonzero matrix element of the compression of  $\rho_1 + \rho_1^*$  to  $\mathcal{H}_{q,t}^{(M)}$ . For a periodic drive proportional to this compressed observable, derive the resonance frequencies

$$\Omega_{\mu\lambda} = |E_\mu - E_\lambda|/\hbar$$

and the relative intensities as squared normalized Macdonald Pieri coefficients. Explain that recovery of  $q, t$  requires known  $\varepsilon$  and  $r$ , resolved partition labels, sufficiently many nondegenerate transitions, and a fixed convention for any discrete parameter ambiguity, and that the prediction concerns the engineered simulator.

**Exercise 14.23** (Macdonald–Jack degeneration). Set  $q = e^{-\delta}$  and  $t = e^{-g\delta}$ ; after the scalar and center-of-mass normalizations in the Calogero–Sutherland limit exercise, prove that the rescaled spectral gaps and transition coefficients converge to the Sutherland energies and Jack Pieri coefficients. State how measured line locations and strengths test this degeneration. Explain that recovery of the deformation parameter requires resolved partition labels and sufficiently many nondegenerate transitions.

**Definition 14.19** (Elliptic shifted factorials and gamma function). For  $|p|, |q| < 1$ , define

$$\theta(z_1, \dots, z_s; p) = \prod_{j=1}^s \theta(z_j; p),$$

$$(a; q, p)_m = \prod_{j=0}^{m-1} \theta(aq^j; p),$$

$$(a_1, \dots, a_s; q, p)_m = \prod_{i=1}^s (a_i; q, p)_m,$$

and

$$\Gamma(z; p, q) = \prod_{j,k \geq 0} \frac{1 - p^{j+1}q^{k+1}/z}{1 - p^j q^k z}.$$

**Definition 14.20** (Deformation hierarchy). For a series  $\sum_m c_m$ , set  $h_m = c_{m+1}/c_m$ . It is *ordinary hypergeometric* when  $h_m$  is rational in  $m$ , and *basic hypergeometric* when  $h_m$  is rational in  $q^m$ . It is *elliptic hypergeometric* when, after writing  $x = q^m$ , its term ratio has the form

$$h(x) = z \frac{\prod_{i=1}^s \theta(a_i x; p)}{\prod_{i=1}^s \theta(b_i x; p)}.$$

Its parameters are *balanced* when

$$\prod_{i=1}^s a_i = \prod_{i=1}^s b_i.$$

By theta quasi-periodicity this is equivalent to

$$h(px) = h(x).$$

**Exercise 14.24** (Elliptic degenerations). Prove

$$\begin{aligned}\theta(pz; p) &= -z^{-1}\theta(z; p), & \Gamma(qz; p, q) &= \theta(z; p)\Gamma(z; p, q), \\ \Gamma(pz; p, q) &= \theta(z; q)\Gamma(z; p, q), & \Gamma(z; p, q)\Gamma(pq/z; p, q) &= 1.\end{aligned}$$

Prove

$$\theta(z; 0) = 1 - z, \quad (a; q, 0)_m = (a; q)_m, \quad \Gamma(z; 0, q) = \frac{1}{(z; q)_\infty}.$$

**Definition 14.21** (Frenkel–Turaev sum). For  $a^2q^{n+1} = bcde$ , define

$$\begin{aligned}_{10}V_9(a; b, c, d, e, q^{-n}; q, p) &= \sum_{m=0}^n \frac{\theta(aq^{2m}; p)}{\theta(a; p)} \\ &\quad \times \frac{(a, b, c, d, e, q^{-n}; q, p)_m}{(q, aq/b, aq/c, aq/d, aq/e, aq^{n+1}; q, p)_m} q^m.\end{aligned}$$

**Exercise 14.25** (Theta addition and Frenkel–Turaev summation). Prove the multiplicative theta addition formula

$$\begin{aligned}\theta(xz, x/z, yw, y/w; p) - \theta(xw, x/w, yz, y/z; p) \\ = \frac{y}{z} \theta(xy, x/y, zw, z/w; p)\end{aligned}$$

by comparing quasi-periodicity and zeros. Use it to telescope the difference between the  $n$ th and  $(n-1)$ st terminating sums and prove

$$_{10}V_9(a; b, c, d, e, q^{-n}; q, p) = \frac{(aq, aq/(bc), aq/(bd), aq/(cd); q, p)_n}{(aq/b, aq/c, aq/d, aq/(bcd); q, p)_n}.$$

Take  $p \rightarrow 0$  and identify the terminating very-well-poised basic hypergeometric summation.

**Definition 14.22** (Elliptic beta kernel). Write

$$\Gamma(tz^{\pm 1}; p, q) = \Gamma(tz; p, q)\Gamma(t/z; p, q), \quad \kappa_{p,q} = \frac{(p; p)_\infty (q; q)_\infty}{2}.$$

For  $|t_j| < 1$  with  $\prod_{j=1}^6 t_j = pq$ , define

$$\mathcal{B}(t_1, \dots, t_6) = \kappa_{p,q} \int_{\mathbb{T}} \frac{\prod_{j=1}^6 \Gamma(t_j z^{\pm 1}; p, q)}{\Gamma(z^{\pm 2}; p, q)} \frac{dz}{2\pi i z}.$$

**Exercise 14.26** (Elliptic beta recurrence). Use the elliptic-gamma difference equations to derive the parameter-shift recurrences of  $\mathcal{B}$ .

**Exercise 14.27** (Elliptic beta integral). Evaluate a residue degeneration for which two parameters multiply to  $pq$  and use analytic continuation to prove

$$\mathcal{B}(t_1, \dots, t_6) = \prod_{1 \leq i < j \leq 6} \Gamma(t_i t_j; p, q).$$

Take  $p \rightarrow 0$  and identify the resulting basic hypergeometric beta integral.

**Definition 14.23** (Elliptic integral operators). For  $|t| < 1$  define

$$(\mathcal{M}_t f)(w) = \frac{\kappa_{p,q}}{\Gamma(t^2; p, q)} \int_{\mathbb{T}} \frac{\Gamma(tw^{\pm 1}z^{\pm 1}; p, q)}{\Gamma(z^{\pm 2}; p, q)} f(z) \frac{dz}{2\pi iz},$$

and

$$\mathcal{D}_s(y, w) = \Gamma\left((pq)^{1/2} s^{-1} y^{\pm 1} w^{\pm 1}; p, q\right).$$

**Exercise 14.28** (Elliptic star–triangle relation). Apply the elliptic beta integral to the internal integration variable and prove the operator identity

$$\mathcal{M}_s \mathcal{D}_{st}(y, \cdot) \mathcal{M}_t = \mathcal{D}_t(y, \cdot) \mathcal{M}_{st} \mathcal{D}_s(y, \cdot)$$

whenever the contours separate reciprocal pole sequences and the parameters satisfy the corresponding modulus bounds. Interpret the kernels as edge weights and  $\mathcal{D}$  as a site weight of a continuous spin model. Prove that the two local triangular partition functions are equal and that the row transfer operators obtained by the two factorizations commute. For a finite periodic strip, deduce equality of the transfer spectra and partition functions under a star–triangle move; state the positivity regime and contour assumptions under which these are statistical-mechanical observables.